DERMOAI

An AI-Assisted Triage and Telemedicine Framework for Extending Dermatological Specialist Services in Resource-Limited Primary Care Settings

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Repository: https://github.com/reponseashimwe/dermoai

DECLARATION

I, Reponse Ashimwe Sibomana, declare that this capstone research report titled "DermoAI: An AI-Assisted Triage and Telemedicine Framework for Extending Dermatological Specialist Services in Resource-Limited Primary Care Settings" is my original work, unless stated, and all external sources have been referenced or cited in my document. This work has not been presented for award of degree or for any similar purpose in any other university.

Signature Date: 15th March 2026

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CERTIFICATION

The undersigned certifies that she has read and hereby recommends for acceptance by African Leadership University a report entitled "DermoAI: An AI-Assisted Triage and Telemedicine Framework for Extending Dermatological Specialist Services in Resource-Limited Primary Care Settings."

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DEDICATION AND ACKNOWLEDGEMENTS

This work is dedicated to every frontline health worker in rural Rwanda who makes difficult clinical decisions daily without the tools they deserve. It is for the patients whose access to specialist care is determined by geography rather than need. May this work contribute, however modestly, to changing that.

I would like to express my sincere gratitude to my supervisor, Samiratu Nthosi, for her guidance, patience, and consistent support throughout this project. Her feedback sharpened both the research direction and the quality of the final work.

I am deeply grateful to Ogore for introducing me to machine learning with clarity, rigour, and genuine enthusiasm. The foundation built in those sessions made this project possible in ways that are difficult to overstate. My thanks also go to Simeon for making the study of ethics not only intellectually serious but genuinely enjoyable. The questions raised in those classes shaped how I thought about algorithmic fairness, data sovereignty, and responsible deployment throughout this work.

To my family, thank you for the unwavering support, the patience during long nights, and the belief that this work was worth finishing. None of this would have been possible without you.

Finally, I would like to thank the medical practitioners who gave their time to participate in the simulated acceptance testing sessions. Their clinical perspective was invaluable and their willingness to engage seriously with an early-stage system was encouraging beyond what formal acknowledgement can capture.

Abstract

Rwanda has fewer than 20 dermatologists serving over 13 million people. General practitioners achieve 58–70% diagnostic accuracy for skin conditions, yet fewer than 6% feel confident independently assessing dermatological disorders, leading to empirical prescribing and antimicrobial resistance. Existing AI tools demonstrate 30–40% performance degradation on darker skin tones due to severe underrepresentation of Fitzpatrick Skin Types V and VI in training datasets.

This study developed and validated DermoAI, a web-based AI-assisted dermatological triage framework enabling frontline health workers in Rwanda to manage routine skin conditions locally while facilitating remote specialist consultation for referral cases. The system employs an EfficientNetB0 convolutional neural network trained on 1,250 augmented FST V–VI images, translating condition predictions into binary Refer or Manage Locally recommendations through conservative rule-based triage mapping integrated with telemedicine capabilities.

Evaluation on a held-out test set of 90 FST V–VI images achieved 76.67% overall accuracy, exceeding the GP diagnostic baseline and the 70% project threshold. The two-stage triage mechanism achieved 95.35% Refer recall with only two critical misclassifications, exceeding the 75% patient safety target. Simulated acceptance testing with three clinicians produced mean scores above 4.0 across all clinical dimensions, with frontline usefulness rated at 4.67 out of 5. An FST accuracy gap of 11.36% passed the defined equity threshold.

These results establish technical feasibility and preliminary clinical utility. Future work should prioritise PASSION dataset integration, condition set expansion beyond the current five classes, and a prospective clinical trial with Rwanda Ministry of Health approval to progress toward regulatory certification.

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List of acronyms/abbreviations

AI	Artificial Intelligence
AMR	Antimicrobial Resistance
API	Application Programming Interface
CDSS	Clinical Decision Support System
CHW	Community Health Worker
CNN	Convolutional Neural Network
DSR	Design Science Research
ERD	Entity Relationship Diagram
FST	Fitzpatrick Skin Type
GP	General Practitioner
GradCAM	Gradient-weighted Class Activation Mapping
ISIC	International Skin Imaging Collaboration
ISO	International Organization for Standardization
JWT	JSON Web Token
LMIC	Low and Middle Income Country
mHealth	Mobile Health
PWA	Progressive Web Application
RWF	Rwandan Franc
SAF	Store-and-Forward
SMS	Short Message Service
WHO	World Health Organization
UML	Unified Modeling Language

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Skin diseases represent a major yet frequently overlooked public health challenge in Low- and Middle-Income Countries (LMICs). With an estimated 4.86 billion new cases each year, they rank as the fourth leading cause of illness worldwide, with the burden especially severe in Sub-Saharan Africa where skin conditions account for 30–70% of all primary care visits (Yakupu et al., 2023).

Rwanda illustrates the severity of this access gap. Fewer than 20 dermatologists serve a population of more than 13 million people, creating a ratio of less than one specialist per million inhabitants (National Institute of Statistics Rwanda, 2022; Ndagijimana et al., 2023). Nearly all specialists are located in Kigali, while more than 70% of the population lives in rural areas. For patients in remote districts, accessing dermatological care may require traveling several hours to the capital, incurring costs that can represent a substantial portion of household income. As a result, access to specialist care is often determined more by geography and financial means than by clinical urgency.

Primary health centers staffed mainly by nurses and general practitioners therefore become the first and often only source of dermatological care for most Rwandans. Although general practitioners achieve diagnostic accuracy of approximately 58–70% for common skin conditions (Freeman & Langan, 2018), only 5.3% of primary care providers in East African settings feel confident diagnosing most dermatological disorders independently (Waitara et al., 2025). This uncertainty frequently leads to empirical prescribing, prolonging patient suffering and contributing to the growing burden of antimicrobial resistance.

Protocol-driven approaches such as symptom checklists have been proposed as low-cost solutions, but they depend on the practitioner correctly recognising the presenting condition before any protocol can be applied. This prerequisite fails in practice because the visual classification step itself is unreliable, as reflected by the fact that fewer than 6% of providers feel confident assessing skin disorders independently. Support is therefore needed for the initial visual classification decision before treatment or referral pathways are considered.

Teledermatology has shown positive results in controlled settings (Amien et al., 2025), but a teleconsult-first workflow cannot resolve the access crisis alone. Without a prior filtering mechanism that separates routine cases from those requiring specialist input, every submitted case consumes scarce specialist time regardless of complexity, simply relocating the bottleneck from waiting rooms to digital queues.

The challenge is further compounded by the lack of decision-support tools adapted for darker skin tones. Many commercially available AI systems exhibit significant performance disparities across skin pigmentation levels, with some achieving diagnostic accuracy as low as 15% on darker skin images (Joerg et al., 2025). When introduced in clinical settings, these tools can widen rather than narrow diagnostic accuracy gaps between light and dark skin tones (Groh et al., 2024), risking amplification of existing health inequities.

Despite advances in individual component technologies, no validated framework currently integrates AI-assisted dermatological triage with telemedicine for Fitzpatrick Skin Types V–VI in African primary care settings. This gap is further confirmed by systematic reviews of AI dermatology tools, which consistently identify the absence of equity-focused, workflow-integrated solutions for African populations (Alipour et al., 2024; Escalé-Besa et al., 2024). This study argues that combining AI-based visual classification with rule-based triage and telemedicine integration provides a more scalable and clinically appropriate solution than either standalone AI diagnosis or teleconsult-first workflows in resource-constrained settings.

1.2 Problem Statement

Rural populations in Rwanda face persistent barriers to accessing appropriate dermatological care due to extreme specialist scarcity and the absence of effective decision support at primary care level. As established in Section 1.1, fewer than 20 dermatologists serve a population of over 13 million people, with nearly all specialists based in Kigali (National Institute of Statistics Rwanda, 2022; Ndagijimana et al., 2023). Rural health centers staffed by nurses and general practitioners have therefore become the main source of dermatological care for most patients, despite limited specialist training and diagnostic support.

In this context, frontline health workers struggle to distinguish routine skin conditions that can be managed locally from those requiring specialist referral. As established in Section 1.1, fewer than 6% of primary care providers in this region feel confident assessing skin disorders independently. This diagnostic uncertainty frequently leads to empirical prescribing practices, including inappropriate use of topical steroids for fungal infections or broad-spectrum antibiotics without diagnostic confirmation. Such practices delay appropriate treatment, increase the risk of preventable complications, and contribute to the growing burden of antimicrobial resistance across Sub-Saharan Africa.

Protocol-driven approaches such as clinical decision algorithms and symptom checklists have been proposed as low-cost alternatives to specialist care. However, these tools presuppose that the practitioner can correctly identify the presenting condition before applying any protocol. When such a small proportion of frontline providers feel confident assessing skin disorders independently, this prerequisite consistently fails in practice (Waitara et al., 2025). The core problem is therefore not the absence of treatment protocols but the absence of reliable visual classification support at the point of first contact.

Store-and-forward teledermatology platforms extend geographic reach but do not resolve this classification gap. They require specialist review of every submitted case, shifting consultation backlogs from physical clinics to digital inboxes without reducing specialist workload (Amien et al., 2025). A teleconsult-first workflow that routes every case to a dermatologist regardless of complexity cannot scale in a health system where fewer than 20 specialists serve over 13 million people. What is needed is a mechanism that filters cases before specialist review, directing only those cases that genuinely require referral while supporting local management of routine conditions.

Commercially available AI tools have not filled this gap. Many demonstrate poor performance on darker skin tones due to underrepresentation in training data, with some achieving diagnostic accuracy as low as 15% on darker skin images (Joerg et al., 2025). Designed for consumer use and disconnected from formal clinical workflows, they are unsuitable for frontline health workers making referral decisions in resource-limited settings.

The combined effect of specialist scarcity, limited diagnostic capability for visual classification at the primary care level, and inadequately adapted AI tools has created a system where access to dermatological care is shaped more by geography and socioeconomic status than by clinical need. The core problem is the absence of a reliable and scalable triage mechanism that enables frontline health workers to distinguish cases manageable locally from those requiring specialist referral — a gap that motivated the development of the DermoAI framework.

1.3 Project's Main Objective

The main objective of this project was to develop and validate DermoAI, a web-based AI-assisted dermatological triage application that enables frontline health workers in rural Rwanda to manage routine skin conditions locally while facilitating remote specialist consultation for cases requiring referral.

1.3.1 List of Specific Objectives

1. Analyze existing dermatological datasets (Fitzpatrick17k and ISIC) to quantify representation gaps for Fitzpatrick Skin Types V–VI and apply targeted augmentation techniques to improve model performance on darker skin types.

2. Develop and evaluate an efficient convolutional neural network capable of identifying common dermatological conditions with at least 70% classification accuracy on Fitzpatrick Skin Types V–VI images.

3. Implement rule-based triage mapping that translates predicted conditions into Refer or Manage Locally recommendations, achieving at least 75% recall for Refer cases on Fitzpatrick Skin Types V–VI, with low-confidence predictions returning Uncertain and escalating to Refer.

4. Implement a telemedicine interface enabling health center staff to connect Refer cases with dermatologists for remote consultation, with integrated case documentation to support system improvement.

5. Evaluate clinical utility through simulated validation with medical consultants, comparing AI triage performance against the documented GP diagnostic baseline of 58–70% (Freeman & Langan, 2018) and assessing workflow feasibility in simulated primary care scenarios.

1.4 Research Questions

1. What is the extent of Fitzpatrick Skin Type V–VI underrepresentation in existing dermatological datasets, and can targeted augmentation measurably improve model performance across skin tone categories?

2. What classification accuracy can an efficient convolutional neural network achieve on Fitzpatrick Skin Types V–VI, and how does this compare to the documented GP diagnostic baseline in primary care settings?

3. Can rule-based triage mapping achieve at least 75% recall for Refer cases on Fitzpatrick Skin Types V–VI, and does an Uncertain fallback for low-confidence predictions reduce the risk of incorrect local management?

4. Can health center staff effectively use the AI triage tool alongside integrated telemedicine capabilities to differentiate cases manageable locally from those requiring referral in simulated Rwandan primary care workflows?

5. How does AI-assisted triage affect specialist workload distribution, and what factors influence health worker acceptance in feasibility testing?

1.5 Project Scope

The research focused on the design and simulated validation of an AI-assisted dermatological triage framework for rural primary care settings in Rwanda, conducted within a fixed ten-week academic timeline. The focus was on technical feasibility, workflow integration, and early indicators of clinical usefulness, rather than live deployment or long-term evaluation.

1.5.1 Content Scope

This study developed an AI-assisted dermatological triage system for common skin conditions in adult primary care settings. The system targeted conditions that represent a substantial proportion of dermatological presentations in Sub-Saharan Africa. Conditions were selected based on training viability within the FST V–VI subset of publicly available datasets, visual distinctiveness supporting reliable classification, and clinical relevance to primary care settings. The full condition selection rationale is documented in Chapter Four.

The target population consisted of adult patients aged 18 years and above presenting at rural primary care facilities with non-emergency dermatological complaints, predominantly within FST V–VI. The system was designed to provide binary triage outputs classified as Refer or Manage Locally rather than autonomous diagnoses. For each case, the system presented a likely condition category, a confidence level, a suggested course of action, and referral pathways for specialist consultation when needed.

1.5.2 Geographical Scope

The geographical focus of the study was limited to public health centers within Gasabo District in Kigali for simulated pilot validation. This district was used as a proxy validation site rather than the intended deployment environment, selected due to logistical feasibility and its combination of urban and peri-urban characteristics providing variation in connectivity infrastructure and health worker digital proficiency. The system was explicitly designed for deployment in rural districts where specialist access constraints are more severe, with portability enabling future deployment across Rwanda's 30 districts and potential adaptation to other Sub-Saharan African contexts facing similar specialist shortages.

1.5.3 Technical Scope

The project implemented a two-stage triage classification system. Stage 1 used a fine-tuned convolutional neural network to classify skin lesions into common dermatological conditions. Stage 2 applied rule-based triage mapping that translated the predicted condition into binary referral recommendations of Refer or Manage Locally based on established clinical risk guidelines. When model confidence fell below the defined threshold, the system returned Uncertain and defaulted to Refer. Keeping classification and triage mapping as separate steps avoided autonomous diagnosis, reflected existing referral practices in primary care, supported clinical oversight, and simplified future regulatory review.

The system presented health workers with clinically interpretable output comprising four elements: triage recommendation, likely condition category, model confidence level, and suggested action. Triage mapping used conservative clinical principles that prioritised patient safety. Conditions flagged as Refer included bacterial infections requiring IV antibiotics, severe drug reactions, suspected malignancies requiring biopsy, and rapidly progressive inflammatory conditions. Conditions mapped to Manage Locally included localised fungal infections, mild-moderate eczema, scabies, and benign nevi without concerning features.

The intended user base comprised frontline health workers, defined as community health workers and general practitioners staffing rural health centers. The system was designed to operate on mid-range Android devices through a Progressive Web Application interface, requiring internet connectivity for image upload, model inference, and telemedicine referral submission. Technical architecture emphasised efficient implementation with a small model storage footprint, inference time optimised for mobile devices, and minimal data requirements during initial installation. The full technical implementation is documented in Chapter Three and Chapter Four.

1.5.4 Temporal Scope

The research was bounded by a ten-week timeline (January to March 2026), structured around iterative Design Science Research cycles. This timeline maintained a realistic scope given solo-investigator constraints and academic resource limitations.

The ten-week period provided sufficient time for dataset analysis, model development and validation, interface implementation, simulated clinical evaluation, and documentation. Longitudinal assessment of sustained adoption and model improvement through continuous data collection would require extended post-graduation studies beyond the current scope.

1.5.5 Methodological Scope

Research Design Framework

The research used Design Science Research methodology, emphasising artifact development and evaluation to solve practical problems. Evaluation combined quantitative and qualitative methods. Quantitative metrics included condition classification accuracy for FST V–VI, triage recall and precision, and inference time. Qualitative assessment involved clinician evaluation of workflow feasibility, clinical utility, output interpretability, and potential barriers to adoption.

Data Collection Approach

Training and validation relied on publicly available datasets, specifically Fitzpatrick17k and the ISIC Archive, with augmentation techniques applied to improve FST V–VI representation. The project did not involve collecting data from Rwandan patients during the ten-week timeline, as doing so would have required Rwanda National Ethics Committee approval, Institutional Review Board clearance, informed consent protocols in Kinyarwanda, and partnership agreements with health facilities. The optional case archival functionality built into the telemedicine interface demonstrated the technical feasibility of future data collection pipelines rather than accumulating production datasets.

Validation Strategy

Validation employed simulated clinical scenarios with medical consultants rather than live patient trials, consistent with expert elicitation approaches used in preliminary feasibility evaluations of clinical decision support systems where the objective is identifying usability and safety concerns rather than achieving statistical power. Consultants role-played as frontline health workers evaluating curated FST V–VI cases from Fitzpatrick17k, generating triage recommendations through the system and providing structured feedback on diagnostic appropriateness, output interpretability, and workflow integration. This approach avoided exposing patients to an unvalidated system during development while allowing preliminary assessment of clinical usefulness.

1.5.6 Regulatory and Clinical Exclusions

The system explicitly excluded:

Autonomous Prescription Capabilities - All treatment recommendations required human physician verification. The system provided triage assessment and suggested actions but did not generate specific medication dosages or treatment protocols.

Override Functionality – Health workers retained full clinical authority to act outside system recommendations; the restriction applied solely to preserving system-generated labels for auditability and model integrity. Final clinical decisions remained the responsibility of health workers and occurred outside the system through standard care pathways.

Live Patient Trials - Validation used historical datasets and simulated clinician workflows rather than real-time patient interactions, ensuring ethical safety during academic development.

Medical Device Certification - The prototype operated as a research artifact demonstrating technical feasibility. Future real-world deployment would require Rwanda Food and Drugs Authority approval, ISO 13485 certification, clinical trial registration, and post-market surveillance protocols.

Pediatric Dermatology - Training data and validation scenarios focused on adult presentations (ages 18+).

Emergency Triage - The system focused on ambulatory primary care settings, not emergency department triage.

1.5.7 Justification for Scope Boundaries

The ten-week timeline was sufficient for demonstrating technical feasibility and workflow integration, while acknowledging that longitudinal outcomes require extended studies beyond academic project timelines. Simulated validation balanced ethical imperatives, specifically avoiding patient risk during unvalidated system testing, with research objectives of demonstrating clinical utility and workflow feasibility, providing preliminary evidence justifying investment in formal clinical trials.

Target conditions selected during dataset analysis provided clinically meaningful coverage while maintaining data quality through adequate representation in public datasets. System design choices preserved data integrity for future model improvement while maintaining clinical safety through existing healthcare oversight mechanisms. DermoAI placed early emphasis on performance for FST V–VI, responding to documented gaps in existing dermatological AI systems. The framework incorporated clinician-mediated data capture mechanisms to support future development of locally relevant datasets. This reduced reliance on externally sourced data and shifted the development approach from extractive to contributive AI models, where data gathered through deployment enriches the system serving the same population.

1.5.8 Project Deliverables

The following deliverables were produced within the project timeline:

- A trained and validated EfficientNetB0 classification model optimised for FST V–VI dermatological images across five condition classes
- A two-stage triage mapping system producing Refer, Manage Locally, and Uncertain outputs with conservative clinical escalation logic
- A FastAPI backend hosting model inference, GradCAM explainability, and integrated telemedicine workflows
- A Next.js Progressive Web Application providing triage, consultation management, and specialist review interfaces
- A consent-gated specialist review queue supporting future model retraining with locally verified clinical labels

1.6 Significance and Justification

1.6.1 Clinical and Public Health Impact

This research addressed persistent challenges in dermatological healthcare delivery across access, workforce capacity, and clinical decision-making. The system was designed to directly support Sustainable Development Goal 3, specifically Target 3.8 on universal health coverage, by extending specialist-level decision support to rural populations currently excluded from timely dermatological care (World Health Organization & World Bank, 2023). By enabling frontline health workers to manage routine conditions locally and connect referral cases to remote specialist oversight, it addressed a geographic access problem where over 70% of Rwanda's population resides in rural areas while specialists remain concentrated in Kigali (National Institute of Statistics Rwanda, 2022).

In Rwanda's context of an extremely limited dermatology workforce, filtering routine cases from specialist queues would free capacity for consultations that genuinely require expert evaluation. The system's interpretable output format provides educational value for health workers, with potential to improve diagnostic decision-making toward levels reported when AI supports non-specialists (Krakowski & Nguyen, 2024).

The framework also addressed antimicrobial resistance by reducing empirical prescribing driven by diagnostic uncertainty. When health workers cannot reliably differentiate between fungal, bacterial, and inflammatory conditions, they frequently prescribe broad-spectrum antibiotics without confirmation or treat fungal infections with steroids that exacerbate them

(World Health Organization, 2021). Evidence-based triage recommendations with clinical context supported more targeted treatment decisions, contributing to Rwanda's National Action Plan on Antimicrobial Resistance.

1.6.2 Data Sovereignty and Algorithmic Justice

Beyond improving access, the system addressed broader concerns related to data representation and algorithmic fairness. Commercial AI tools exhibit 30–40% lower accuracy on darker skin tones (Daneshjou et al., 2022; Groh et al., 2021), directly attributable to severe underrepresentation where Fitzpatrick Skin Type VI constitutes only 3.97% of images in widely used datasets despite representing populations exceeding one billion people globally (Alipour et al., 2024).

DermoAI's data collection framework was designed to be verified by clinicians and to ensure that information gathered in Rwanda remained within the country's health information systems. This supported development of locally relevant models that reflect Rwandan disease prevalence patterns and phenotypic diversity. By establishing minimum performance thresholds, specifically at least 75% recall on Refer cases for FST V–VI, as acceptance criteria, the research prioritised equitable accuracy as a prerequisite for deployment. This addressed the concern that AI systems intended to improve healthcare access could paradoxically worsen disparities.

1.6.3 Methodological and Theoretical Contribution

This project demonstrated a replicable framework for responsible AI deployment in low-resource healthcare settings. The bias mitigation protocol used stratified validation across Fitzpatrick Skin Types with minimum performance thresholds, providing a template for fairness-aware medical AI development where dataset representation gaps are severe. Integrating AI triage with telemedicine connectivity maintained clinical accountability while capturing verified ground truth for model improvement, demonstrating that clinician-verified case archival can build training datasets incrementally through routine clinical workflows rather than requiring extractive data collection.

1.6.4 Policy and Implementation Implications

These methodological contributions have practical implications for health policy. This research contributes evidence relevant to Rwanda's Digital Health Strategy and aligns with national priorities for expanding access to specialist services in underserved areas. By demonstrating technical feasibility and initial clinical utility through simulated validation, the research provides justification for government consideration of pilot clinical trials and phased national rollout.

The framework's replicable nature has implications beyond Rwanda. Approximately 23 countries in Sub-Saharan Africa report dermatologist-to-population ratios below 1:500,000

(Tiwari et al., 2022), facing similar challenges of specialist concentration in urban centers and inadequate decision support for rural primary care. The modular architecture, comprising an adaptable AI model, flexible triage mapping rules, and a portable telemedicine interface, enables adaptation to different national contexts with varying disease prevalence patterns and regulatory frameworks.

1.6.5 Expected Beneficiaries

Primary beneficiaries of future deployment would include rural patients across Rwanda's 30 districts currently lacking access to dermatological specialist care, frontline health workers gaining diagnostic confidence through AI decision support, and specialist dermatologists experiencing reduced low-complexity case burden through intelligent filtering.

Rwanda's Ministry of Health would gain an evidence base for digital health strategy and a model for addressing specialist shortages in other clinical domains. More broadly, this research contributes to the growing body of evidence on equitable AI deployment in LMICs, providing a methodological approach that could inform similar projects across Sub-Saharan Africa.

1.7 Research Budget

The research budget supported a realistic simulated validation while managing costs appropriate for an academic project. While free-tier cloud services provide basic functionality, commercial services ensured reliability during critical validation phases, particularly during simulated clinician testing and model retraining workflows.

Table 1.1: Research Budget Estimation

Item	**Description**	**Est. Cost (RWF)**
Google Colab Pro	GPU Cloud Compute	33,000
Vercel	Frontend Hosting	0
Render standard	Fast API Hosting	36,000
Supabase	Database & Auth	0
Mista IO	SMS API Credits	10,000
Domain Name	Vercel Subdomain	0
Contingency	Miscellaneous	20,000
TOTAL		99,000 RWF

- Google Colab Pro: Essential for efficient batch retraining and model fine-tuning. Free-tier GPU access is unreliable with frequent disconnections.

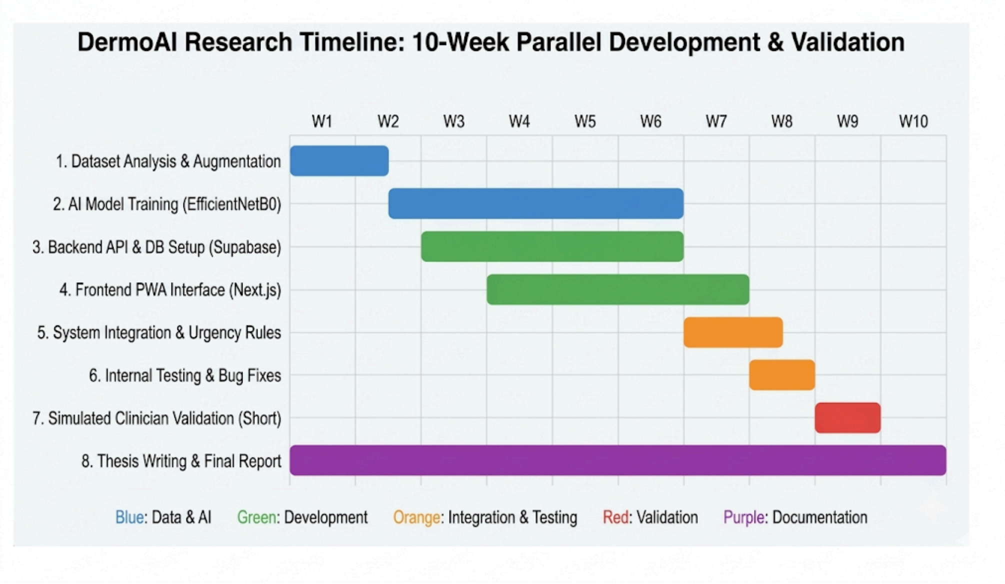
- Vercel: Hosted the Next.js Progressive Web Application frontend. The free tier was sufficient for prototype deployment and testing.
- Render Standard: Hosted the FastAPI backend responsible for model inference and API endpoints during validation testing.
- Supabase: Free tier sufficient for pilot storage needs (~500 case records). Daily backups are not critical at the research stage.
- Mista IO: Covered SMS notifications for simulated Refer case alerts during validation testing with medical consultants. This tests the technical feasibility of the alert system that would notify physicians of cases requiring referral in real deployment.
- Domain Name: Free Vercel subdomain (e.g., dermoai-rwanda.vercel.app) adequate for pilot phase. Custom domain deferred to production.
- Contingency: Buffer for currency fluctuations, additional mobile data bundles for field testing, or API usage overages.

Funding Source: Personal investment.

1.8 Research Timeline

Project execution occurred over a ten-week period (January–March 2026), structured around iterative Design Science Research cycles. To maximize efficiency, the timeline followed a parallel development strategy in which model training and interface construction progressed simultaneously, balancing technical rigor with realistic expectations for a solo-investigator project.

Figure 1.1: Research Timeline Gantt Chart



Key Task Dependencies and Sequencing

- Model development began after completion of dataset preparation and augmentation during the early phase of the project. Training and evaluation were conducted iteratively as the dataset pipeline was finalized.
- Application development proceeded in parallel with model training, using mock data interfaces to establish the frontend and backend infrastructure before the AI model was fully integrated.
- The trained AI model and the application interface were integrated during the later development phase, enabling implementation of the rule-based triage mapping system.
- System validation was conducted near the end of the project using curated historical cases with three to five medical consultants, allowing targeted feedback on workflow feasibility and clinical usability. Ethical considerations governing data use, algorithmic fairness, clinical safety, and responsible deployment are addressed in Chapter Three, Section 3.8.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter examines academic and technical literature on artificial intelligence in dermatology, telemedicine implementation in resource-limited settings, and task shifting models for expanding specialist capacity. The review focuses specifically on interventions relevant to Sub-Saharan African contexts, where infrastructure constraints, specialist shortages, and phenotypic representation gaps create unique implementation challenges distinct from those addressed in Western healthcare settings.

Literature was systematically identified through searches of PubMed, IEEE Xplore, and Google Scholar using keywords including dermatology AI, teledermatology Africa, CNN architectures medical imaging, algorithmic bias dark skin, and task shifting dermatology. The review examined approximately 40 peer-reviewed publications from 2017-2025, focusing on empirical studies of implemented systems in LMICs. Theoretical frameworks and Western-only deployments received less emphasis.

The review was structured to first establish a historical context for AI in dermatology, then critically examine existing technological solutions and their limitations in African settings. Subsequent sections analyze related work on deep learning architectures suitable for mobile deployment, algorithmic fairness challenges specific to darker skin tones, telemedicine implementation patterns in resource-limited environments, and task shifting models that enable non-specialist physicians to extend specialist capacity. Individual technical components have been validated separately. However, integrated frameworks combining AI-assisted triage with telemedicine connectivity for African populations remain largely absent from the reviewed literature. This gap confirms the novelty and necessity of the DermoAI framework developed in this study.

2.2 Historical Background of the Research Topic

The intersection of dermatology and artificial intelligence has evolved dramatically over the past decade, progressing from rule-based expert systems toward sophisticated deep learning approaches while simultaneously confronting critical questions of algorithmic fairness and clinical deployment feasibility. This historical progression reveals both the technical maturation of AI diagnostic capabilities and the growing recognition that accuracy metrics alone prove insufficient for equitable healthcare deployment.

Early Automation Attempts and Technical Limitations

Early automated dermatological diagnosis systems, developed primarily in the 1990s and early 2000s, relied on feature engineering approaches where domain experts manually defined visual characteristics that computer systems should detect. These rule-based systems attempted to codify diagnostic heuristics such as color variation patterns, border irregularity metrics, and asymmetry measures into algorithmic form. However, these approaches struggled with the visual complexity and subtle variations of skin conditions. They achieved limited clinical utility and failed to scale beyond narrowly defined use cases. The manual feature definition process could not capture the nuanced pattern recognition that experienced clinicians develop through years of practice, and systems performed poorly when encountering conditions outside their predefined rule sets.

The 2017 Deep Learning Breakthrough

The field experienced a paradigm shift in 2017 when Esteva et al. published landmark research in Nature demonstrating that Convolutional Neural Networks (CNNs) trained on large image datasets could classify skin cancer with accuracy matching certified dermatologists. Using a dataset of 129,450 clinical images, their deep learning model achieved performance comparable to 21 board-certified dermatologists across multiple skin cancer classification tasks (Esteva et al., 2017). This breakthrough demonstrated that data-driven approaches could learn relevant visual features directly from examples rather than requiring manual specification, triggering intensive research and commercial development of AI dermatology applications. The success suggested that specialist-level diagnostic capability might be deliverable through widely accessible smartphone applications, potentially democratizing access to expert-quality diagnosis.

The Fairness Crisis and Representation Gaps

However, a fairness crisis emerged as these models moved from controlled research settings toward real-world deployment. Critical examinations by Daneshjou et al. (2022) and Groh et al. (2021) revealed that the high accuracy claimed in academic publications degraded sharply for non-white populations. Systematic analysis of training datasets showed severe underrepresentation of darker skin tones. In the widely used Fitzpatrick17k dataset, Skin Type VI (deeply pigmented skin) constituted only 3.97% of images despite representing populations exceeding one billion people globally (Alipour et al., 2024). Models trained on these biased datasets demonstrated 30-40% performance degradation when tested on darker skin tones, with particularly concerning failures in detecting malignancies that appear visually different on pigmented skin.

This revelation prompted fundamental reconsideration of AI deployment ethics in healthcare. Research between 2017 and 2019 had focused primarily on accuracy metrics, with minimal attention to demographic performance variation. Since 2020, the field has shifted toward explicitly evaluating algorithmic fairness, with growing consensus that AI medical devices must demonstrate equitable performance across all populations before deployment. Work by Benčević et al. (2024) further showed that simple algorithmic adjustments like class weighting prove insufficient to address representation gaps. The only effective solution involves collecting and integrating genuinely diverse, demographically representative training data. This finding has profound implications for AI development in Africa, where both existing datasets lack representation and infrastructure for systematic local data collection remains limited.

Evolution Toward Task Shifting Models

Parallel to technical development, clinical implementation research has evolved from AI replacement paradigms toward AI augmentation frameworks. Early enthusiasm suggested AI might replace dermatologists entirely, but practical deployment revealed that the most valuable application involves assisting non-specialist physicians (Krakowski & Nguyen, 2024). Meta-analyses demonstrate that while AI achieves 87% sensitivity comparable to specialists, the greatest impact occurs when AI helps general practitioners, improving their diagnostic accuracy by 13-28 percentage points (Salinas et al., 2024). This finding aligns with World Health Organization task shifting models, where trained non-specialists supported by appropriate tools extend specialist capacity in contexts facing severe health workforce shortages (World Health Organization, 2016).

This approach proves particularly relevant in regions like Sub-Saharan Africa where specialist shortages cannot be resolved through training programs alone, given the decades required to attain certified dermatologists.

2.3 Overview of Existing Systems

Modern dermatological support systems generally fall into four categories: consumer-facing symptom checkers, clinical decision support systems (CDSS), teledermatology platforms, and protocol-based triage tools. When assessed for implementation in Rwanda's public health sector, each category has distinct limitations that prevent effective deployment at scale.

2.3.1 VisualDx and Clinical Decision Support

VisualDx is widely regarded as a leading clinical decision support system for dermatology, used by clinicians in the United States and Europe as a reference-based diagnostic support platform (VisualDx, 2024). The platform offers extensive libraries of medical images organized by diagnosis, along with differential diagnosis tools.

However, using VisualDx in African primary care faces fundamental barriers. The platform requires annual subscription fees ranging from $400-800 per user, prohibitively expensive for resource-constrained health systems. The system assumes stable, high-bandwidth internet connectivity unavailable in many rural facilities.

Most critically, VisualDx functions as a reference library requiring significant medical literacy, whereas Rwandan health centers are often staffed by nurses or general practitioners with limited dermatology training who need more directive decision support. The platform design assumes users already have diagnostic frameworks to guide their reference searches. This assumption fails in contexts where foundational dermatological training is limited.

2.3.2 Google Derm Assist and Consumer AI

Consumer-facing tools like Google DermAssist use proprietary black-box algorithms to provide skin analysis directly to the general public through smartphone applications.

However, these consumer tools create ethical risks when deployed in low-resource clinical settings. They are typically trained on proprietary datasets that cannot be externally audited for bias.

Daneshjou et al. (2022) showed that many state-of-the-art models exhibit significant performance degradation on diverse skin tones, with accuracy drops of 30-40% for darker phenotypes. Recent studies further demonstrate that AI-generated dermatology training images show only 10.2% dark skin representation and achieve merely 15% diagnostic accuracy for intended conditions (Joerg et al., 2025). These systems lack the telemedicine bridge needed to connect rural patients to verified specialists for diagnosis confirmation and treatment. They function as isolated diagnostic tools disconnected from clinical accountability structures, which raises concerns about their suitability for integration into healthcare delivery systems.

2.3.3 Store and Forward Teledermatology

Traditional store-and-forward (SAF) platforms currently represent the standard for remote dermatological diagnosis in African settings. Studies from Mali demonstrated that SAF teledermatology can achieve over 96% appropriate management rates when image quality is adequate (Faye et al., 2018). More recent implementations in South Africa demonstrate 55.1% of cases resolved remotely with median response times of 20.8 minutes (Amien et al., 2025).

Despite documented effectiveness, SAF systems suffer from critical scalability limitations. They remain fundamentally labor-intensive, requiring specialists to individually review every submitted case regardless of complexity or urgency. A South African implementation handling referrals from 395 health facilities still required specialist review of every case, demonstrating the scalability constraints of this approach (Amien et al., 2025).

In Rwanda's context with fewer than 20 dermatologists serving 13 million people, this creates the same bottleneck in digital form that exists physically. The technology shifts the queue from physical waiting rooms to digital inboxes without reducing the cognitive burden on specialists who must still differentiate routine cases from those requiring expert evaluation.

2.3.4 Protocol-Based Dermatology Triage Systems

Protocol-based dermatology triage systems represent one of the most widely used non-specialist decision support approaches in resource-limited healthcare settings. These systems typically use structured symptom checklists or clinical decision trees to guide health workers toward treatment or referral recommendations. The World Health Organization has promoted such algorithms as part of task shifting strategies to extend specialist capacity in low-resource environments (World Health Organization, 2016).

When applied by clinicians with moderate dermatological training, decision trees can improve diagnostic consistency and reduce inappropriate prescribing (Freeman & Langan, 2018). However, these systems depend on the practitioner correctly recognizing the visual characteristics of the presenting condition before selecting the appropriate branch of the decision tree.

In settings where frontline providers report low confidence in dermatological diagnosis, this prerequisite becomes a critical limitation. Waitara et al. (2025) found that only 5.3% of primary care providers in East African settings feel comfortable diagnosing most skin disorders independently. Under these conditions, protocol-based systems frequently fail because the initial visual classification step required to activate the algorithm is itself unreliable.

These findings suggest that decision support tools that assist with the visual classification stage are therefore necessary before protocol-based treatment pathways can function effectively in low-resource African settings. This limitation motivates the exploration of AI-assisted image classification as a triage support mechanism, not as a replacement for established clinical protocols, but as the prerequisite layer that makes those protocols actionable.

2.4 Review of Related Work

This section examined academic literature on three main pillars of the DermoAI solution: deep learning architectures for mobile deployment, algorithmic equity in dermatological AI, and telemedicine implementation in resource-limited African contexts.

2.4.1 Deep Learning Architectures for Mobile Triage

The use of AI on mobile devices in resource-limited environments receives strong support from recent technical literature. Early models like ResNet50 required substantial computing power, but Srinivasu et al. (2021) showed that lightweight architectures, especially MobileNetV2, can achieve classification accuracy above 85% with inference times as low as 0.09 seconds. This matters for Rwanda, where users predominantly have mid-range smartphones with limited processing power and intermittent connectivity.

Comparative studies by Kausar et al. (2021) found that ensemble methods combining multiple models can reach 98% accuracy but require cloud infrastructure for real-time operation. The literature suggests that hybrid approaches using lightweight models for immediate offline triage and cloud-based ensembles for specialist verification work best in settings with inconsistent connectivity (Gessert et al., 2020). Recent systematic reviews confirm that mobile-based AI tools can achieve diagnostic accuracy comparable to in-person consultations when properly validated for specific deployment contexts (Escalé-Besa et al., 2024).

2.4.2 The Data Gap and Algorithmic Bias

Multiple studies identify underrepresentation of African skin tones as the primary barrier to safe AI deployment in dermatology. Alipour et al. (2024) systematically reviewed 54 public datasets and found stark disparities. In the standard Fitzpatrick17k dataset, Skin Type VI (deeply pigmented skin) accounts for only 3.97% of images. The consequences of this gap are severe. Daneshjou et al. (2022) found that models trained on standard datasets see 30-40% performance drops when tested on diverse skin images. More troubling, recent evidence shows that when AI assistance is provided to primary care physicians, diagnostic accuracy gaps between light and dark skin tones actually widen rather than narrow (Groh et al., 2024). This suggests that biased AI tools may amplify rather than reduce existing healthcare disparities.

Importantly, Benčević et al. (2024) showed that standard algorithmic corrections such as class weighting or oversampling prove insufficient to address representation gaps. The only effective solution is to collect and integrate diverse, representative data validated by clinicians familiar with disease presentation variations across skin tones. This literature supports DermoAI's approach of prioritizing local data collection mechanisms rather than relying solely on algorithmic adjustments to correct fundamentally biased training distributions.

2.4.3 Telemedicine, Augmented Intelligence, and Task Shifting

Literature on augmented intelligence suggests that AI works best when it assists human judgment rather than replacing it. Salinas et al. (2024) conducted a meta-analysis showing that while AI sensitivity (87%) rivals that of specialists, the biggest impact happens when AI helps generalists, improving their diagnostic accuracy by 13-28 percentage points (Krakowski & Nguyen, 2024). These findings suggest that AI systems may be particularly valuable when used to assist general practitioners during the initial visual classification stage of dermatological assessment, where diagnostic uncertainty is highest and where protocol-based approaches alone have been shown to fall short (Waitara et al., 2025). In African contexts, this aligns with the task shifting model promoted by WHO, where trained non-specialists supported by appropriate tools extend specialist capacity to underserved populations.

Freeman and Langan (2018) documented how training non-specialists in dermatology can improve diagnostic rates from 41% to 81% when combined with decision support tools. This confirms that equipping general practitioners with AI-assisted triage represents a valid clinical strategy for addressing the dermatologist shortages documented by Tiwari et al. (2022), who noted that fewer than one dermatologist serves every million people across multiple African countries. Recent studies from Kenya further validate this approach, showing that 94.7% of primary care providers express interest in teledermatology consultation services when integrated with their existing workflows (Waitara et al., 2025).

2.4.4 mHealth Implementation in Rwanda

Literature specific to Rwanda confirms the country's readiness for mobile health interventions. Baratali et al. (2023) highlighted Rwanda's extensive network of over 58,000 Community Health Workers and 85% mobile network coverage as a solid foundation for digital health initiatives. The country has demonstrated successful implementation of

SMS-based interventions for COVID-19 case management, achieving high sustainability and scalability metrics.

Critically, Nkurunziza et al. (2022) validated the specific workflow implemented in DermoAI. Their study showed that CHWs in rural Rwanda successfully captured and transmitted wound images via mobile devices for remote specialist diagnosis, achieving diagnostic concordance rates sufficient for clinical decision-making. This operational validation suggests the feasibility of the developed workflow within Rwanda's existing healthcare infrastructure and that frontline workers possess the technical capacity to use image-based telemedicine tools effectively.

2.4.5 Synthesis and Implications for DermoAI

The reviewed literature converges on four findings directly relevant to DermoAI's design. First, AI-assisted triage is most effective when supporting generalist health workers rather than replacing specialists, with diagnostic accuracy improvements of 13-28 percentage points reported when AI assists non-specialists (Salinas et al., 2024; Krakowski & Nguyen, 2024). Second, the primary barrier to reliable AI triage in Sub-Saharan Africa is not architectural but representational: FST VI constitutes only 3.97% of publicly available training images despite representing over one billion people globally (Alipour et al., 2024), and algorithmic corrections alone cannot resolve this gap (Benčević et al., 2024). Third, protocol-based triage tools consistently fail in low-confidence diagnostic settings because visual classification is a prerequisite for protocol activation, not a downstream task (Waitara et al., 2025; Freeman & Langan, 2018). Fourth, telemedicine workflows have been validated in Rwandan community health worker settings with sufficient diagnostic concordance for clinical use (Nkurunziza et al., 2022).

No reviewed framework integrates all four elements for dermatological triage serving FST V-VI populations. DermoAI addressed this gap by coupling efficient on-device classification with human-in-the-loop telemedicine verification, ensuring both technical performance and clinical accountability while establishing pathways for continuous improvement through locally verified data collection.

2.5 Strengths and Weaknesses of Existing Systems

Weaknesses in African contexts render existing systems nearly ineffective for equitable deployment. They rely on datasets where darker skin tones remain statistically underrepresented (Alipour et al., 2024), leading to dangerous misdiagnoses for Rwandan patients. They assume constant 4G/5G connectivity unavailable in rural settings, and most fundamentally they lack feedback loops for continuous improvement, functioning as static products that do not learn from local populations or adapt to regional disease prevalence patterns.

As Yotsu et al. (2023) noted in their study of mHealth applications in resource-limited settings, effective systems require integration with local healthcare worker workflows. These features remain systematically absent from major Western platforms designed for high-resource environments. The gap between technical capability and contextual appropriateness represents the primary barrier to equitable AI deployment in African dermatology. Table 2 summarizes the key features and limitations of existing dermatological support systems compared to the developed DermoAI framework.

Table 2.1: Comparison of Existing Dermatological Solutions vs. DermoAI

Feature	**VisualDx / Clinical CDSS**	**Google Derm Assist / Consumer AI**	**Traditional Teledermatology (SAF)**	**Protocol-Based Triage Tools**	**DermoAI (developed)**
Primary Technology	Knowledge Base / Reference Library	Proprietary "Black Box" Deep Learning	Email / WhatsApp (Store-and-Forward)	Symptom Checklists / Decision Trees	Interpretable CNN-based triage with telemedicine verification
Target Audience	Specialists & Doctors	General Public (Consumers)	Dermatologists	Nurses & General Practitioners	Nurses & General Practitioners
Data Representation	Diverse but Western-centric	Proprietary (Unknown Bias)	N/A (Human diagnosis)	N/A (Rule-based)	Optimized for Fitzpatrick V & VI
Diagnostic Model	Reference / Educational	Autonomous Output	Specialist Review	Rule-based Protocol	Triage (Refer/Manage Locally)
Key Weakness in Rwanda	High Cost & Bandwidth Heavy	Algorithmic Bias & No Medical Link	Slow Turnaround & Labor Intensive	Requires prior visual recognition ability	Unvalidated in real-world clinical settings

2.6 General Comment and Conclusion

This literature review reveals critical gaps in research on AI fairness, dermatological triage, and African healthcare delivery. Considerable research exists on AI architectures for skin disease classification, and separate research exists on teledermatology implementation in African settings. However, the reviewed literature lacks integrated frameworks that combine AI-assisted triage with telemedicine connectivity while explicitly addressing algorithmic bias for African phenotypes.

Beyond technical gaps, the literature reveals unresolved tensions around AI deployment in African healthcare contexts. First, data sovereignty concerns create tension between the need for diverse training data and the risk of extractive data practices where African patient information enriches Western commercial models without local benefit (Alipour et al., 2024). Second, regulatory frameworks remain uncertain. While the FDA has approved AI diagnostic tools, African regulatory bodies lack established pathways for evaluating algorithmic bias and contextual appropriateness (Krakowski & Nguyen, 2024). Third, debates persist around appropriate deployment models, with some researchers advocating autonomous AI diagnosis to maximize access while others emphasize human-in-the-loop approaches that maintain clinical accountability (Salinas et al., 2024). This study navigated these tensions through three approaches: prioritizing local data sovereignty, using a conservative triage-focused design that avoids autonomous diagnosis, and implementing explicit human oversight mechanisms.

The reviewed literature on protocol-based triage systems confirmed that the absence of reliable visual classification at the point of first contact, not the absence of treatment protocols, is the core barrier to appropriate dermatological care in low-resource settings (Waitara et al., 2025; Freeman & Langan, 2018). This finding directly justified the ML-based classification approach in DermoAI, not as a replacement for clinical protocols but as the prerequisite layer that makes those protocols actionable. The literature on algorithmic bias further confirmed that this classification layer must be explicitly optimised for FST V-VI to avoid amplifying existing disparities rather than reducing them (Groh et al., 2024; Benčević et al., 2024).

Current literature confirms that the foundational technology is mature and ready for deployment. Efficient CNN architectures can run on mid-range smartphones with accuracy sufficient for clinical triage (Srinivasu et al., 2021). Telemedicine workflows have been validated in Rwandan healthcare settings (Nkurunziza et al., 2022). However, the data gap remains the primary obstacle to safe deployment. No validated system in the literature provides both accurate AI triage optimized for Fitzpatrick Skin Types V and VI and sustainable pipelines for collecting verified local data.

This study addressed that specific gap through an integrated approach combining AI triage explicitly optimized for African skin phenotypes, telemedicine connectivity ensuring clinical safety through specialist oversight, and optional case archival establishing pathways for continuous improvement using locally verified data. Chapter Three translates these findings into concrete technical architecture and implementation strategy.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter describes how the gaps identified in Chapter Two were addressed through the design and development of the DermoAI system. The chapter outlines the methodological framework guiding system development, describes the conceptual and technical structure of the developed solution, and explains the ethical principles governing its intended deployment. Emphasis is placed on aligning technical feasibility with clinical safety, equity, and workflow constraints relevant to dermatological care in Rwanda.

3.2 Research Design

This research followed a Design Science Research methodology, selected for its suitability in developing and evaluating technological artifacts that solve identified socio-technical problems. In this study, the artifact is an AI-assisted dermatological triage system integrated with telemedicine workflows, designed to operate within the constraints of Rwanda's public healthcare context. The DSR framework supported iterative refinement of the DermoAI artifact through continuous interaction between design decisions, evaluation outcomes, and contextual constraints identified in the literature. System requirements were derived from prior analysis of dermatology service delivery in Sub-Saharan Africa, focusing on specialist scarcity, phenotypic representation gaps in existing datasets, and infrastructure limitations affecting rural healthcare facilities. These requirements informed architectural decisions prioritising efficient inference, connectivity-aware design, and conservative clinical escalation mechanisms. Evaluation activities were embedded throughout development, combining quantitative performance assessment with qualitative analysis of workflow feasibility and clinical appropriateness.

3.2.1 Conceptual System Model

The conceptual system model underlying DermoAI integrates three core components into a single operational framework: AI-assisted triage, telemedicine oversight, and continuous data-driven improvement. Existing systems typically address these requirements separately rather than integrating them, and no reviewed framework combined equitable AI performance on darker skin tones, clinical accountability through human oversight, and sustainable mechanisms for generating locally verified datasets.

Component 1: AI-Assisted Triage at Primary Care Level

At the core of the model is an AI-assisted triage mechanism designed to support rather than replace clinical decision-making at the primary care level. The system employed a two-stage classification approach, illustrated in Figure 3.1. In the first stage, dermatological images captured at rural health centers are processed through a convolutional neural network optimised for Fitzpatrick Skin Types V and VI. In the second stage, conservative rule-based clinical guidelines translate predicted conditions into binary triage recommendations of Refer or Manage Locally. This separation enables structured referral guidance without attempting autonomous diagnosis, ensuring alignment with primary care workflows and regulatory expectations.

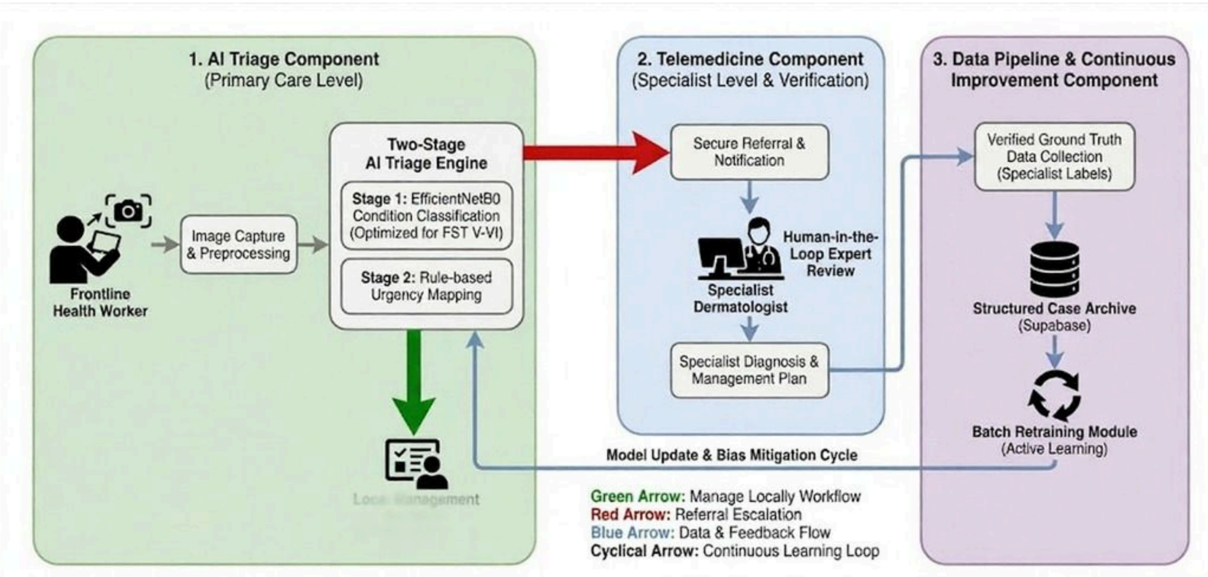
Component 2: Telemedicine Oversight and Clinical Accountability

Clinical oversight is incorporated through an integrated telemedicine component that introduces structured human verification into the triage workflow. Cases flagged as Refer are routed to dermatology specialists for review via a secure web interface. Specialist assessments support immediate patient management decisions while also producing verified clinical labels, ensuring that accountability for diagnostic and treatment decisions remains with licensed clinicians rather than with the AI system.

Component 3: Data Pipeline for Continuous Learning

A data pipeline component captures specialist-verified outcomes and stores them as high-quality reference labels. These verified cases are periodically aggregated and used for batch model retraining during subsequent development cycles. By embedding clinician verification within routine referral workflows, the system supports gradual performance improvement while reducing reliance on externally sourced datasets, responding directly to the equity and data sovereignty challenges discussed in Chapter Two.

Figure 3.1: DermoAI Conceptual System Model



3.2.2 Dataset Description and Data Preparation

Two publicly available dermatological datasets were used in this study. The Fitzpatrick17k dataset served as the primary training source, containing 16,577 images spanning 112 conditions with explicit FST annotations validated by board-certified dermatologists (Groh et al., 2021), filtered to a FST V-VI subset of 2,155 images. The ISIC Archive was analysed as a supplementary source but was not incorporated due to insufficient valid FST label coverage at only 1.91% of images, confirming Fitzpatrick17k as the only viable primary training source. Full dataset preparation, augmentation, and condition selection are documented in Chapter Four.

3.2.3 Triage Model Design

The two-stage triage architecture described in Section 3.2.1 was operationalised through specific technical implementation choices. Public dermatological datasets label clinical conditions rather than urgency levels. Since DermoAI focuses on triage rather than diagnosis, the system separates condition classification from triage assessment.

Stage 1: Condition Classification

Stage 1 evaluated two convolutional neural network architectures suitable for mobile inference: MobileNetV2 and EfficientNetB0. MobileNetV2 served as the baseline implementation due to its efficiency on CPU-based devices and documented performance on mobile medical imaging tasks (Srinivasu et al., 2021). EfficientNetB0 was evaluated as a comparative candidate given its documented performance advantages on small, imbalanced datasets (Tan & Le, 2019). Final architecture selection was determined by performance on FST V-VI validation sets, with the selected model trained using transfer learning from ImageNet pre-trained weights and fine-tuned on augmented dermatological datasets with emphasis on Fitzpatrick Skin Types V and VI.

Stage 2: Triage Mapping

Stage 2 applied rule-based mapping to translate predicted conditions into binary triage recommendations of Refer or Manage Locally based on established dermatological risk guidelines. This conservative approach prioritizes sensitivity for Refer cases over specificity, reflecting the clinical principle that false negatives pose greater patient safety risks than false positives (Krakowski & Nguyen, 2024).

Because dermatological urgency may depend on clinical history, symptom progression, or systemic signs not visible in images, the system treats triage outputs as advisory risk assessments that must be interpreted alongside patient history and clinical examination by the health worker. The AI classification provides a structured first-pass signal, while the health worker retains full authority to act on clinical context not captured by the image alone (Waitara et al., 2025).

When model confidence falls below the predefined threshold of 0.45, the system returns an Uncertain output that automatically escalates the case for specialist review. Additionally, when any Refer class probability exceeds 0.60, the system forces a Refer recommendation regardless of the primary prediction. The design threshold of 0.45 was refined to 0.35 during production deployment based on observed confidence distributions with real-world image quality, as documented in Chapter Five.

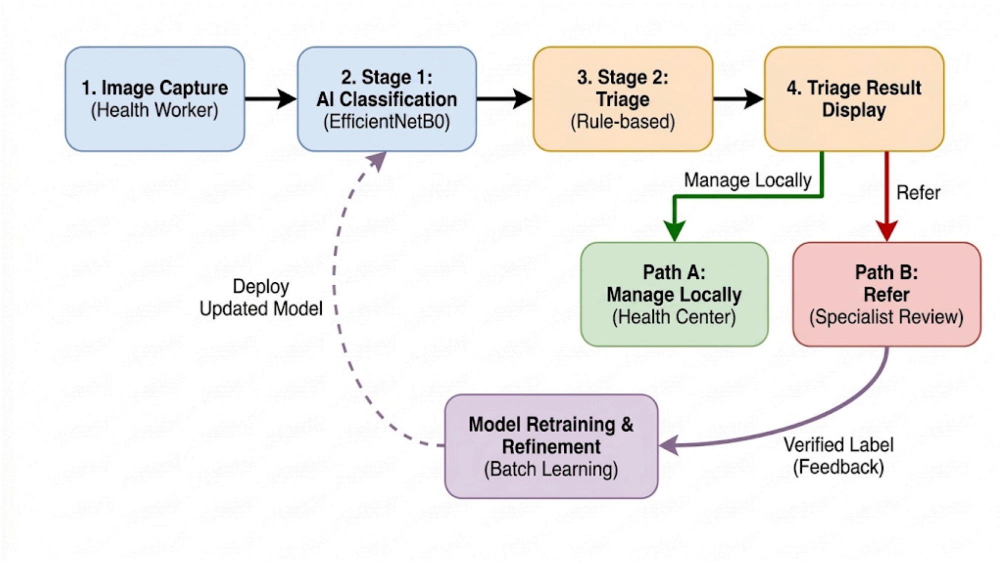
3.2.4 Validation Strategy and Success Criteria

System validation was conducted using simulated clinical scenarios rather than live patient trials. This approach avoided ethical and regulatory challenges associated with real patient data collection while allowing assessment of technical feasibility and workflow integration. Validation used a purposive sampling strategy in which three to five medical consultants with dermatology experience evaluated curated cases representing common FST V-VI conditions, ensuring that feedback reflected clinical expertise relevant to the target deployment context. This sample size is consistent with expert elicitation approaches used in preliminary feasibility evaluations of clinical decision support systems, where the objective is identifying usability and safety concerns rather than achieving statistical power (Kushniruk & Patel, 2004).

The evaluation dataset consisted of held-out images from the Fitzpatrick17k dataset stratified by skin type and condition category, ensuring representative coverage of FST V-VI presentations across the five target conditions. Quantitative evaluation focused on recall for Refer cases within Fitzpatrick Skin Types V and VI, with a minimum acceptable recall threshold of 75% required for deployment consideration. Additional metrics included overall accuracy, precision for Refer classifications, and balanced F1-scores across skin type groups.

Qualitative evaluation involved medical consultants assessing the clinical appropriateness of triage outputs, interpretability of system feedback, and usability within simulated frontline workflows. System success required that quantitative performance thresholds were met and that qualitative feedback confirmed the system's perceived utility and operational feasibility.

Figure 3.2: Two-Stage Triage and Telemedicine Loop Workflow



3.3 Functional and Non-Functional Requirements

The following tables define the requirements that guided the design and implementation of DermoAI. Functional requirements describe what the system must do. Non-functional requirements describe how the system must perform.

Table 3.1: Functional Requirements

Req ID	Feature	Description
FR1	User Authentication	Allow health workers and specialists to register and log in using email and password, with role-based access control restricting features to authorised roles.
FR2	Image Upload and Triage	Allow health workers to upload one or more skin lesion images per consultation and receive a triage recommendation for each image.
FR3	Two-Stage Triage Output	Classify each image using the EfficientNetB0 model and apply rule-based triage mapping to produce a Refer, Manage Locally, or Uncertain recommendation accompanied by a confidence level and suggested action.
FR4	GradCAM Explainability	Generate a GradCAM attention overlay for each non-Uncertain prediction, highlighting the image regions that influenced the triage recommendation.
FR5	Specialist Notification and Review	Automatically send an SMS notification to on-call specialists when a consultation resolves to Refer, and shall provide specialists with a review interface to assign verified diagnostic labels to consented cases.
FR6	Telemedicine Consultation	Enable health workers and specialists to initiate live video consultations for Refer cases requiring real-time clinical discussion, using WebSocket-based video sessions.

Table 3.2: Non-Functional Requirements

Req ID	Requirement	Description
NFR1	Clinical Safety	Low-confidence predictions defaults to Refer rather than Manage Locally. Any prediction where a Refer-class probability exceeds the override threshold shall force a Refer recommendation regardless of the primary classification output.

NFR2	Equity	Achieve at least 75% recall for Refer cases on Fitzpatrick Skin Types V and VI as a minimum condition for deployment consideration.
NFR3	Security and Privacy	All data transmission occurs over HTTPS. Patient images are stored with access restricted to authorised users through JWT-based authentication. The system complies with Rwanda's Law on Protection of Personal Data and Privacy (2021), with data hosted on African-region servers.
NFR4	Usability	The four-element triage output format is interpretable by health workers without specialist dermatology training, as assessed through simulated acceptance testing.
NFR5	Maintainability	The system maintains separation between the classification model and triage mapping rules, enabling independent updates to clinical escalation logic without retraining the model.
NFR6	Portability	Accessible through a Progressive Web Application on mid-range Android devices without requiring native application installation.

3.4 System Architecture

The system was built using a modern client-server architecture optimised for mobile-first deployment in resource-constrained environments, illustrated in Figure 3.3. The architecture was designed to minimise connectivity requirements for core triage operations while enabling cloud-based telemedicine escalation for Refer cases.

3.4.1 Frontend Client Layer

The frontend was implemented as a Progressive Web Application using the Next.js framework. Next.js was selected for its Server-Side Rendering capabilities, which significantly improve initial page load times on 3G networks common in rural Rwanda. The framework also provides built-in image optimisation and code splitting, reducing bandwidth requirements. Key features include an image capture interface, triage results display, telemedicine submission form, and real-time teleconsultation via WebSocket integration.

3.4.2 AI Backend Server Layer

The inference server was implemented using FastAPI, chosen for its high performance and native async support, enabling handling of multiple concurrent triage requests efficiently. The classification model is loaded once at application startup and held in memory for rapid inference, avoiding the latency of repeated model loading. All endpoints require authentication tokens validated through Supabase, ensuring that only authorised health workers can access the system. Image uploads are size-limited to prevent denial-of-service attacks and reduce bandwidth consumption on mobile networks.

3.4.3 Database Layer

Supabase provides a PostgreSQL database with built-in authentication and real-time subscriptions. Supabase allows selection of a data region, enabling compliance with Rwandan data protection requirements by hosting data within African servers rather than distant data centres.

3.4.4 Communication Layer

The frontend communicates with the backend through RESTful API endpoints over HTTPS, ensuring secure transmission of clinical data, protecting patient confidentiality, and aligning with standard healthcare data security practices.

Refer case notifications are delivered via SMS using Mista IO, a pan-African telecommunications platform with presence in Rwanda. SMS was selected because it works on basic mobile phones without requiring data connectivity or smartphone ownership. SMS messages are sent only for Refer cases to avoid notification fatigue among specialists. Real-time teleconsultation between health workers and specialist dermatologists is facilitated through WebSocket-based video sessions using LiveKit, enabling live clinical review of flagged cases without requiring patients to travel to urban facilities.

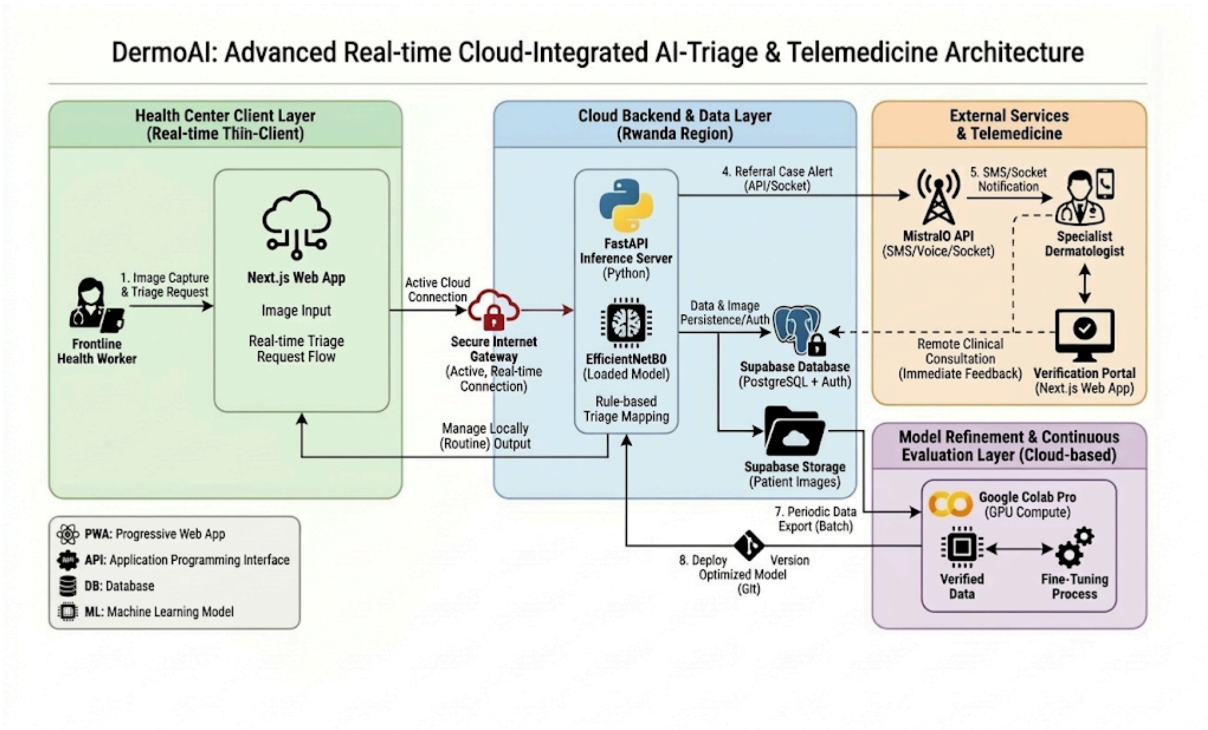
3.4.5 Retraining Module

Model retraining occurs in Google Colab Pro notebooks rather than as part of the production system. This separation keeps the operational system efficient and stable. Verified labels accumulated in the Supabase database are periodically exported and used to fine-tune the classification model through transfer learning. Newly trained models undergo validation against held-out test sets stratified by Fitzpatrick Skin Type before deployment to ensure performance thresholds are maintained across all skin tone groups.

3.4.6 System Integration and Data Flow

The end-to-end data flow follows the three-component architecture described in Section 3.2.1: health worker image capture triggers AI triage through the FastAPI backend, Refer cases route automatically to specialist notification and optional LiveKit teleconsultation, and specialist-verified outcomes are stored for future model retraining. Full workflow detail is illustrated in Figure 3.3.

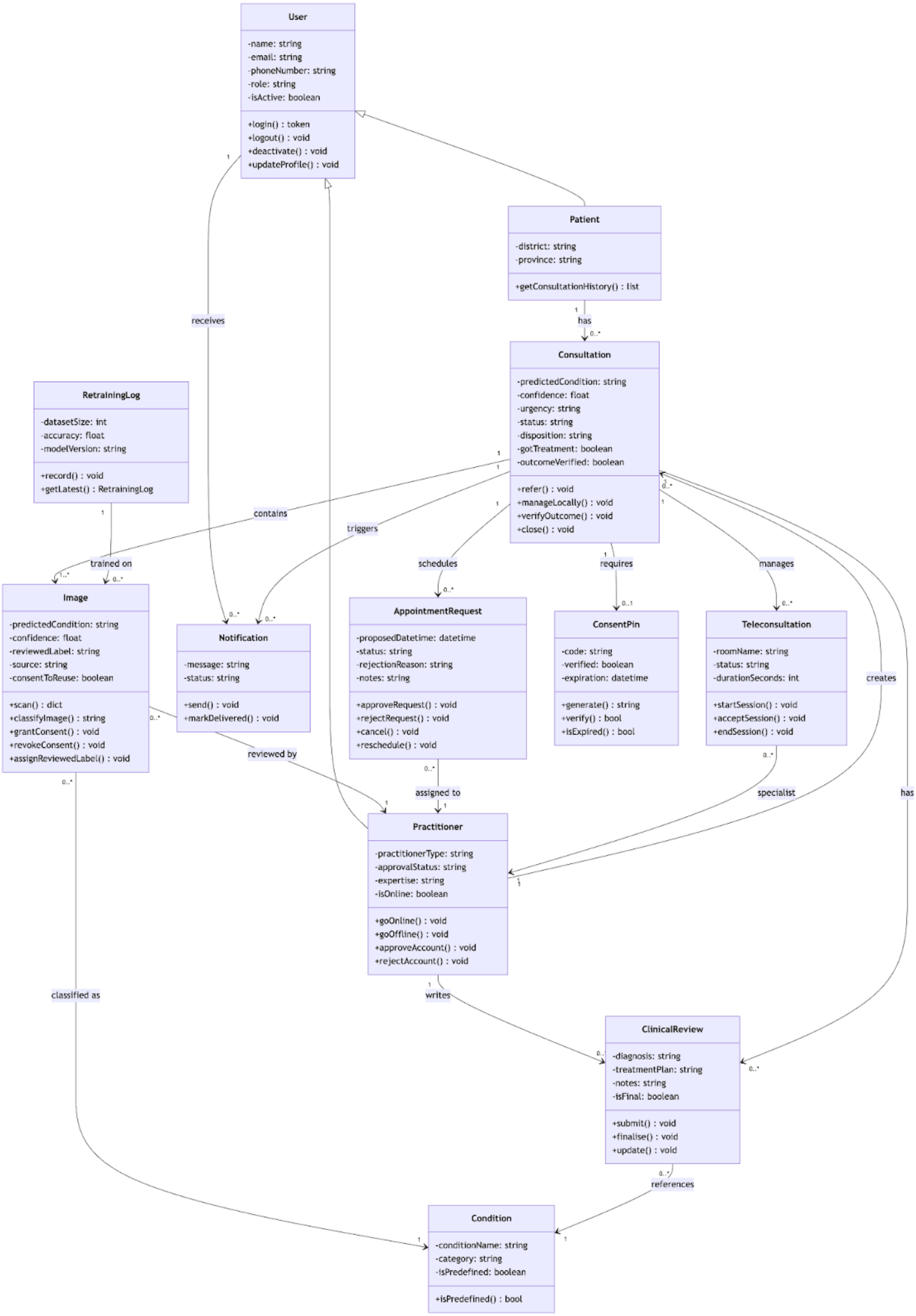
Figure 3.3: System Architecture



3.5 Class Diagram

The system was designed following object-oriented principles with modular separation of concerns across distinct components, illustrated in Figure 3.4. The class diagram follows a domain-driven structure organised into four conceptual groups. The user domain comprises User as the base class for authentication and role-based access control, with Patient and Practitioner extending it through inheritance to capture role-specific attributes and behaviours. The consultation core includes Consultation as the central coordinating entity, linking to Image for clinical photograph management and AI triage output, ClinicalReview for structured clinician documentation, Notification for SMS delivery via Mista IO, and ConsentPin for ethical patient consent management. The AI and image domain captures Image with its predicted condition, confidence score, and consent status, alongside Condition for the five classifiable dermatological conditions and RetrainingLog for auditing model improvement cycles. The scheduling domain includes AppointmentRequest and Teleconsultation for managing remote specialist access via LiveKit. The design emphasises loose coupling — Notification failures do not block clinical workflows, triage outputs are captured as Image attributes independently of clinical review, and retraining is decoupled from the operational system through a standalone RetrainingLog entity.

Figure 3.4: System Class Diagram

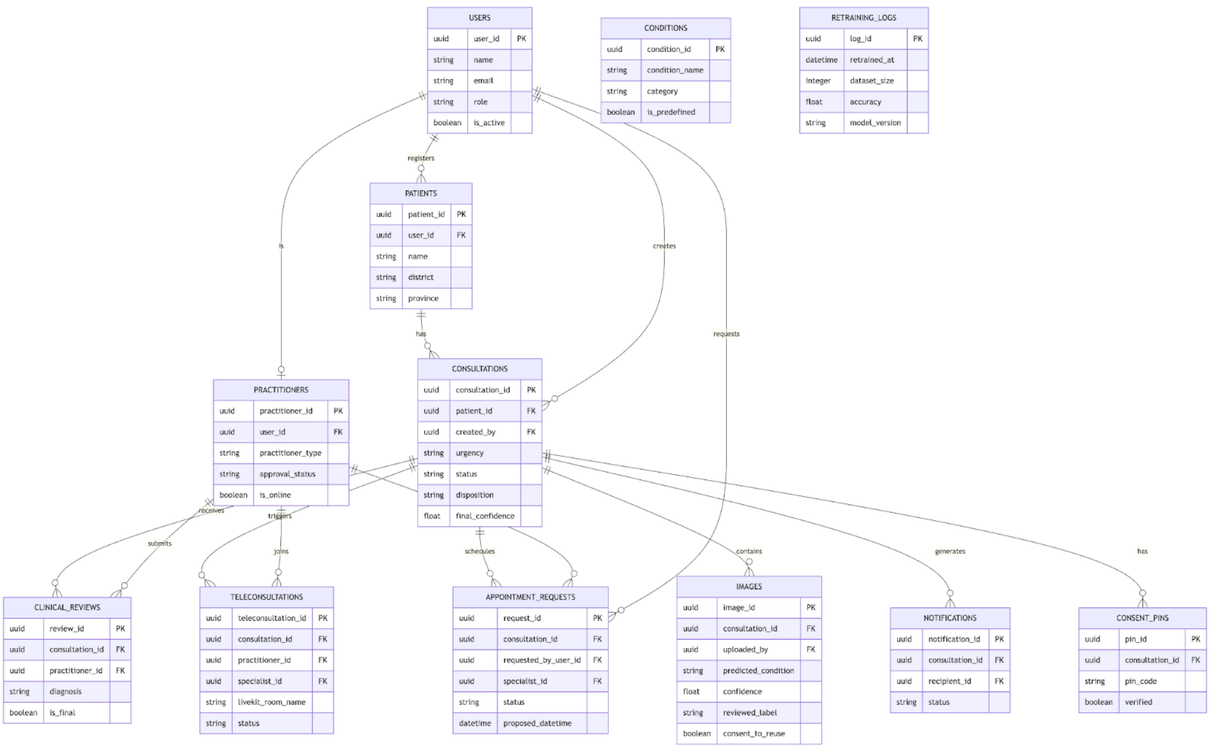


3.6 UML Diagrams (ERD)

3.6.1 Entity Relationship Diagram

The Entity Relationship Diagram details the database schema supporting both operational functionality and future model refinement, illustrated in Figure 3.5. The schema tracks users, consultations, images, specialist reviews, and VerifiedData records that feed the continuous learning pipeline. The design balances normalisation for data integrity with selective denormalisation for query performance in resource-constrained environments.

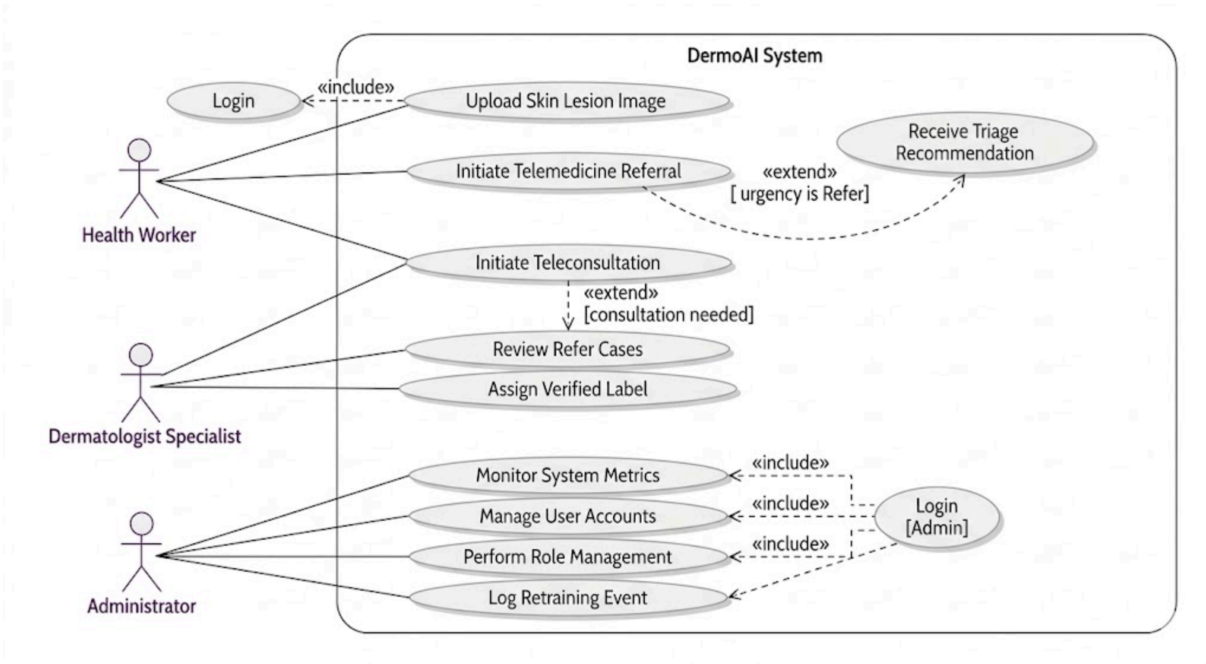
Figure 3.5: Entity Relationship Diagram (ERD)



3.6.2 Use Case Diagram

The primary system actors are health workers, specialist dermatologists, and administrators. Health workers interact with the system to upload skin lesion images, receive triage recommendations, view GradCAM explanations, and initiate telemedicine referrals for Refer cases. Specialists interact through the review queue to assess flagged cases, assign verified diagnostic labels, and initiate or receive live teleconsultation sessions. Administrators interact with the ML metrics dashboard to monitor prediction distributions and log retraining events. These interactions collectively operationalise the triage, referral, and continuous improvement workflows described in Section 3.2.2.

Figure 3.6: Use Case Diagram



3.7 Development Tools

3.7.1 Languages, Frameworks and Libraries

Python 3.10+ served as the primary language for backend services and machine learning components, providing the necessary libraries for deep learning through TensorFlow 2.x with Keras, API development through FastAPI, and data manipulation through NumPy and Pandas. The Next.js frontend used TypeScript for type safety and improved developer experience. TensorFlow 2.x with Keras was selected for its mature deployment support, extensive documentation, and proven track record in medical imaging applications. Next.js provided React-based frontend development with built-in mobile performance optimisations. FastAPI provides high-performance API development with automatic OpenAPI documentation generation. LiveKit provided the open-source WebRTC infrastructure supporting real-time video teleconsultation.

3.7.2 Development Environment and Version Control

Visual Studio Code provided the integrated development environment, chosen for its extensive extension ecosystem supporting Python, TypeScript, and debugging capabilities. Google Colab Pro notebooks provided GPU-accelerated computing essential for model training, with the Pro tier selected for its longer continuous runtime and priority access to faster T4 and V100 GPUs. Git provided version control for both code and model checkpoints, enabling reproducible experiments.

3.7.3 Data, Storage and Deployment Infrastructure

Supabase provided a PostgreSQL database with built-in user authentication and real-time subscription capabilities. The Fitzpatrick17k dataset served as the primary training source due to its explicit skin type labelling validated by board-certified dermatologists. The ISIC Archive was analysed as a supplementary training source but was not incorporated into the final training pipeline due to insufficient valid FST label coverage, as documented in Chapter Four. Vercel provided serverless deployment for the Next.js application with automatic scaling and zero-configuration HTTPS. The FastAPI backend was deployed on Render, selected for its straightforward deployment workflow and support for persistent model storage. Patient images were stored in Cloudinary (Cloudinary, 2024) to improve database performance and enable efficient CDN distribution for faster image loading on mobile networks.

3.8 Ethical Considerations

This project adhered to strict ethical guidelines regarding medical data, patient safety, and algorithmic fairness. The ethical framework is structured around five core pillars.

3.8.1 Patient Privacy and Data Protection

Although the pilot used public, anonymised datasets like Fitzpatrick17k, the system architecture follows Privacy by Design principles. All data transmission occurs over HTTPS with end-to-end encryption. Future real-world deployment would require strict compliance with Rwanda's Law on Protection of Personal Data and Privacy (2021) to protect sensitive medical imagery. The database schema supports role-based access control, ensuring that patient data is accessible only to authorised healthcare providers involved in direct care.

3.8.2 Algorithmic Fairness and Bias Mitigation

The core ethical obligation is to reduce bias rather than amplify existing healthcare disparities. Using validation sets across different skin type groups ensures that models are not released unless they perform fairly, with a minimum threshold of 75% recall on Refer cases for FST V-VI. This threshold prevents the deployment of discriminatory technology that would worsen health equity. Model performance is monitored across skin type categories, and any degradation in performance for underrepresented groups triggers mandatory revalidation before continued use.

3.8.3 Safety and Clinical Accountability

To prevent misuse and maintain clinical accountability, the system explicitly labels all outputs as risk assessments rather than medical diagnoses. This framing clarifies that the system provides decision support rather than definitive diagnostic conclusions. The design mandates physician-oversight workflows, requiring human verification for all Refer cases before any treatment action is taken. Health workers retain full authority to override system recommendations based on clinical judgement and patient context not captured by image analysis alone.

The system implements formal fallback protocols to mitigate the risk of incorrect triage outputs influencing frontline care decisions. When model confidence falls below the predefined threshold of 0.45, the system returns an Uncertain output that automatically escalates the case to Refer. Additionally, when the probability assigned to any Refer class exceeds 0.60, the system forces a Refer recommendation regardless of the primary prediction. These rules ensure that ambiguous and borderline cases default to the safer clinical outcome. The design threshold of 0.45 was refined to 0.35 during production deployment based on observed confidence distributions with real-world image quality, as documented in Chapter Five.

The conservative triage mapping intentionally accepts higher false-positive rates to minimise the risk of missing cases that require specialist attention. This design choice prioritises patient safety over system efficiency, acknowledging that unnecessary specialist reviews pose less risk than delayed diagnosis of serious conditions.

3.8.4 Informed Use and Limitations Disclosure

Health workers using the system receive training on its intended use, technical limitations, and scenarios where human judgement should override algorithmic recommendations. The interface displays confidence levels alongside triage recommendations, enabling users to assess reliability and exercise appropriate caution with low-confidence predictions. System limitations, including reduced performance on conditions underrepresented in training data and potential degradation with image quality issues, are documented and communicated to all users during onboarding.

3.8.5 Operational Assumptions

The system was designed on the assumption that frontline health workers would have access to mid-range Android smartphones with functional cameras, and that health centers would maintain periodic internet connectivity sufficient for telemedicine referral submission. Variations in real-world image capture conditions, including lighting, framing, and camera quality, may affect model performance in ways not captured by the validation dataset. The system's confidence display was designed to support health workers in recognising when image quality may be insufficient for reliable triage.

The telemedicine component assumed that dermatology specialists in urban centres would have the capacity and institutional support to review flagged Refer cases remotely. Without reliable specialist engagement, the telemedicine referral pathway cannot function as designed. Securing specialist participation in a real deployment would require formal agreements with the Ministry of Health and district health authorities, which falls outside the scope of this research project.

CHAPTER FOUR: SYSTEM IMPLEMENTATION

4.1 Implementation and Coding

4.1.1 Introduction

This chapter documents the implementation of DermoAI, covering the machine learning pipeline, backend API, and frontend application. The focus is on the technical decisions made at each stage including dataset preparation, condition selection, model architecture, training configuration, and deployment. Performance results and clinical evaluation are presented in Chapter Five.

4.1.2 Tools and Technologies

The machine learning pipeline was built in Python 3.11 using TensorFlow 2.19 with the Keras API for model training and serialisation. EfficientNetB0 pretrained on ImageNet served as the base architecture. Data preparation used pandas, NumPy, and scikit-learn. Image augmentation used the Albumentations library and GradCAM explainability was implemented through a dedicated service module using TensorFlow's gradient tape. All training experiments ran on GPU-accelerated compute via Google Colab Pro with experiments tracked in Jupyter Notebooks.

The backend API was built using FastAPI with SQLAlchemy and asyncpg connecting to a PostgreSQL database hosted on Supabase, selected for its African-region data hosting option to support compliance with Rwanda's Law on Protection of Personal Data and Privacy (2021). Patient images were stored and delivered through Cloudinary's content delivery network. Real-time teleconsultation used LiveKit, SMS referral notifications used the Mista IO API, and the server was deployed on Render using Uvicorn.

The frontend was built in Next.js 16 with TypeScript, with Tailwind CSS version 4 for styling and React Query for server-state synchronisation. The application was configured as a Progressive Web Application with Service Worker support. JWT tokens with refresh rotation handled authentication and role-based access control was enforced at both API and routing levels. The frontend was deployed on Vercel. The live application is accessible at https://dermo.vercel.app and the API documentation at https://dermoai-24lz.onrender.com/docs.

4.2 Data Collection and Exploratory Data Analysis

4.2.1 Dataset Description

Two publicly available dermatological datasets were used in this project. The Fitzpatrick17k dataset served as the primary training source due to its explicit Fitzpatrick Skin Type annotations validated by board-certified dermatologists (Groh et al., 2021). The full dataset contains 16,577 images spanning 112 dermatological conditions and is publicly accessible at https://github.com/mattgroh/fitzpatrick17k.

The FST V-VI subset of 2,155 images was obtained by filtering on fitzpatrick_scale values of 5 and 6, representing the target population for this study. An integrity check confirmed perfect alignment between the filtered CSV entries and images on disk with zero missing files. The dataset contains two FST label columns: fitzpatrick_scale, assigned by the dataset authors, and fitzpatrick_centaur, assigned by a separate automated tool. A disagreement rate of 53.1% was observed between the two columns, and fitzpatrick_scale was adopted as the primary label throughout all filtering, stratification, and fairness evaluation steps on the basis that it represents the human-validated annotation.

The ISIC Archive, containing 549,591 images, was analysed as a potential supplementary source but found to carry valid FST labels on only 1.91% of images, the majority of which were benign melanocytic lesions sourced from a single Western institution. This confirmed Fitzpatrick17k as the only viable primary training source, consistent with the representation gap documented by Alipour et al. (2024). A label quality consideration noted during preparation was that condition labels in Fitzpatrick17k derive from online dermatology atlases rather than histologically confirmed diagnoses, with only 69% of images assessed as clearly diagnostic by board-certified dermatologists (Groh et al., 2021). This limitation informed the conservative condition selection approach documented in Section 4.3.1.

4.2.2 Exploratory Data Analysis

Exploratory data analysis examined the dataset structure, FST distribution, class imbalance, and clinical category balance within the FST V-VI subset to inform the condition selection and augmentation strategy in Section 4.3.

The raw metadata structure of the filtered subset is presented in Figure 4.1, confirming the key columns used throughout the pipeline: md5hash as the image identifier, fitzpatrick_scale as the primary FST label, label as the fine-grained condition name, and three_partition_label as the clinical priority category.

Figure 4.1: Dataset metadata preview showing the FST V-VI subset structure

	md5hash	fitzpatrick_scale	fitzpatrick_centaur	label	nine_partition_label	three_partition_label	qc	
0	45f7fe0e10214e32e890cad9d29d4811	6	5	kaposi sarcoma	malignant dermal	malignant	NaN	https://www.dermaamin.com/
1	ddcad677b7b1e9084f3f51a8e026aa8d	5	5	hidradenitis	inflammatory	non-neoplastic	NaN	https://www.dermaamin.com/
2	1e119546f5bc2b9165bb10ddd7fe5f69	5	4	xeroderma pigmentosum	genodermatoses	non-neoplastic	NaN	https://www.dermaamin.com/
3	4c3f795cf8eb72b946f9bd2642cf23c1	6	5	melanoma	malignant melanoma	malignant	NaN	https://www.dermaamin.com/
4	b09233673fc585369e723ec841ed0acb	5	3	actinic keratosis	malignant epidermal	malignant	NaN	https://www.dermaamin.com/

The FST distribution analysis revealed an internal imbalance between the two target skin types. FST V comprised 1,527 samples (70.9%) and FST VI comprised 628 samples (29.1%), as illustrated in Figure 4.2. This disproportion carried forward into the training data and directly influenced the fairness evaluation conducted in Chapter Five.

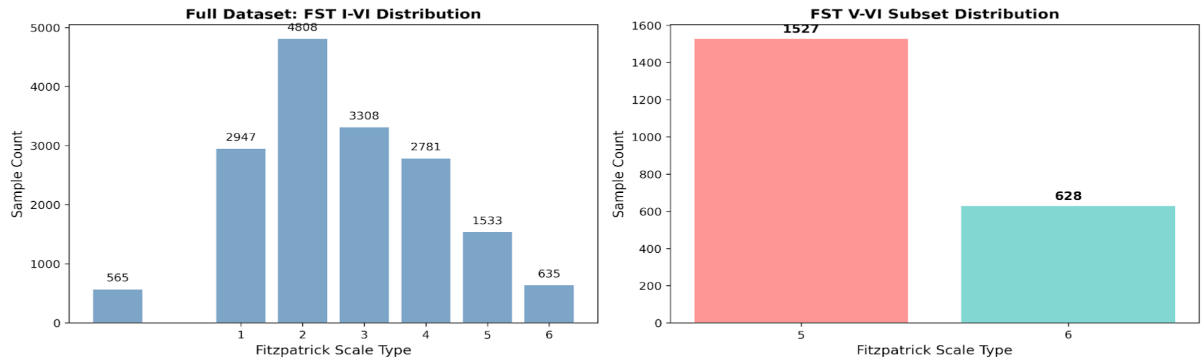


Figure 4.2: FST distribution across the full Fitzpatrick17k dataset and the filtered FST V-VI subset

Condition-level analysis revealed severe long-tail class imbalance across the 112 conditions, with the most common condition carrying 127 samples and the rarest carrying a single sample, yielding a 127:1 maximum-to-minimum ratio. The top 20 conditions by sample count are presented in Figure 4.3.

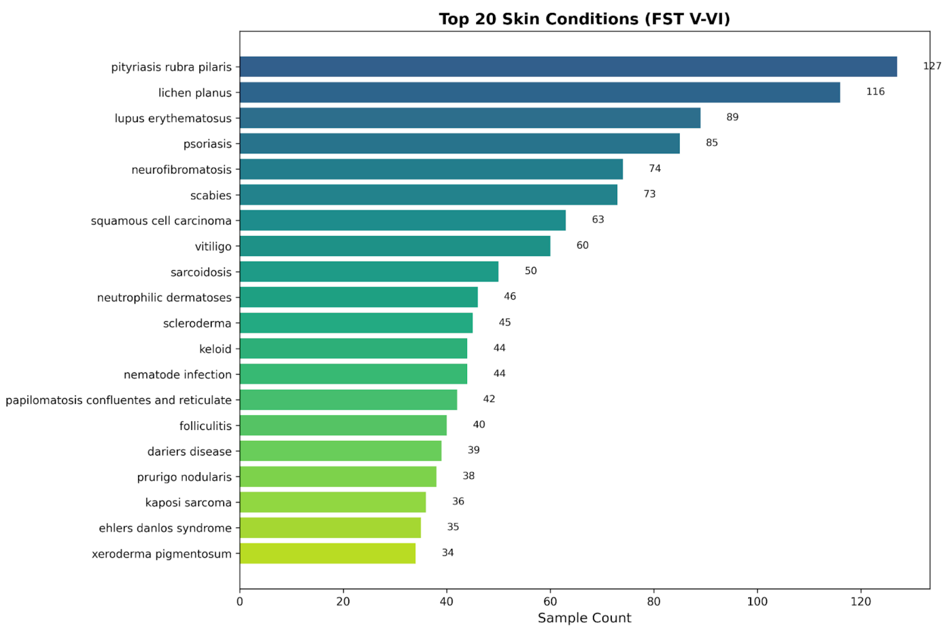


Figure 4.3: Top 20 conditions by sample count in the FST V-VI subset

The three-partition clinical priority analysis showed that 81.0% of FST V-VI samples were non-neoplastic, 9.6% were malignant, and 9.4% were benign. Within the malignant subset, several conditions carried fewer than five samples, presenting a patient safety risk that informed the condition exclusion decisions described in the following section.

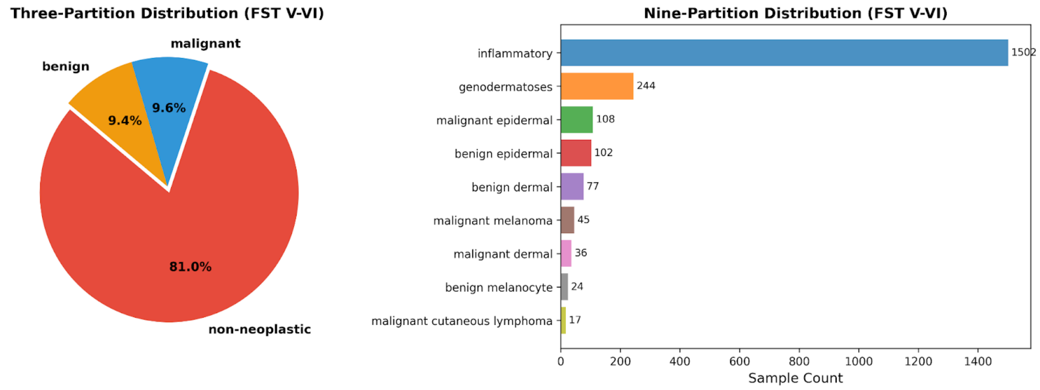


Figure 4.4: Three-partition and nine-partition clinical category distributions in FST V-VI subset

Table 4.1 consolidates the key quantitative findings from the EDA. These characteristics collectively informed every decision made in the data preparation and model design stages that follow.

Table 4.1: EDA summary metrics for the FST V-VI subset

Metric	**Value**
Total FST V-VI images	2,155
FST V samples	1,527 (70.9%)
FST VI samples	628 (29.1%)
Total conditions	112
Mean samples per condition	19.2
Median samples per condition	11.0
Maximum samples	127
Minimum samples	1
Max/min imbalance ratio	127:1
Rare conditions under 10 samples	52 (46.4%)
Malignant samples	206 (9.6%)
Non-neoplastic samples	1,746 (81.0%)
Benign samples	203 (9.4%)

4.3 Feature Engineering and Data Preparation

4.3.1 Condition Selection

The 112-condition taxonomy was reduced to a final five-class system through a tiered decision framework evaluating training viability, patient safety risk, and clinical relevance to Rwanda's primary care context. The framework applied the following thresholds: conditions with a training split of 50 or more images were considered independently viable; conditions with 35 to 49 training images were evaluated further based on malignancy status or Rwanda-specific priority; conditions below 21 training images were excluded or grouped. Table 4.2 documents the outcome of this framework for all eight candidate conditions that reached the evaluation stage.

Table 4.2: Condition selection decision framework

Condition	FST V-VI Samples	Train Split	Decision	Reason
Pityriasis rubra pilaris	127	76	Retained	Sufficient samples, visually distinctive erythrodermic plaques, strong REFER signal
Lupus erythematosus	89	53	Retained	Sufficient samples, cutaneous flares on FST V-VI warrant specialist REFER
Psoriasis	85	51	Retained	Sufficient samples, common papulosquamous presentation in primary care
Neurofibromatosis	74	44	Retained	Visually distinctive nodules and café-au-lait patches, stable MANAGE LOCALLY triage
Scabies	73	44	Retained	Highest clinical priority for Rwanda, hyperendemic in primary care settings
Lichen planus	116	70	Excluded	Visual overlap with psoriasis on FST V-VI consistently degraded all class recall empirically
Squamous cell carcinoma	63	38	Excluded	Malignancy — false negative risk unacceptable at this sample size
Vitiligo	60	36	Excluded	Below minimum viability threshold after FST V-VI filtering

Notably, lichen planus met the sample count threshold but was excluded following empirical training trials in which its inclusion consistently reduced recall across all other classes, an outcome attributable to the well-documented visual overlap between lichen planus, psoriasis, and cutaneous lupus on deeply pigmented skin (Daneshjou et al., 2022). The deliberate exclusion of squamous cell carcinoma means the five-class system contains no malignant conditions, ensuring that classification errors carry manageable clinical consequences rather than the life-threatening implications of a missed malignancy. The five retained conditions produced a working dataset of 448 images from which stratified splits were constructed.

4.3.2 Augmentation Pipeline

Data augmentation was applied exclusively to the training set to address the class imbalance identified in Section 4.2.2 and improve model generalisation across the range of image capture conditions expected in rural Rwandan clinic settings. Prior to augmentation, all images were converted to RGB, resized to 224×224 pixels, and saved at JPEG quality 95 to establish a consistent input format. The full augmentation pipeline implemented using the Albumentations library is shown in Figure 4.5.

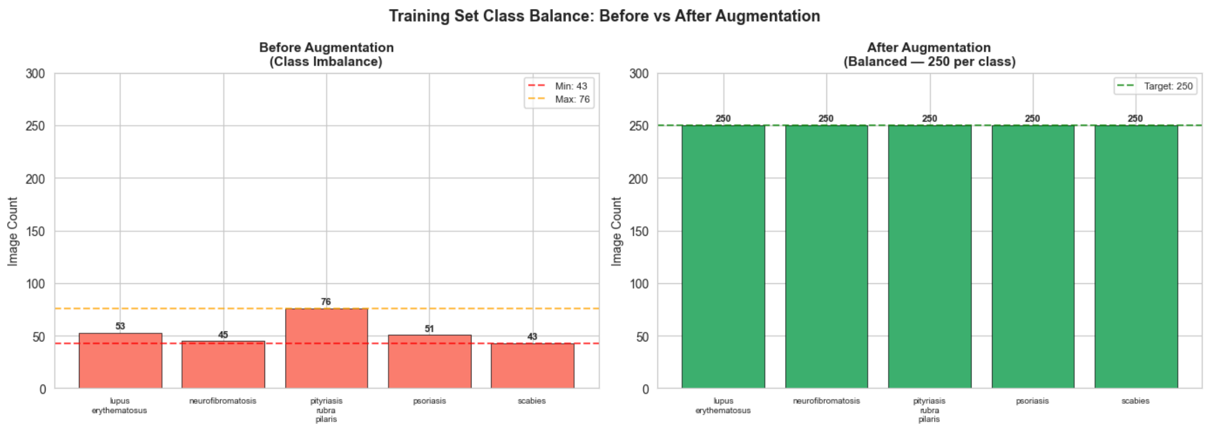
Figure 4.5: Augmentation pipeline: Albumentations transforms applied to the training set

```
augmentation_pipeline = A.Compose([

    # --- Geometric transforms (all validated for skin lesions) ---
    A.HorizontalFlip(p=0.5),
    A.VerticalFlip(p=0.5),
    A.Rotate(limit=45, p=0.7),
    A.ShiftScaleRotate(
        shift_limit=0.1,
        scale_limit=0.1,
        rotate_limit=45,
        border_mode=0,
        p=0.5
    ),
    A.RandomResizedCrop(
        size=(224, 224),
        scale=(0.85, 1.0),
        p=0.3
    ),

    # --- Colour transforms (within literature-validated ranges) ---
    A.RandomBrightnessContrast(
        brightness_limit=0.15,
        contrast_limit=0.15,
        p=0.7
    ),
    A.HueSaturationValue(
        hue_shift_limit=20,       # safe for FST V-VI: stays within realistic skin tone range
        sat_shift_limit=20,
        val_shift_limit=15,
        p=0.5
    ),

    # --- Quality variation (mimics rural clinic phone camera conditions) ---
    A.GaussianBlur(blur_limit=(3, 5), p=0.2),
    A.GaussNoise(var_limit=(5, 20), p=0.2),

    A.Resize(height=224, width=224),
])
```

The HueSaturationValue hue shift was deliberately limited to ±20 degrees to remain within clinically realistic FST V-VI skin tone boundaries, and brightness and contrast variation was capped at ±15% to simulate rural clinic lighting variability without distorting diagnostically relevant lesion features. Gaussian blur and noise transforms were included specifically to replicate mid-range smartphone camera characteristics common in the intended deployment context. The augmentation target was set at 250 images per class, a threshold above the minimum of 200 per class documented as sufficient for transfer learning fine-tuning in medical imaging (Tan & Le, 2019).

Because raw class sizes differed, augmentation factors varied per class: pityriasis rubra pilaris required the lowest factor at 3.29×, while scabies required the highest at 5.81×, a disparity that directly connects to scabies being the most challenging class in the evaluation presented in Chapter Five. As illustrated in Figure 4.6, the pipeline expanded the raw training set from an imbalanced range of 43 to 76 images per class to a perfectly balanced 1,250-image training set. No augmentation was applied to the validation or test sets.

Figure 4.6: Training set class balance before and after augmentation



4.3.3 Stratified Split

A stratified 60/20/20 split was applied using a composite stratification key formed by concatenating the final class label with the FST binary label, producing 10 stratification groups covering each of the five conditions across FST V and FST VI. This preserved both class distribution and the FST ratio simultaneously across all three sets. The split was implemented in two steps: a first split separating 60% as training and 40% as a temporary holdout, followed by a second 50/50 split of the holdout to produce equal validation and test sets. Data leakage between splits was verified by confirming zero overlap in md5hash values across all three sets prior to augmentation. Table 4.3 confirms that the 71:29 FST ratio identified in the EDA was preserved across all three sets.

Table 4.3: Stratified split verification showing FST ratio preservation

Split	**Total**	**FST V**	**FST VI**	**PRP**	**Lupus**	**Psoriasis**	**Neurofibromatosis**	**Scabies**
Train	268 (60%)	201 (75.0%)	67 (25.0%)	76	53	51	45	43
Val	90 (20%)	67 (74.4%)	23 (25.6%)	26	18	17	14	15

Test	90 (20%)	68 (75.6%)	22 (24.4%)	25	18	17	15	15

4.4 Model Architecture and Training

4.4.1 Model Architecture

DermoAI used EfficientNetB0 pretrained on ImageNet as the base feature extractor. EfficientNetB0 was selected due to its compound scaling approach, which balances network depth, width, and resolution simultaneously, providing stronger feature extraction capability on small imbalanced datasets without a prohibitive parameter cost (Tan & Le, 2019). All input images were resized to 224×224 pixels, a standard dimension for EfficientNetB0 and consistent with the high variability in raw image dimensions identified during EDA where mean width was 523 pixels and mean height was 451 pixels. A custom classification head was attached comprising GlobalAveragePooling2D, a Dense layer of 256 units with ReLU activation and L2 regularisation, Dropout at 0.3, a second Dense layer of 128 units with ReLU activation and L2 regularisation, Dropout at 0.15, and a five-class softmax output layer. The complete model summary is shown in Figure 4.7, confirming a total of 4,411,048 parameters with 361,477 trainable during Phase 1 feature extraction.

Figure 4.7: DermoAI_EfficientNetB0 model summary

Model Summary:

Model: "DermoAI_EfficientNetB0"

Layer (type)	Output Shape	Param #
input_layer_3 (InputLayer)	(None, 224, 224, 3)	0
efficientnetb0 (Functional)	(None, 7, 7, 1280)	4,049,571
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 1280)	0
dense_3 (Dense)	(None, 256)	327,936
dropout_2 (Dropout)	(None, 256)	0
dense_4 (Dense)	(None, 128)	32,896
dropout_3 (Dropout)	(None, 128)	0
dense_5 (Dense)	(None, 5)	645

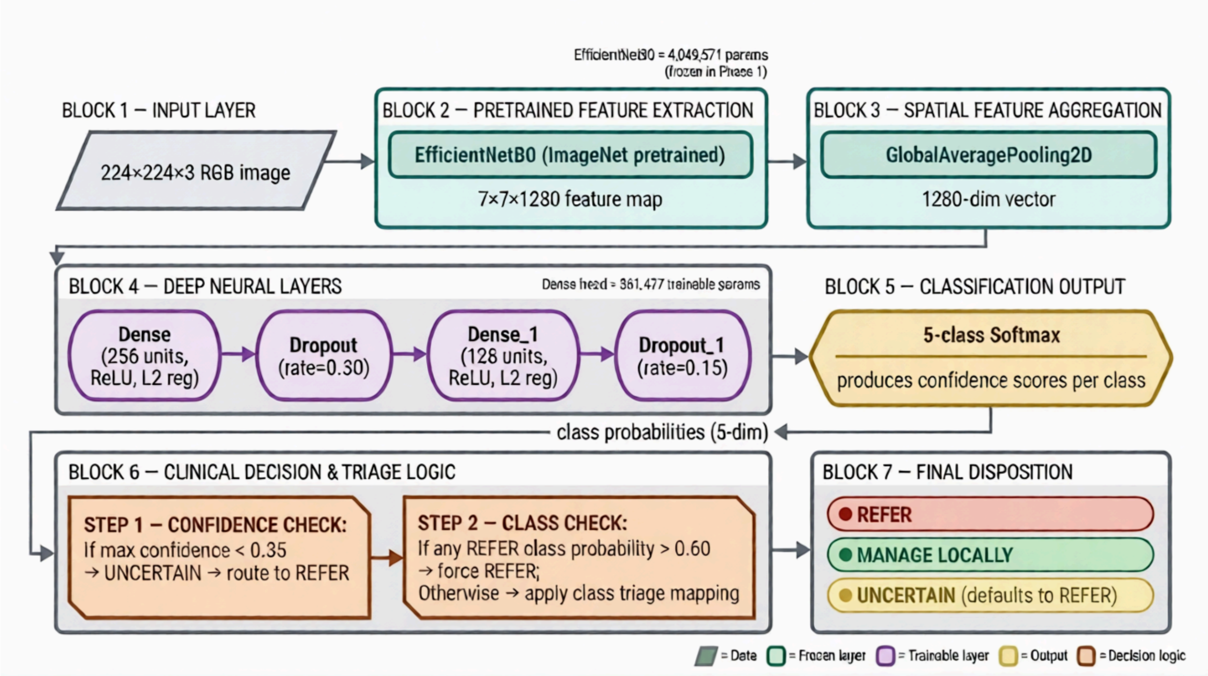
Total params: 4,411,048 (16.83 MB)

Trainable params: 361,477 (1.38 MB)

Non-trainable params: 4,049,571 (15.45 MB)

The trained classifier fed directly into a two-stage triage decision mechanism that translated condition predictions into clinical referral recommendations before presenting results to the health worker, as illustrated in Figure 4.8. Stage 1 checked whether the maximum class probability fell below the confidence threshold, routing low-confidence predictions to UNCERTAIN and escalating them to REFER as a safety default. Stage 2 checked whether any REFER-class probability exceeded the override threshold and forced a REFER recommendation regardless of the primary prediction. This separation between condition classification and triage mapping kept the system aligned with primary care workflows and avoided autonomous diagnosis.

Figure 4.8: DermoAI two-stage architecture



4.4.2 Training Configuration

Training followed a two-phase transfer learning strategy designed to maximise feature extraction from the limited 1,250-image training set. Phase 1 froze the EfficientNetB0 base and trained only the classification head, allowing the model to learn dermatological feature mappings without disrupting ImageNet-pretrained convolutional weights. Phase 2 unfroze the last 30 layers of the base and fine-tuned at a reduced learning rate, allowing upper convolutional layers to adapt to FST V-VI dermatological features while preserving lower-level representations. Training used a batch size of 16, producing 79 steps per epoch from the 1,250-image training set. This step count was used to calculate the total decay steps for the CosineDecay learning rate schedule in both phases. The complete configuration for both phases is documented in Table 4.4.

Table 4.4: Two-phase training configuration

Configuration	Phase 1 Feature Extraction	Phase 2 Fine-Tuning
Base model	Frozen — 0 trainable layers	Last 30 layers unfrozen
Trainable parameters	361,477	1,857,637

Batch size	16	16
Steps per epoch	79	79
Epochs	50	40
Initial learning rate	3×10^{-4}	5×10^{-5}
LR schedule	CosineDecay to 1×10^{-7}	CosineDecay to 1×10^{-7}
Optimizer	Adam	Adam
Loss function	Focal Loss γ=2.0, α=0.25, smoothing=0.1	Same
Early stopping patience	20 epochs on val_loss	20 epochs on val_loss
Lupus erythematosus weight	3.0×	4.0×
Pityriasis rubra pilaris weight	1.5×	4.0×
Neurofibromatosis weight	2.0×	2.0×
Psoriasis weight	1.5×	2.0×
Scabies weight	2.5×	3.0×

Asymmetric class weights were applied in both phases to reflect clinical risk asymmetry, penalising misclassification of REFER conditions more heavily than MANAGE LOCALLY conditions, and to prioritise recall for REFER classes consistent with the conservative safety-first design described in Section 3.2.3. REFER class weights were elevated further in Phase 2 to maximise referral recall in line with the 75% safety target defined in Section 3.2.4. The CosineDecay learning rate schedule was chosen to produce smooth continuous learning rate reduction without abrupt transitions that could destabilise training on a small dataset.

4.4.3 Loss Function and Inference Implementation

Focal loss was used throughout both training phases in place of standard cross-entropy to address the challenge of hard and easy examples within the small training set. The focusing parameter gamma of 2.0 down-weighted easy examples during training, forcing the model to concentrate on difficult and ambiguous cases, particularly conditions with visual overlap on FST V-VI skin. Label smoothing at 0.1 was applied to prevent overconfident predictions and reduce the train-to-validation generalisation gap. The focal loss function was registered with Keras serialisation using the register_keras_serializable decorator to allow model reloading at inference time without requiring the training loss to be present in memory, as shown in Figure 4.9.

Figure 4.9: focal_loss_with_smoothing registered with Keras serialisation

```python
import tensorflow.keras.backend as K

@tf.keras.utils.register_keras_serializable(package='DermoAI')
def focal_loss_with_smoothing(y_true, y_pred, gamma=2.0, alpha=0.25, smoothing=0.1):
    """
    Focal Loss with Label Smoothing for multi-class classification.

    Registered with Keras serialization so model can be saved and
    reloaded without 'could not locate function' errors.

    Args:
        gamma:    Focusing parameter — down-weights easy examples (default: 2.0)
        alpha:    Balance parameter for class weighting (default: 0.25)
        smoothing: Label smoothing factor — prevents overconfidence (default: 0.1)
    """
    num_classes = tf.cast(tf.shape(y_true)[-1], tf.float32)
    y_true_smooth = y_true * (1.0 - smoothing) + (smoothing / num_classes)
    epsilon = K.epsilon()
    y_pred = tf.clip_by_value(y_pred, epsilon, 1.0 - epsilon)
    cross_entropy = -y_true_smooth * tf.math.log(y_pred)
    loss = alpha * tf.pow(1.0 - y_pred, gamma) * cross_entropy
    return tf.reduce_mean(tf.reduce_sum(loss, axis=-1))

FOCAL_GAMMA = 2.0
FOCAL_ALPHA = 0.25
```

A critical preprocessing requirement for EfficientNetB0 is that input pixel values must be in the range [-1, 1], supplied via efficientnet.preprocess_input rather than standard [0, 1] normalisation. An earlier implementation normalising to [0, 1] collapsed all class probabilities to approximately 0.2, effectively disabling the classifier. The corrected preprocessing function is shown in Figure 4.10, with an inline comment documenting this requirement explicitly.

Figure 4.10: _preprocess function showing EfficientNetB0 pixel range requirement

```python
def _preprocess(img: Image.Image) -> np.ndarray:
    """Resize and preprocess image to model input shape (1, 224, 224, 3).

    EfficientNetB0 was trained with efficientnet.preprocess_input which maps
    pixel values [0, 255] → [-1, 1]. Dividing by 255 produces [0, 1] which is
    out-of-distribution and collapses all predictions to ~0.2.
    """
    from keras.applications.efficientnet import preprocess_input
    img = img.resize(INPUT_SIZE)
    arr = np.array(img, dtype=np.float32)          # keep [0, 255]
    arr = preprocess_input(arr)                     # → [-1, 1]
    return np.expand_dims(arr, axis=0)
```

The production inference service implemented the full two-stage triage routing logic within the predict_with_details function, shown in Figure 4.11. Two threshold values govern the routing: CONFIDENCE_THRESHOLD set to 0.35 and REFER_OVERRIDE_THRESHOLD set to 0.60. The confidence threshold was lowered from the training value of 0.45 to 0.35 in production after observing that real-world uploads with varied lighting and framing produced systematically lower confidence scores than the curated Fitzpatrick17k training images. At inference, test-time augmentation was applied using 10 prediction passes with random horizontal flip, rotation, zoom, and contrast adjustments to stabilise confidence estimates before final class probabilities were computed. The function returns the predicted condition, confidence score, urgency decision, all class probabilities, triage stage identifier, and model version metadata in a single response object.

Figure 4.11: predict_with_details method implementing two-stage triage logic

```python
def predict_with_details(image_url: str) -> dict:
    predictions = _get_predictions(image_url)
    max_confidence = float(np.max(predictions))
    predicted_idx = int(np.argmax(predictions))

    triage_stage = None

    # Stage 1: Low confidence → UNCERTAIN
    if max_confidence < CONFIDENCE_THRESHOLD:
        predicted_condition = "UNCERTAIN"
        confidence = max_confidence
        triage_stage = "STAGE_1_LOW_CONFIDENCE"
    else:
        # Stage 2: REFER override check
        refer_override = False
        for refer_class in REFER_CLASSES:
            refer_idx = CLASS_NAMES.index(refer_class)
            if predictions[refer_idx] > REFER_OVERRIDE_THRESHOLD:
                predicted_condition = refer_class
                confidence = float(predictions[refer_idx])
                refer_override = True
                triage_stage = "STAGE_2_REFER_OVERRIDE"
                break

        if not refer_override:
            predicted_condition = CLASS_NAMES[predicted_idx]
            confidence = max_confidence
            triage_stage = "NORMAL_PREDICTION"

    urgency = classify_urgency(predicted_condition, confidence)

    return {
        "predicted_condition": predicted_condition,
        "confidence": round(confidence, 4),
        "urgency": urgency,
        "all_probabilities": {
            CLASS_NAMES[i]: round(float(predictions[i]), 4)
            for i in range(len(CLASS_NAMES))
        },
        "model_version": MODEL_VERSION,
        "model_date": MODEL_DATE,
        "triage_stage": triage_stage,
    }
```

4.4.4 Explainability Implementation

GradCAM explainability was implemented as a dedicated service module to generate gradient-weighted class activation maps for every non-UNCERTAIN prediction. Activation maps were computed over the top_conv layer of EfficientNetB0 using TensorFlow's gradient tape, weighted by the global average pooled gradients of the predicted class output with respect to the convolutional feature maps. The resulting heatmap was resized to 224×224 pixels, converted to a jet colormap, and superimposed on the original image using alpha blending at a ratio of 0.4 heatmap to 0.6 original image. The overlay was returned as a base64-encoded PNG stored alongside the prediction in the API response. The generate_gradcam_for_prediction function implementing this pipeline is shown in Figure 4.12.

Figure 4.12: generate_gradcam_for_prediction implementing GradCAM explainability

```python
def generate_gradcam_for_prediction(
    # Load image (same as ml_service.py)
    from app.services.ml_service import _load_image

    image = _load_image(image_url)
    gradcam_image, metrics = generate_gradcam(model, image, class_idx, layer_name)
    gradcam_base64 = gradcam_to_base64(gradcam_image)

    return {
        "gradcam_base64": gradcam_base64,
        "gradcam_metrics": metrics,
        "class_idx": class_idx,
    }
```

When multiple images were uploaded to a single consultation, the aggregate_predictions function applied majority voting over individual condition predictions using Python's Counter class and computed mean confidence across all uploaded images to produce a single consolidated urgency recommendation, as shown in Figure 4.13.

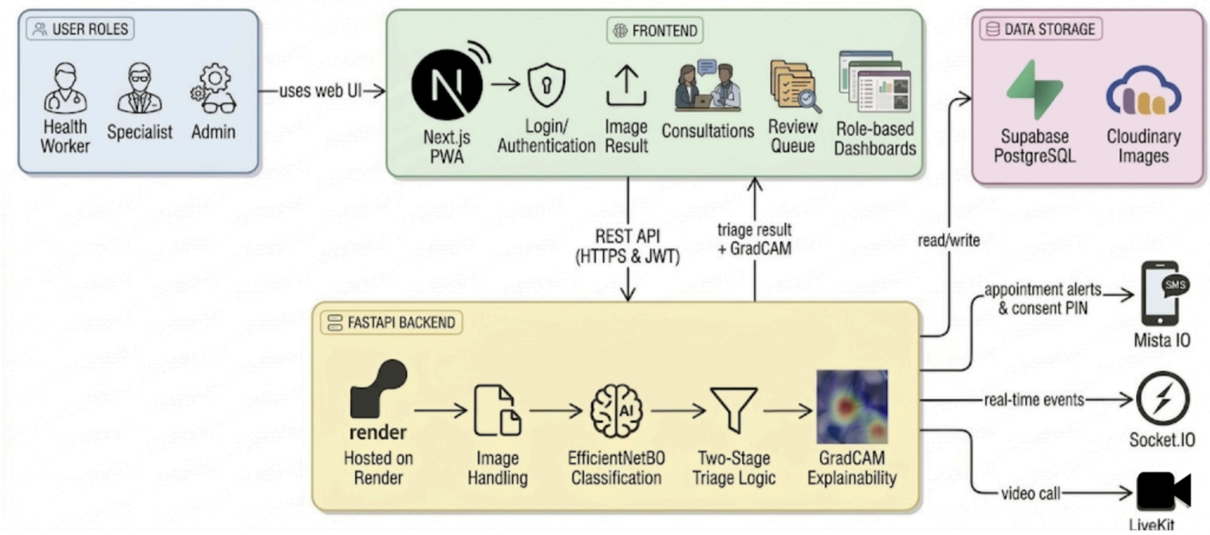
Figure 4.13: aggregate_predictions for single consultation

```python
def aggregate_predictions(images: list[dict]) -> dict[str, str | float | None]: 1mo Reponse MVP
    if not images:
        return {
            "final_predicted_condition": None,
            "final_confidence": None,
            "urgency": None,
        }

    conditions = [
        img["predicted_condition"]
        for img in images
        if img.get("predicted_condition")
    ]
    confidences = [
        img["confidence"]
        for img in images
        if img.get("confidence") is not None
    ]

    if not conditions:
        return {
            "final_predicted_condition": None,
            "final_confidence": None,
            "urgency": None,
        }

    counter = Counter[Any](conditions)
    final_condition = counter.most_common(1)[0][0]
    final_confidence = (
        round(sum(confidences) / len(confidences), 4) if confidences else 0.0
    )
    urgency = "REFER" if final_condition == "UNCERTAIN" else classify_urgency(final_condition, final_confidence)

    return {
        "final_predicted_condition": final_condition,
        "final_confidence": final_confidence,
        "urgency": urgency,
    }
```

4.5 System Features Implementation

The DermoAI system integrates the trained model into a full clinical workflow spanning three user roles: health workers who submit images and receive triage recommendations, specialists who review flagged cases and assign verified labels, and administrators who monitor system performance. The complete architecture connecting all components is illustrated in Figure 4.14.

Figure 4.14: DermoAI system architecture showing all integrated components



4.5.1 Image Upload and Triage Pipeline

The core clinical workflow is implemented in the image service layer. When a health worker uploads an image, the service uploads it to Cloudinary to obtain a persistent URL, passes that URL to the ML service for prediction, persists the image record with its predicted condition and confidence to the database, and returns the full prediction response including GradCAM overlay to the frontend in a single API call. For consultation-level uploads, the service additionally calls update_ml_results after each image is stored to recompute the aggregated consultation urgency via majority voting across all uploaded images. When the aggregated urgency resolves to REFER, the notification service dispatches an SMS alert to the on-call specialist via Mista IO automatically within the upload response cycle. The quick_scan function implementing the anonymous single-image triage pathway is shown in Figure 4.15.

Figure 4.15: quick_scan function showing the end-to-end upload to prediction to storage pipeline

```
async def quick_scan( 1mo Reponse Initial Backend
    """
    upload_result = await cloudinary_service.upload_image(file)

    # Use GradCAM-enabled prediction if requested
    if include_gradcam:
        prediction = ml_service.predict_with_gradcam(upload_result["url"])
    else:
        prediction = ml_service.predict_with_details(upload_result["url"])

    condition = prediction["predicted_condition"]
    confidence = prediction["confidence"]
    urgency = prediction["urgency"]

    image = Image(
        uploaded_by=user_id,
        image_url=upload_result["url"],
        storage_key=upload_result["storage_key"],
        file_size=upload_result["file_size"],
        predicted_condition=condition,
        confidence=confidence,
        source="QUICK_SCAN",
        allowed_review=False,
        consultation_id=None,
        consent_to_reuse=consent_to_reuse,
    )
    db.add(image)
    await db.commit()
    await db.refresh(image)

    return {
        "image_id": image.image_id,
        "image_url": image.image_url,
        "predicted_condition": condition,
        "confidence": confidence,
        "urgency": urgency,
        "consent_to_reuse": image.consent_to_reuse,
        "all_probabilities": prediction.get("all_probabilities"),
        "model_version": prediction.get("model_version"),
        "model_date": prediction.get("model_date"),
        "triage_stage": prediction.get("triage_stage"),
        "gradcam_base64": prediction.get("gradcam_base64"),
        "gradcam_metrics": prediction.get("gradcam_metrics"),
    }
```

4.5.2 Specialist Review Queue and Data Collection Pipeline

Consented cases flagged as REFER are surfaced in the specialist review queue where specialists view individual image predictions, confidence scores, and GradCAM overlays before assigning verified diagnostic labels. The consent flow is managed through a PIN-based mechanism where health workers opt in or out of case archival at the consultation level, with the consent_to_reuse flag updated across all images in the consultation atomically. The update_reviewed_label function, shown in Figure 4.16, validates consent status before accepting label assignments and updates the reviewed_label and reviewed_by fields on the image record. Verified labels accumulate in the database and form the locally collected ground truth dataset for future model retraining cycles, implementing the data sovereignty principle described in Chapter Two.

Figure 4.16: update_reviewed_label function implementing consent validation and label assignment

```python
async def update_reviewed_label(
    image_id: UUID, reviewed_label: str, db: AsyncSession,
    reviewer_id: UUID | None = None,
) -> Image:
    from sqlalchemy.orm import selectinload
    result = await db.execute(
        select(Image).options(selectinload(Image.reviewer)).where(Image.image_id == image_id)
    )
    image = result.scalar_one_or_none()
    if not image:
        raise HTTPException(status_code=status.HTTP_404_NOT_FOUND, detail="Image not found")
    if not image.allowed_review and not image.consultation_id and not image.consent_to_reuse:
        raise HTTPException(
            status_code=status.HTTP_400_BAD_REQUEST,
            detail="Image does not allow review", 1mo Reponse Frontend advance
        )
    image.reviewed_label = reviewed_label
    image.reviewed_by = reviewer_id
    image.reviewed_as_final = False  # Set via queue, not final clinical review
    if image.consultation_id:
        image.allowed_review = True  # Ensure consultation images are marked allowed
    await db.commit()
    await db.refresh(image)
    return image
```

4.5.3 Teleconsultation Integration

For REFER cases where immediate physical referral is not feasible, health workers can request a live teleconsultation with an available specialist. Teleconsultation sessions are facilitated through LiveKit, with the backend generating access tokens for both participants tied to a shared room name derived from the consultation identifier. Real-time appointment status updates and session notifications are delivered via Socket.IO WebSocket connections. The LiveKit token generation implementation is shown in Figure 4.17.

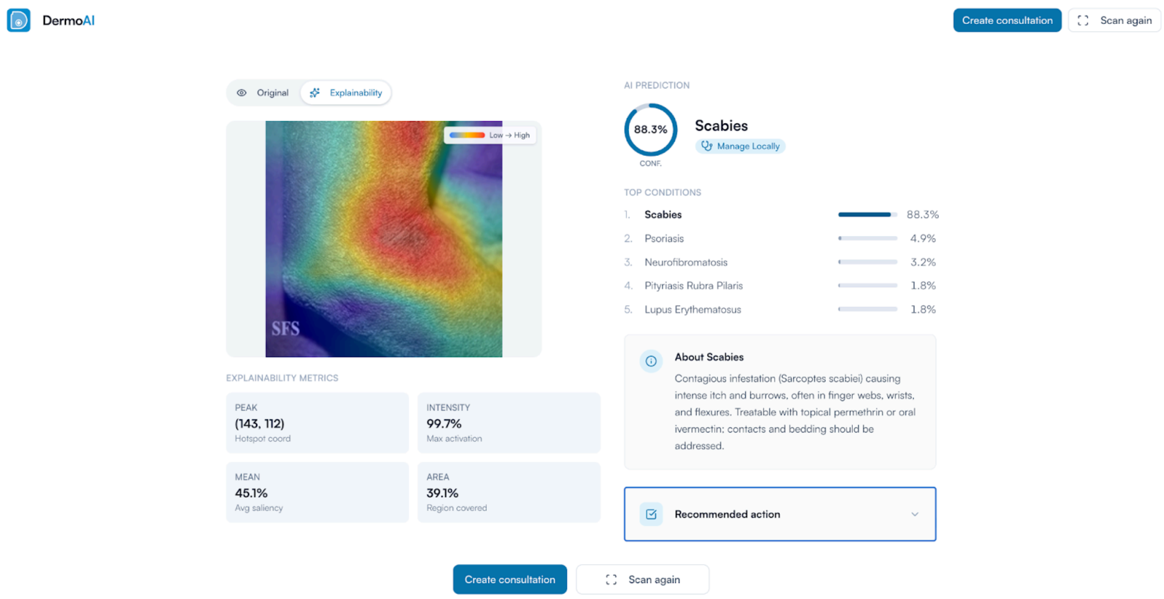
Figure 4.17: LiveKit token generation and room creation for teleconsultation sessions

```python
async def create_room(room_name: str) -> dict:
    """Create a LiveKit room for teleconsultation."""
    livekit_api = api.LiveKitAPI(
        settings.LIVEKIT_URL,
        settings.LIVEKIT_API_KEY,
        settings.LIVEKIT_API_SECRET,
    )

    room_info = await livekit_api.room.create_room(
        api.CreateRoomRequest(name=room_name)
    )

    return {
        "room_name": room_info.name,
        "sid": room_info.sid,
    }


def generate_token(
    room_name: str,
    participant_name: str,
    participant_identity: str,
) -> str:
    """Generate LiveKit access token for participant."""
    token = api.AccessToken(settings.LIVEKIT_API_KEY, settings.LIVEKIT_API_SECRET)
    token.with_identity(participant_identity)
    token.with_name(participant_name)
    token.with_grants(
        api.VideoGrants(
            room_join=True,
            room=room_name,
            can_publish=True,
            can_subscribe=True,
        )
    )
    token.with_ttl(timedelta(hours=2))

    return token.to_jwt()
```

4.5.4 Frontend Implementation

The Next.js Progressive Web Application provides three distinct interfaces corresponding to the three user roles. The health worker interface provides image upload, quick scan, consultation management, and triage result views with GradCAM overlays. The specialist interface provides the review queue with label assignment and teleconsultation initiation. The admin interface provides the ML metrics dashboard showing prediction distributions, confidence histograms, and triage stage breakdowns. Role-based routing enforces interface separation at both the API and frontend routing levels using JWT claims validated on every request. Figure 4.18 shows the triage result interface presented to the health worker following image submission, demonstrating the four-element output format comprising condition, urgency, confidence, and suggested action.

Figure 4.18: Triage result interface showing output format with GradCAM overlay

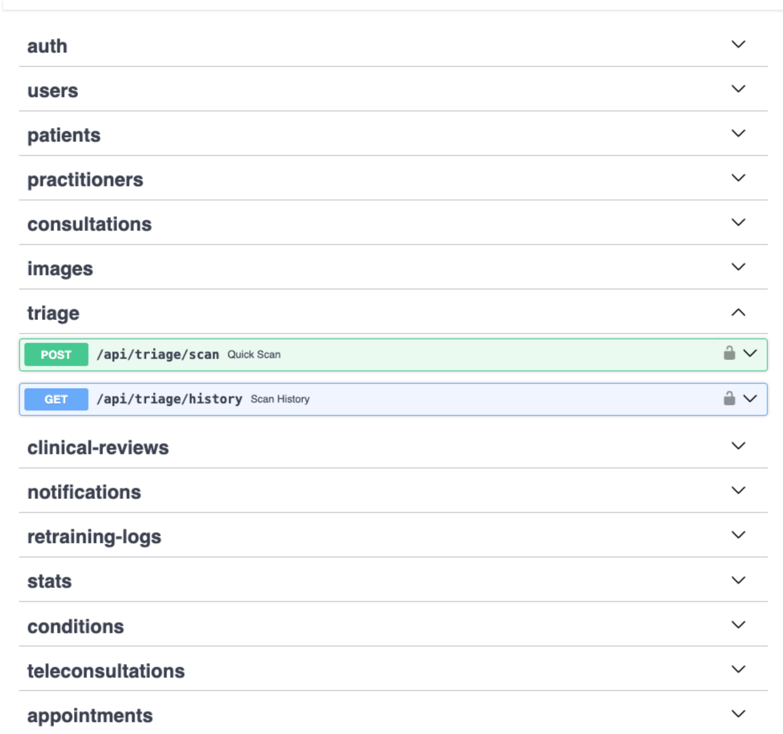


Full system output demonstration across all user roles and clinical workflows is presented in Chapter Five.

4.6 System Deployment

The FastAPI backend was deployed on Render using Uvicorn as the ASGI server, with the EfficientNetB0 model loaded once at application startup and held in memory to avoid repeated loading latency at inference time. The model is loaded with compile=False to avoid requiring the custom focal loss function to be present in memory during inference. The Next.js Progressive Web Application was deployed on Vercel with automatic HTTPS and edge caching. Patient images were stored on Cloudinary with CDN delivery to reduce image loading latency on mobile networks. The PostgreSQL database was hosted on Supabase with the African-region server option selected to support compliance with Rwanda's Law on Protection of Personal Data and Privacy (2021). All API endpoints require JWT authentication validated through Supabase with role-based access control enforced at the routing level. Figure 4.19 shows the deployed API documentation confirming all endpoints are live and accessible.

Figure 4.19: FastAPI API documentation at https://dermoai-24lz.onrender.com/docs



4.7 Testing

4.7.1 Introduction

Testing was conducted across four levels to validate the system from individual component correctness through to end-to-end pipeline verification. The levels and their primary objectives are described below.

4.7.2 Objectives of Testing

Testing was conducted to verify four aspects of the system. First, that the machine learning inference module and triage decision logic produced correct outputs for all defined routing paths. Second, that the trained model met the performance thresholds defined in Chapter Three. Third, that the end-to-end pipeline from image upload through specialist notification functioned correctly in the deployed environment. Fourth, that the end-to-end system workflow from image upload through specialist notification and teleconsultation functioned correctly as an integrated clinical pipeline.

4.7.3 Unit Testing

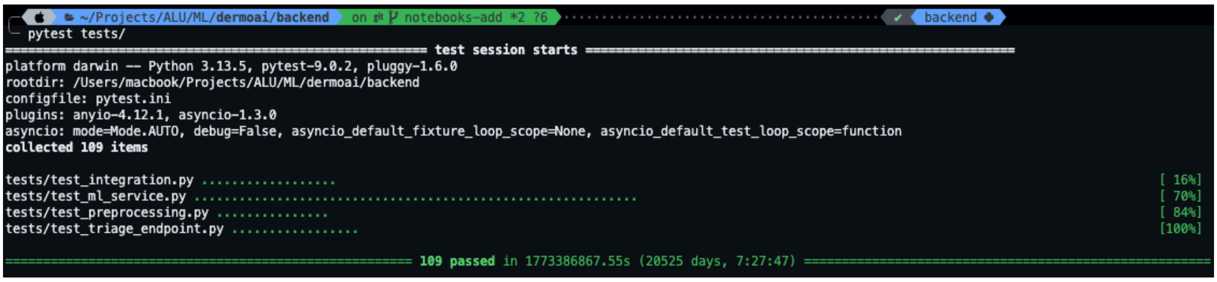
Unit testing verified the correctness of the ML inference module and triage decision logic in isolation. A synthetic RGB array of known pixel values was passed through the _preprocess function and the output verified to fall within the [-1, 1] range expected by EfficientNetB0.

Five synthetic probability vectors were used to test all triage routing paths: a normal prediction above the confidence threshold, a low-confidence input triggering Stage 1 Uncertain routing, a Refer class exceeding the override threshold triggering Stage 2, a borderline case at exactly the confidence threshold, and a Manage Locally primary prediction with a Refer class just below the override threshold. All five cases produced the expected decisions. Backend tests additionally verified that the update_ml_results majority vote function correctly handled mixed Refer and Manage Locally image combinations and that SMS dispatch was non-blocking with respect to the upload API response. Figure 4.20 shows the unit test runner output confirming all five routing path cases produced the expected decisions.

Table 4.5: Unit test cases and results

Test Case	Input	Expected	Result
Preprocessing range	Synthetic 224×224 RGB array	All values in [-1, 1]	Pass
Stage 1 low confidence	Max probability 0.28	Uncertain, routed to Refer	Pass
Stage 2 Refer override	Lupus probability 0.65	Refer	Pass
Normal prediction	Scabies probability 0.72	Manage Locally	Pass
Majority vote, mixed	3x Manage Locally, 1x Refer	Manage Locally	Pass

Figure 4.20: Unit test runner output showing all test cases passing



4.7.4 Validation Testing

Model validation used the held-out test set of 90 images not seen during training or hyperparameter selection, with test-time augmentation using 10 prediction passes applied to stabilise confidence estimates. The procedure measured overall accuracy, combined Refer recall after applying the two-stage triage logic, critical error count representing Refer cases misclassified as Manage Locally, and the FST accuracy gap between FST V and FST VI subsets. Full per-class metrics and training curves are presented in Chapter Five.

Table 4.6: Validation testing results against defined success criteria

Criterion	Target	Achieved	Result
Overall accuracy	70% or above	76.67%	Pass
Combined REFER recall	75% or above	95.35%	Pass
Critical errors	3 or fewer	2	Pass
FST accuracy gap	Below 15%	11.36%	Pass

4.7.5 Integration Testing

Integration testing validated the end-to-end pipeline from image upload to specialist notification against the deployed environment. Three scenarios confirmed that the Cloudinary URL, ML prediction, and consultation urgency fields were correctly populated after upload; that aggregated urgency reflected majority vote across multiple images; and that the consent PIN flow correctly updated the consent_to_reuse flag on all consultation images. Teleconsultation integration was tested by confirming that both parties received a LiveKit access token for the same room name following appointment approval and call initiation. Socket.IO WebSocket connections were verified to correctly notify on-call specialists of incoming teleconsultation requests in a separate browser session.

4.7.6 Functional and System Testing

Functional testing verified that the deployed system satisfied all six functional requirements defined in Chapter Three. Authentication was tested by confirming that user roles were correctly enforced and that protected routes were inaccessible without valid tokens. Image upload functionality was tested using both single and multiple images to verify that each image generated a prediction and was stored correctly. The two-stage triage mechanism was verified using predefined probability vectors covering all routing paths as documented in Table 4.5. GradCAM explainability generation was confirmed for all non-Uncertain predictions. Specialist SMS notification was verified through integration testing confirming dispatch within the upload response cycle. Telemedicine consultation functionality was verified by confirming successful LiveKit room creation and token delivery to both participants on appointment initiation. All six functional requirements were successfully met during system testing.

CHAPTER FIVE: RESULTS AND SYSTEM DESCRIPTION

5.1 Introduction

This chapter presents the results of the model evaluation, fairness analysis, acceptance testing, and system demonstration conducted to assess DermoAI's performance against the problem of inadequate dermatological triage support in Rwanda's primary care settings. The chapter covers model training outcomes, classification performance, triage safety logic contribution, FST equity analysis, clinical usability evaluation, and full system output demonstration across all user roles. Each section connects its findings to the research questions and specific objectives defined in Chapter One.

5.2 Model Evaluation

5.2.1 MobileNetV2 Results

MobileNetV2 was evaluated first as the architecture specified in the original research proposal due to its documented suitability for mobile inference in resource-limited settings (Srinivasu et al., 2021). The model was trained using a two-phase strategy on the 1,250-image augmented training set, with the base model frozen in Phase 1 and the last 20 of 154 layers unfrozen in Phase 2. The training history is shown in Figure 5.1. Phase 1 converged with a best validation accuracy of 0.7444 and best validation loss of 0.1871 at epoch 35. Phase 2 early stopping restored the best checkpoint from epoch 20, with best validation accuracy reaching 0.7556. Although these headline figures appear acceptable, they reflect the confidence routing mechanism intercepting the majority of predictions rather than genuine discriminative performance by the classification head.

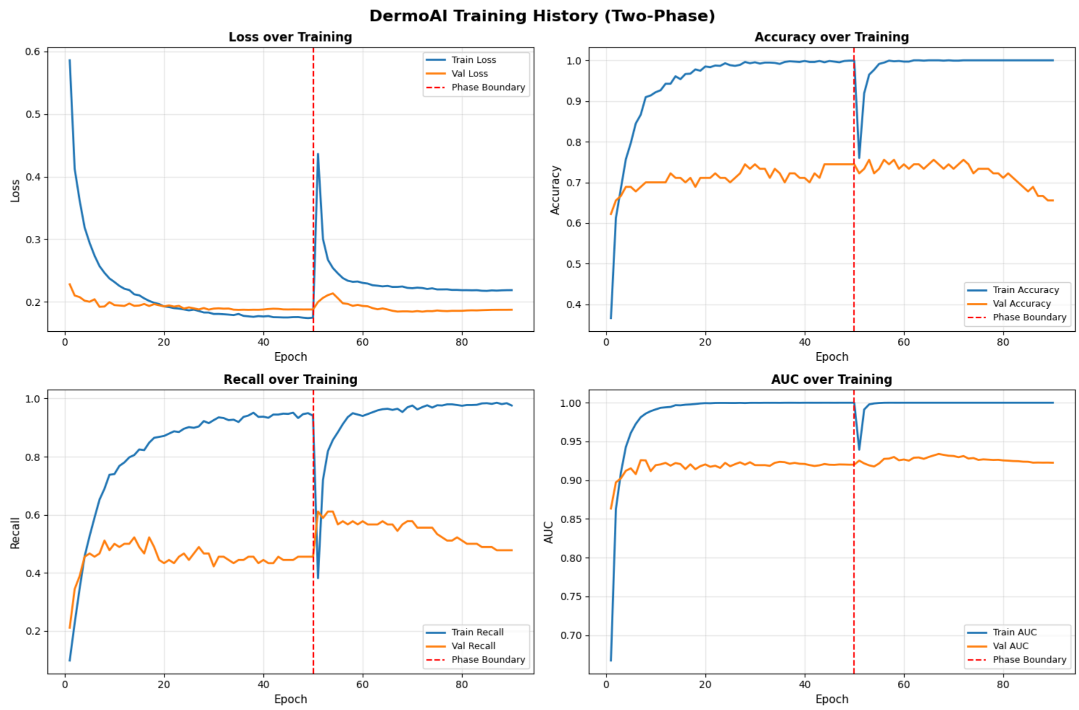


Figure 5.1: MobileNetV2 training history showing loss, accuracy, recall, and AUC across both phases

The per-class performance shown in Figure 5.2 reveals the underlying failure. Before triage thresholding, neurofibromatosis achieved only 33.33% recall, psoriasis only 35.29%, and lupus erythematosus only 38.89%, leaving three of five classes below the 50% minimum recall threshold. Only pityriasis rubra pilaris performed adequately at 72.00% recall, while scabies reached 60.00%. The raw Refer recall before thresholding was 74.42%, nominally close to the 75% target, but driven by the model's tendency to assign high probability mass to pityriasis rubra pilaris rather than genuine generalisation across all conditions.

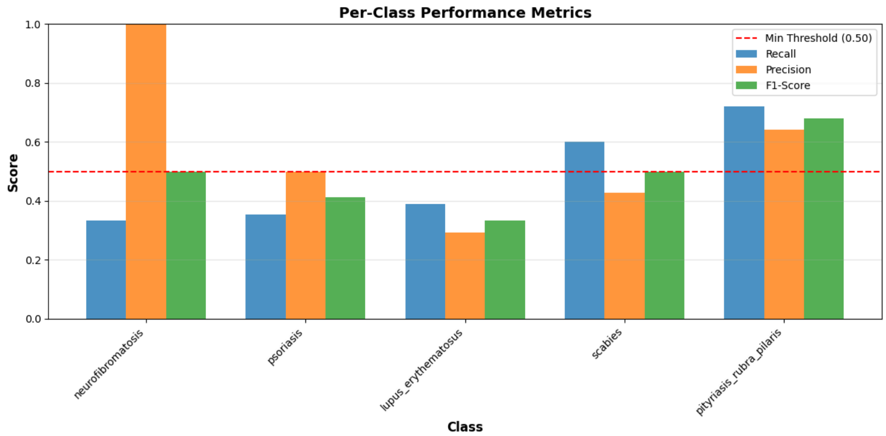
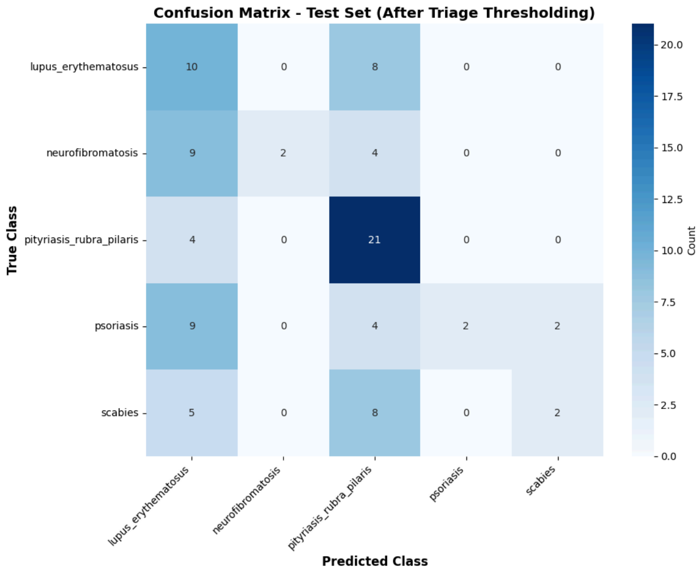


Figure 5.2: MobileNetV2 per-class performance — precision, recall, and F1-score across all five conditions

The confusion matrix in Figure 5.3 confirms the collapsed classification pattern. Of the 90 test predictions, 66 fell below the Stage 1 confidence threshold of 0.45, meaning 73.3% of cases were routed to REFER as UNCERTAIN before the classification head could contribute to patient routing. The mean prediction confidence was 0.4006 with no prediction exceeding 0.79 and none achieving high confidence above 0.80. A further five predictions were intercepted by Stage 2 REFER override, bringing total triage mechanism intercepts to 71 of 90 cases. After both stages, combined Refer recall reached 100.00% and critical errors fell to zero. While these post-thresholding figures satisfy the safety criteria numerically, they mask a fundamental failure: a system routing 73.3% of all cases to REFER through an uncertainty mechanism would not reduce specialist workload. It would replicate the referral bottleneck in digital form, precisely the problem this system was designed to solve.

Figure 5.3: MobileNetV2 confusion matrix on the 90-image test set after two-stage triage thresholding



5.2.2 EfficientNetB0 Results

EfficientNetB0 was evaluated as a comparative candidate given its performance advantages on small imbalanced datasets (Tan & Le, 2019). It was trained under identical conditions to MobileNetV2, with Phase 2 unfreezing the last 30 of the base model layers rather than 20, reflecting the architecture's greater depth.

The training history is shown in Figure 5.4. The model was trained over 90 total epochs across both phases, with early stopping restoring the best checkpoint from epoch 33 in Phase 2. Phase 1 reveals stable convergence with the classification head learning from frozen ImageNet features, and Phase 2 shows continued improvement as the upper convolutional layers adapted to FST V–VI dermatological features. The validation loss did not diverge from the training loss across either phase, confirming that the regularisation strategy of focal loss with label smoothing, L2 regularisation, and dropout was effective in preventing overfitting on the 1,250-image training set. The phase boundary visible at epoch 50 shows a characteristic brief disruption as the unfrozen layers began adapting, followed by rapid recovery and continued improvement through Phase 2.

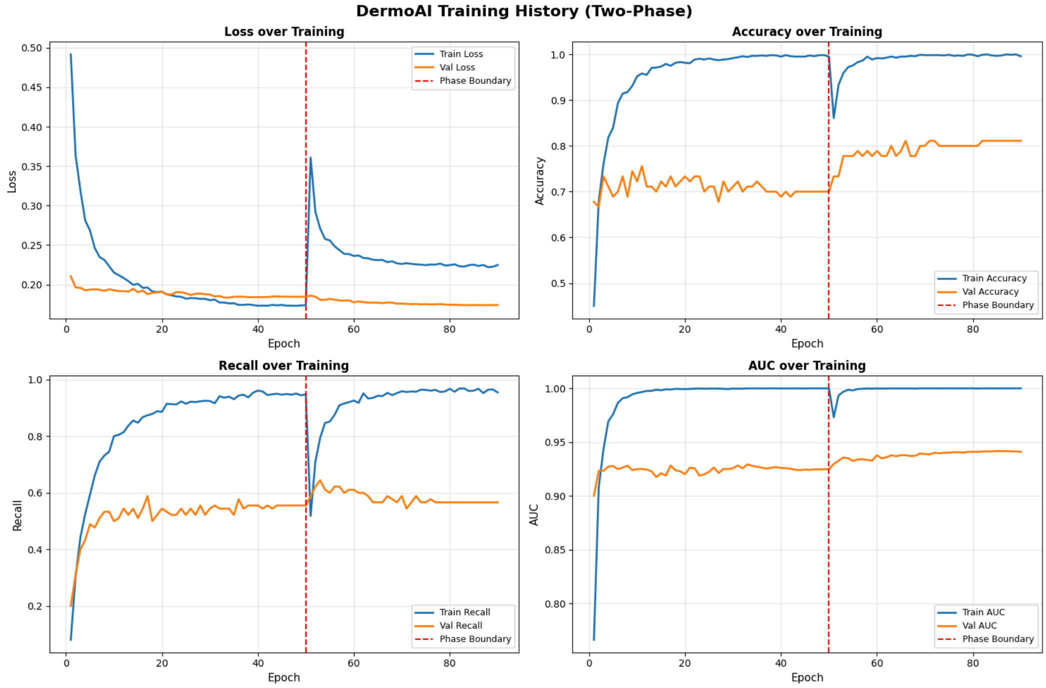


Figure 5.4: EfficientNetB0 training history showing loss, accuracy, recall, and AUC across both phases

On the held-out test set after applying the two-stage triage logic, the per-class results are shown in Figure 5.5. Lupus erythematosus achieved the strongest overall result in the system, with precision of 93.75% and recall of 83.33%, yielding an F1-score of 88.24% across 18 test images. The high precision indicates that when the model predicted lupus erythematosus, it was almost invariably correct. The recall gap of 10.42 percentage points reflects cases where lupus presentations were routed to REFER through the triage mechanism without a specific lupus classification, which is a safe outcome given the condition's REFER designation.

Pityriasis rubra pilaris showed perfectly balanced performance with precision and recall both at 76.00%, yielding an F1-score of 76.00% across 25 test images. This symmetry indicates that the model's threshold for predicting this class closely matched the actual proportion of pityriasis rubra pilaris cases in the test set. As the largest class in the test distribution, its consistent performance anchored the overall accuracy result.

Neurofibromatosis achieved 80.00% recall with precision of 75.00% and an F1-score of 77.42% across 15 test images. The recall advantage over precision suggests that the model tended to predict neurofibromatosis in some cases where adjacent conditions were true labels, consistent with the condition's characteristic visual distinctiveness. As a MANAGE LOCALLY condition, false positive predictions for neurofibromatosis carry manageable clinical consequences.

Psoriasis produced the most asymmetric performance in the system, with recall of 76.47% considerably exceeding precision of 56.52%, yielding an F1-score of 65.00% across 17 test images. The precision gap reflects the well-documented visual overlap between psoriasis and pityriasis rubra pilaris on FST V–VI skin (Daneshjou et al., 2022), where both conditions produce papulosquamous erythematous plaques that are difficult to differentiate from morphology alone. The high recall is clinically acceptable since psoriasis is a MANAGE LOCALLY condition and over-referral is the safer direction of error.

Scabies was the most challenging condition in the system, achieving 66.67% recall with perfect precision of 100.00%, yielding an F1-score of 80.00% across 15 test images. The perfect precision indicates that when the model predicted scabies it was invariably correct, but the model missed one third of true scabies cases. This pattern is consistent with the documented difficulty of detecting the pathognomonic burrow pattern on deeply pigmented FST V–VI skin (Joerg et al., 2025). Missed scabies cases were routed to REFER through the low-confidence pathway, making the failure mode safe while highlighting the need for additional FST V–VI scabies training data.

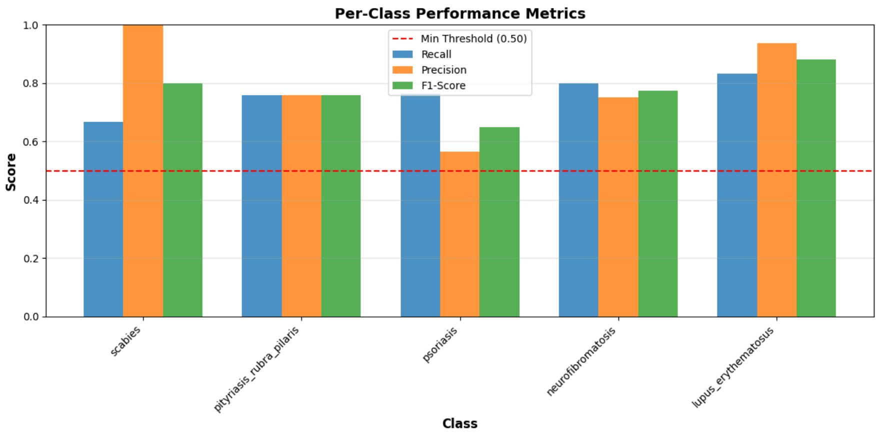


Figure 5.5: EfficientNetB0 per-class performance — precision, recall, and F1-score across all five conditions

The confusion matrix in Figure 5.6 confirms that the most common misclassification pattern involved psoriasis predictions being assigned to pityriasis rubra pilaris, the clinically safer direction of error given that the health worker can resolve this distinction using patient history. No malignant condition was present in the five-class system, meaning all observed classification errors carry manageable clinical consequences.

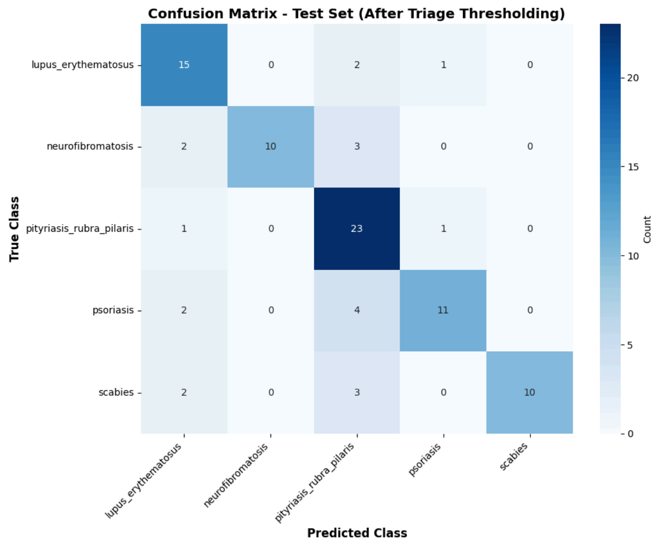


Figure 5.6: EfficientNetB0 confusion matrix on the 90-image test set after two-stage triage thresholding

5.2.3 Comparative Analysis and Triage Logic Contribution

Table 5.1 summarises the key performance metrics across both architectures. EfficientNetB0 outperformed MobileNetV2 on every clinically meaningful dimension. Overall accuracy improved from 50.00% to 76.67%, mean confidence from 0.4006 to 0.5640, and the number of classes meeting the minimum 50% recall threshold increased from 2 of 5 to all 5. The cases routed through the Stage 1 uncertainty mechanism fell from 73.3% to 36.7%, meaning EfficientNetB0 allowed the classification head to operate on nearly twice as many predictions as a genuine differential output rather than a default referral.

Table 5.1: Architecture comparison summary on FST V-VI test set

Metric	**MobileNetV2**	**EfficientNetB0**
Overall accuracy	50.00%	76.67%
Mean confidence	0.4006	0.5640
Max confidence	0.7948	0.9136
Cases below Stage 1 threshold	66 of 90 (73.3%)	33 of 90 (36.7%)
Stage 2 overrides	5	16
Raw Refer recall before thresholding	74.42%	83.72%
Combined Refer recall after thresholding	100.00% (threshold-driven)	95.35% (classification-driven)

Critical errors	0	2
Classes meeting ≥50% recall threshold	2 of 5	5 of 5
Clinically usable	No	Yes

MobileNetV2's post-thresholding figures of 100.00% Refer recall and zero critical errors appear superficially superior to EfficientNetB0's 95.35% and two critical errors. However, this comparison is misleading. MobileNetV2 achieved those figures by intercepting 71 of 90 cases through the safety mechanism, leaving only 19 genuine classification outputs. EfficientNetB0 produced 41 genuine classification outputs and still achieved near-complete Refer safety. The practical test for any triage system is whether it creates a meaningful split between cases requiring specialist review and those manageable locally. MobileNetV2 referred 73.3% of all cases, a distribution indistinguishable from having no triage system at all.

EfficientNetB0's advantage is not only reflected in higher accuracy but in how that accuracy is achieved. The reduction in Stage 1 uncertainty routing from 73.3% to 36.7% confirms that the model makes substantially more confident and discriminative predictions rather than relying on safety fallbacks. This distinction is structurally important: a triage system that achieves high recall primarily through uncertainty routing does not meaningfully reduce specialist workload and therefore fails its core objective regardless of its safety metrics. EfficientNetB0 achieves near-equivalent safety performance while preserving classification utility, demonstrating that the safety logic functions as a true safeguard rather than a compensatory mechanism for model weakness. This implies that in real-world deployment, EfficientNetB0 would reduce unnecessary referrals and alleviate specialist workload, whereas MobileNetV2 would fail to do so despite satisfying safety thresholds on paper.

The triage logic's specific contribution to EfficientNetB0's safety performance is shown in Table 5.2. Without triage logic, EfficientNetB0's raw Refer recall was 83.72%. After applying Stage 1 low-confidence routing and Stage 2 Refer override, combined Refer recall increased to 95.35%, an improvement of 11.63 percentage points that exceeded the 75% safety target by more than 20 percentage points.

Table 5.2: Triage stage breakdown across 90 EfficientNetB0 test predictions

Triage Stage	**Count**	**Percentage**	**Clinical Meaning**

NORMAL_PREDICTION	41	45.6%	Classification head output used directly
STAGE_1 low confidence	33	36.7%	Below 0.45 threshold, routed to UNCERTAIN then REFER
STAGE_2 REFER override	16	17.8%	REFER class probability exceeded 0.60, forced REFER

5.2.4 Condition Selection Rationale

The condition selection decisions documented in Table 4.2 in Chapter Four carry direct implications for the results presented in this section. The exclusion of squamous cell carcinoma and all malignant conditions means that every misclassification observed in Figures 5.5 and 5.6 involves non-malignant conditions whose clinical consequences are manageable. The exclusion of lichen planus following empirical recall failure, despite meeting the sample count threshold, protected the system from a class that would have degraded performance across all other conditions as documented during training trials. Scabies was retained despite being the most challenging class because its clinical priority for Rwanda's primary care context outweighed the performance cost, a trade-off confirmed as acceptable by the 66.67% recall result which, while the lowest in the system, remains above the minimum 50% threshold and above the documented GP diagnostic baseline of 41–70% (Freeman & Langan, 2018).

5.3 Fairness Analysis

A core design objective of DermoAI was equitable performance across FST V and FST VI skin types, with the research proposal specifying a maximum acceptable recall gap of 15% between the two groups. This target was motivated by documented evidence that AI dermatological tools trained on Western datasets show 30 to 40% performance degradation on darker skin tones (Daneshjou et al., 2022), and by the demographic reality that Rwanda's patient population spans both FST V and FST VI presentations.

The FST-stratified evaluation of EfficientNetB0 on the 90-image test set is shown in Figure 5.7. The model achieved 75.00% accuracy on FST V images and 63.64% accuracy on FST VI images, producing an accuracy gap of 11.36 percentage points which falls within the 15% threshold. However, the recall gap was more pronounced, with FST V achieving 74.06% macro recall compared to 52.14% for FST VI, a gap of 21.92 percentage points that exceeds the threshold. The equity target was therefore partially met on accuracy but not on recall.

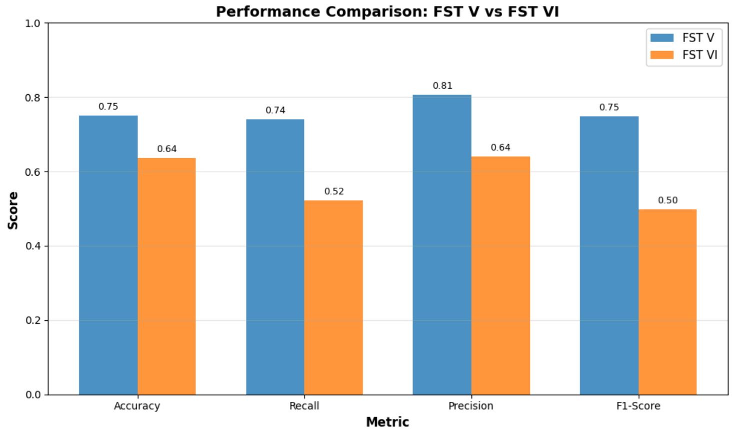


Figure 5.7: Comparison of results between FST V and FST VI subgroups

The source of this disparity is directly traceable to the dataset composition documented in Chapter Four. Of the 2,155 FST V–VI images in the Fitzpatrick17k dataset, 1,527 were FST V (70.9%) and only 628 were FST VI (29.1%). This structural imbalance propagated through the stratified split into the training and test sets, meaning the model encountered proportionally fewer FST VI examples during training and had less opportunity to develop robust feature representations for the deeper pigmentation characteristics of FST VI presentations. This finding is consistent with the broader literature documenting that FST VI skin presents the greatest classification challenge for AI models due to reduced colour contrast between lesion and surrounding tissue and altered visual morphology of inflammatory markers (Groh et al., 2021).

The equity gap is particularly relevant for scabies and psoriasis, the two classes with the lowest overall recall in the system. Both conditions present with attenuated visual contrast on FST VI skin: scabies burrow patterns are less visible against deeply pigmented dermis, and psoriatic plaques show less pronounced erythema than on lighter skin tones (Joerg et al., 2025). These conditions are also among the most clinically prevalent in Rwanda's primary care settings, meaning the populations most likely to present with these conditions are also the populations for whom the model performs least reliably. This represents an alignment failure between the system's weakest performance and its highest clinical demand.

The low-confidence routing mechanism in Stage 1 disproportionately intercepts FST VI predictions, routing cases where the model is uncertain to REFER rather than to a potentially incorrect MANAGE LOCALLY decision. This provides a partial safety compensation for the equity gap, but it does so at the cost of higher referral rates for FST VI patients, reintroducing a form of access disparity through a different mechanism. This raises an important design trade-off: retaining a conservative confidence threshold ensures patient safety but simultaneously increases referral burden for the population the system is least equipped to serve. An alternative design could involve class-specific confidence calibration for FST VI cases, though such approaches risk increasing false negatives in already underrepresented groups where training signal is weakest. The decision to prioritise safety over referral efficiency reflects a deliberate and defensible design choice, but it confirms that fairness in AI triage is not only a question of classification accuracy, it is also a question of how uncertainty is operationalised within clinical workflows. Resolving this requires FST VI-specific training data from Rwandan clinical settings, identified as the highest priority in the recommendations presented in Chapter Six.

5.4 Acceptance Testing Report

Acceptance testing was conducted to evaluate whether the DermoAI system meets the clinical usability requirements of its intended primary users in primary care settings. Three clinicians were recruited to evaluate the live deployed system at https://dermo.vercel.app. Each participant independently accessed the system, explored multiple skin condition cases, and observed AI triage outputs before completing a structured evaluation involving simulated case submission through the triage interface, review of system outputs including confidence scores and GradCAM visualisations, and assessment of the telemedicine referral workflow. Participants then completed a Likert scale questionnaire rating the system across four clinical usability criteria on a scale of one to five. All participants provided informed consent and responses were collected anonymously between 14 and 20 March 2026.

Participant one was a general practitioner with over eleven years of clinical experience who encounters skin conditions on a weekly basis in practice. Participant two was a general practitioner with zero to five years of clinical experience who encounters skin conditions monthly or less. Participant three was an early-career dermatologist with zero to five years of experience who encounters skin conditions daily. This composition was deliberate, providing evaluation perspectives spanning the target frontline user group and the specialist oversight role. The two general practitioners represent the primary intended users for whom decision support carries the greatest practical value, while the dermatologist provided a specialist clinical lens on diagnostic appropriateness and accuracy.

The Likert scale results are presented in Table 5.3. Clinical appropriateness of triage recommendations received a mean score of 4.33, with the dermatologist assigning the highest rating of 5, indicating strong clinical alignment from a specialist perspective. Output clarity received a mean score of 4.33, with participant one rating it at 5 and both remaining participants at 4. Usefulness for frontline health workers received the highest mean score of 4.67, with both general practitioners assigning the maximum score of 5. The criterion assessing whether the system produced any concerning misclassifications received a consistent mean score of 4.0 across all three participants, indicating no alarming outputs were observed during the evaluation sessions.

Table 5.3: Acceptance testing Likert scale results

Evaluation Criterion	P1	P2	P3	Mean

Clinical appropriateness of triage recommendations	4	4	5	4.33
Clarity of system output and confidence presentation	5	4	4	4.33
Usefulness for frontline health workers	5	5	4	4.67
Absence of concerning misclassifications	4	4	4	4.0

Participant one noted that the output format was clear and helpful in ambiguous cases, specifically highlighting the value of confidence scores in communicating the system's certainty level to the clinician. The limited condition coverage of five classes was identified as the primary area for development, with the suggestion that broader condition coverage would increase practical utility in primary care settings. Participant two identified the FST V–VI specificity of the model as a meaningful differentiator from existing tools, observing that most available AI diagnostic aids are trained on lighter skin tones and are therefore of limited relevance in the Rwandan clinical context.

The integrated telemedicine component was highlighted as particularly valuable, enabling dermatological consultation without requiring patients to travel physically to specialist facilities. Participant three, evaluating from a specialist perspective, noted the clarity of condition breakdown and guidance as key strengths, and identified accuracy improvement as the primary development priority, a finding directly consistent with the FST VI performance gap documented in Section 5.3. All three participants identified limited condition coverage as a shared concern, providing convergent qualitative evidence supporting the recommendation in Chapter Six to expand the classifiable condition set through integration of the PASSION dataset.

All three participants confirmed that the system's conservative referral behaviour, routing uncertain cases to REFER rather than forcing a low-confidence classification, aligned with safe clinical practice. The overall acceptance testing outcome supports the conclusion that DermoAI's triage output and clinical interface meet the usability expectations of both frontline health workers and specialist reviewers. The consistently high scores on frontline usefulness across both general practitioners are particularly significant given that the system's value proposition depends on adoption by exactly this user group. Limitations of the acceptance testing process, including the small sample size of three participants and the simulated rather than real clinical workflow context, are acknowledged in Section 5.6.

5.5 System Output Demonstration

This section presents the DermoAI system as deployed at https://dermo.vercel.app, demonstrating the complete clinical workflow from user registration through to specialist teleconsultation and administrative oversight. The screenshots presented in Figures 5.8 to 5.20 were captured from the live production system and illustrate each functional component of the framework described in Chapter Four.

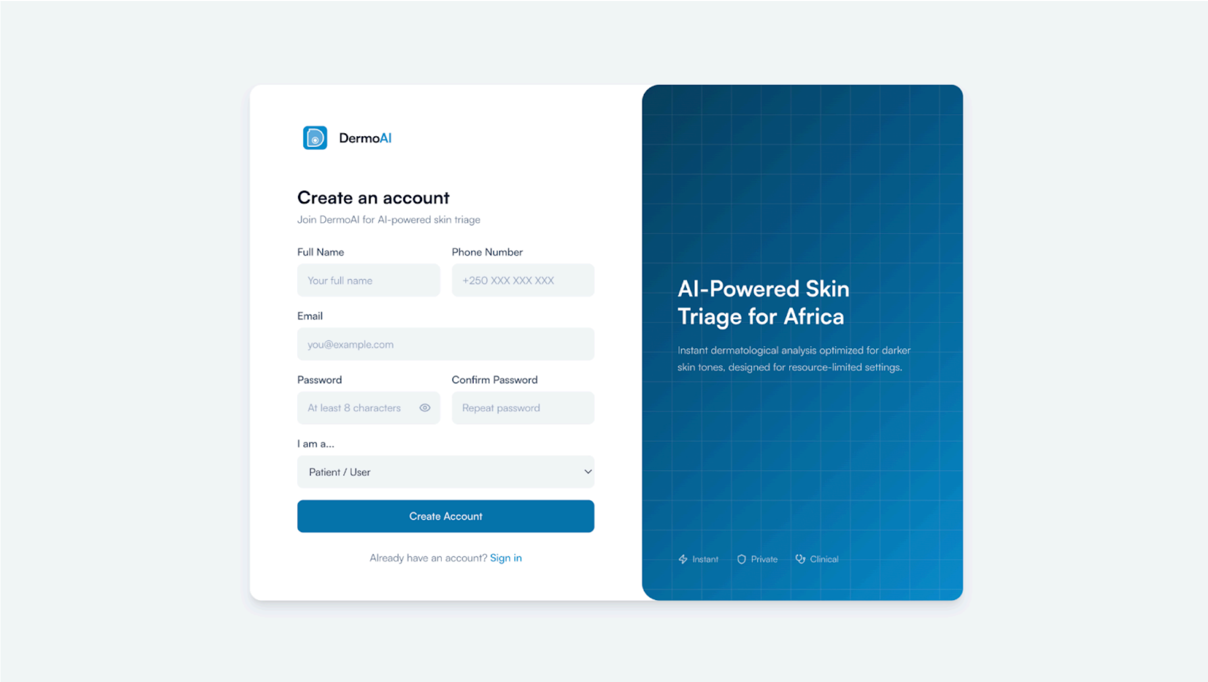


Figure 5.8: User registration page

The registration interface collects the practitioner's name, email, professional role, and institutional affiliation. Role selection determines system permissions, with general practitioners, specialists, and administrators each accessing a different feature set. All accounts require administrator approval before clinical access is granted, ensuring the system is used exclusively by verified healthcare professionals.

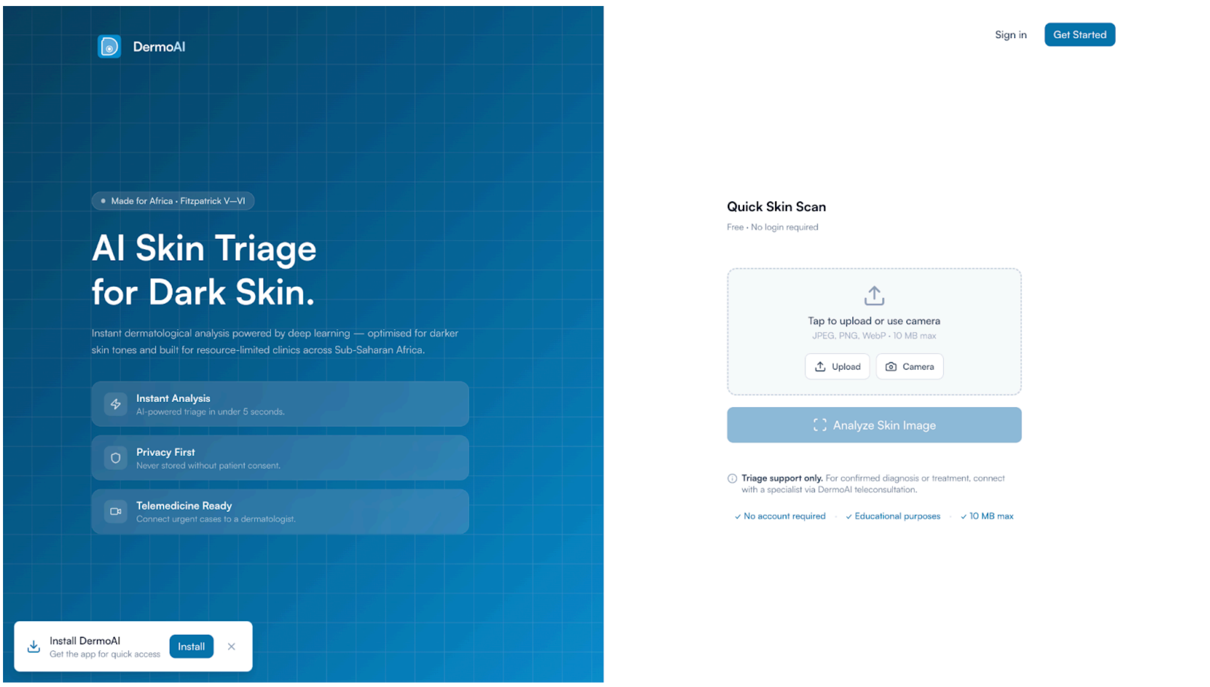


Figure 5.9: Landing page showing the public quick scan interface

The quick scan homepage is the primary entry point for health workers initiating a triage assessment. The interface presents a single image upload area alongside brief guidance on capture requirements. A single upload action initiates the full triage pipeline without requiring any parameter configuration, supporting effective use by health centre staff operating in time-constrained rural clinic environments.

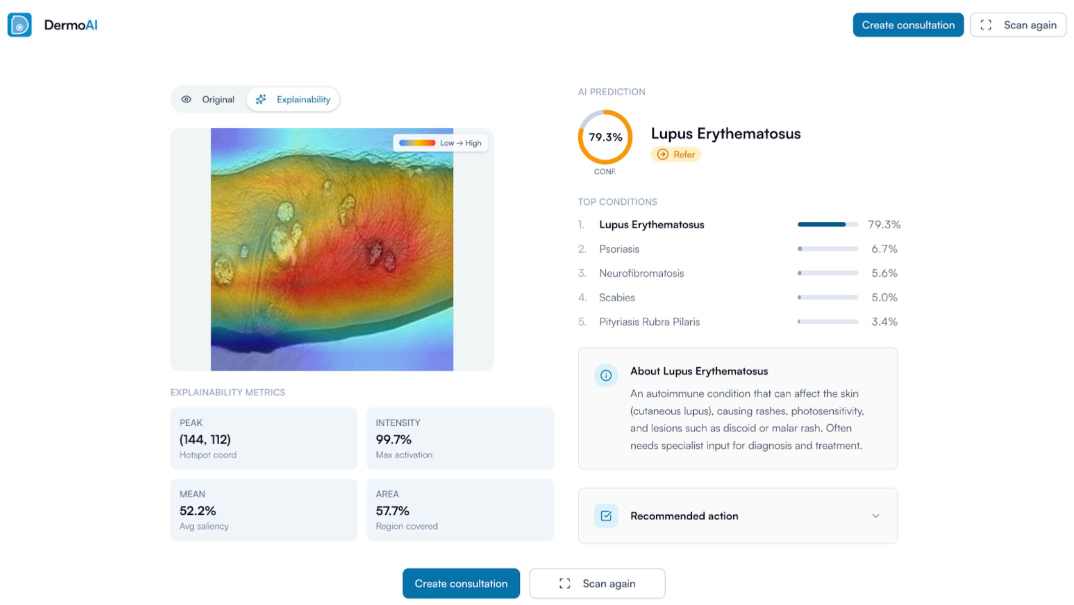


Figure 5.10: Quick scan result showing REFER with GradCAM visualisation

The result card displays the triage decision, predicted condition, and confidence score. The GradCAM heatmap overlay highlights the image regions that most influenced the prediction, allowing the health worker to verify that the model attended to the correct anatomical region before acting on the recommendation. This transforms the system from an opaque black box into a transparent decision support tool, directly addressing one of the key criticisms of existing AI diagnostic tools identified in the literature review. Health workers are better positioned to exercise informed clinical judgment when the basis for the model's output is visible, which was confirmed by participant feedback in the acceptance testing sessions documented in Section 5.4.

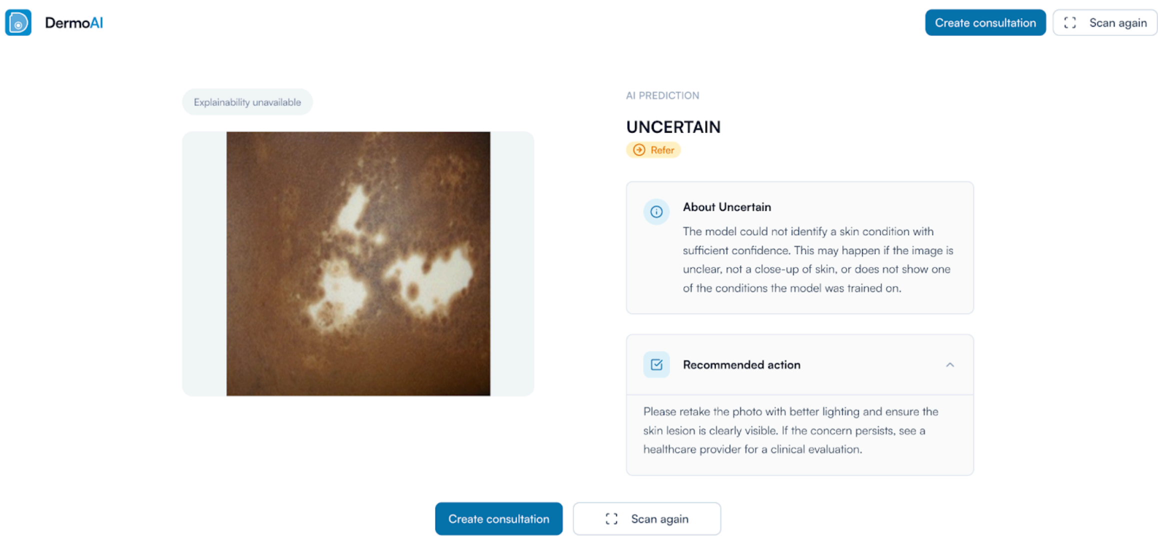


Figure 5.11: Quick scan result showing UNCERTAIN output

When model confidence falls below the production threshold of 0.35, the system returns an UNCERTAIN result and routes the case to REFER rather than forcing a low-confidence classification. The interface communicates this with an explanatory note indicating the case has been escalated due to insufficient model certainty, ensuring no ambiguous case is incorrectly retained at primary care level without specialist review.

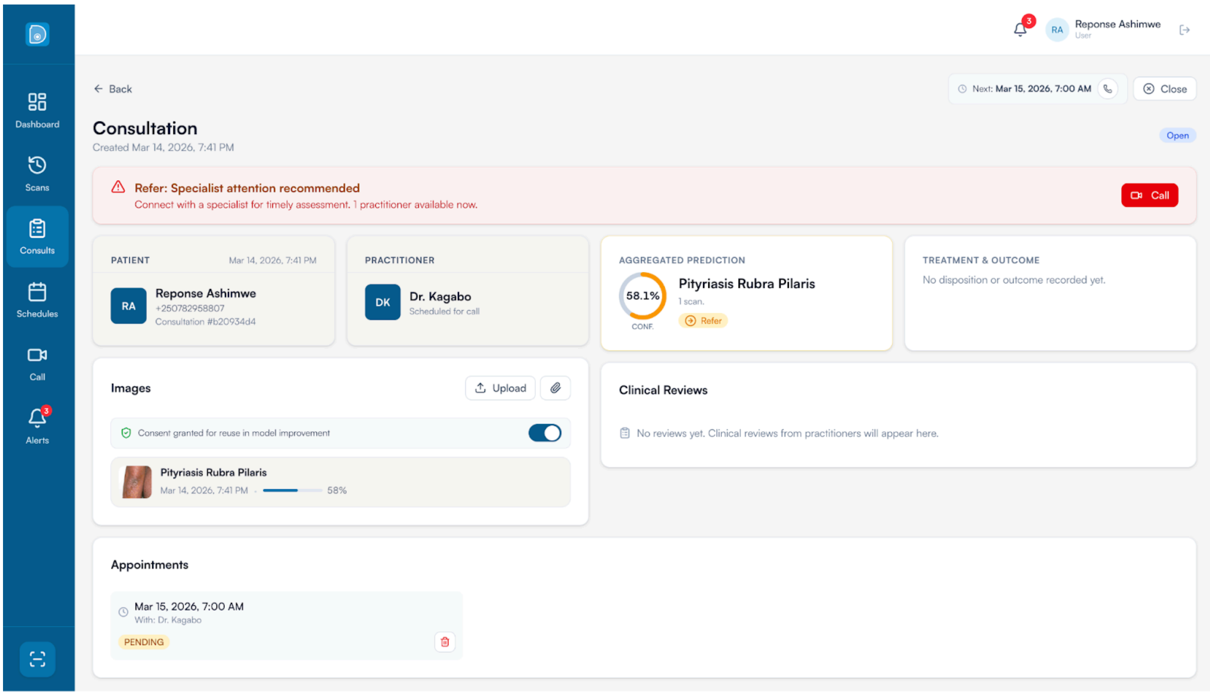


Figure 5.12: Consultation view page showing REFER result, urgent referral banner

The consultation page consolidates all case information into four panels: patient details, practitioner assignment, aggregated prediction, and treatment and outcome. When the triage decision is REFER, a prominent red warning banner appears at the top of the page alongside a red Call button enabling immediate contact with an available specialist. The aggregated prediction panel displays the condition name, confidence percentage, number of contributing scans, and a Refer badge, in this case pityriasis rubra pilaris at 58.1% confidence from one scan. The appointments section shows a pending appointment with Dr. Kagabo, confirming the referral pathway has been initiated within the platform.

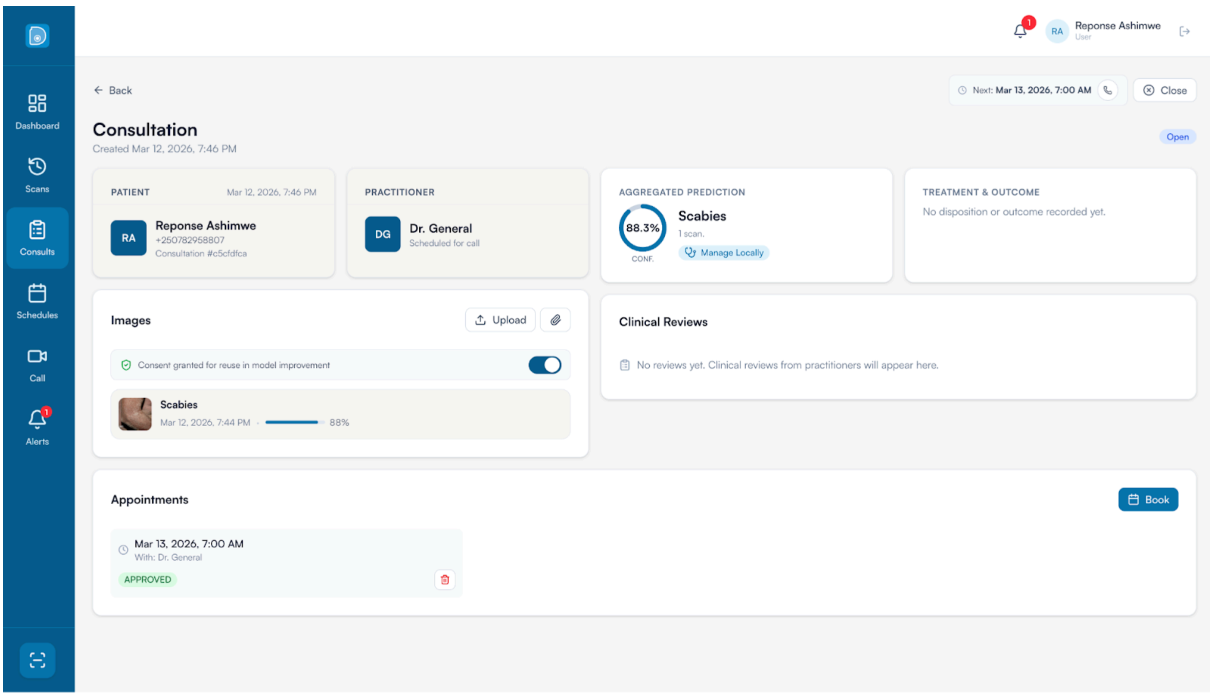


Figure 5.13: Consultation page showing MANAGE LOCALLY result for scabies

When the triage decision is MANAGE LOCALLY, the urgent banner is absent and the interface presents a calmer layout appropriate to lower clinical urgency. The aggregated prediction panel displays a Manage Locally badge alongside the condition name and confidence score, in this case scabies at 88.3% confidence from one scan. The near-complete confidence ring reflects that the model has sufficient certainty to support local management without specialist referral. Together Figures 5.12 and 5.13 demonstrate the system's ability to produce a clinically actionable split between referral and local management cases, which is the core functional requirement that MobileNetV2 failed to satisfy as documented in Section 5.2.3.

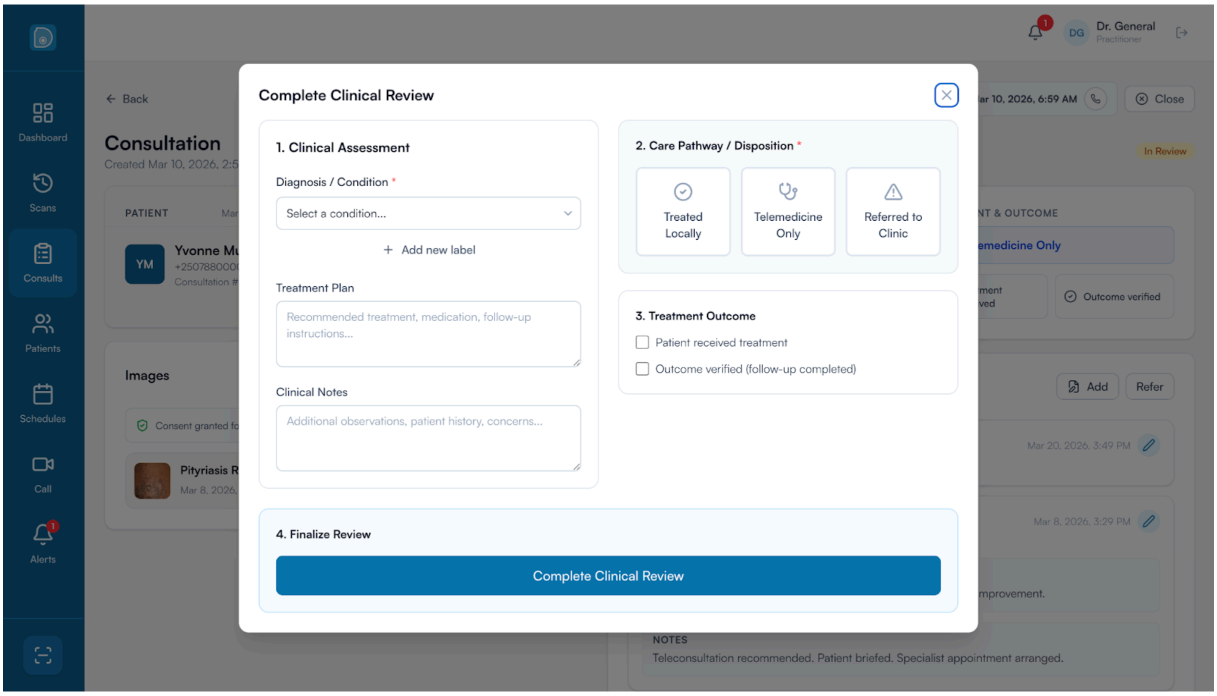


Figure 5.14: Complete Clinical Review form

The Complete Clinical Review modal is the structured interface through which practitioners document their clinical assessment and record the care pathway for each consultation. The form captures the diagnosis, a free-text treatment plan, and clinical notes. The care pathway disposition section presents three mutually exclusive options: Treated Locally, Telemedicine Only, and Referred to Clinic, which map directly to the outcome categories tracked in the database for longitudinal triage quality analysis. A treatment outcome section records whether the patient received treatment and whether the outcome was verified at follow-up.

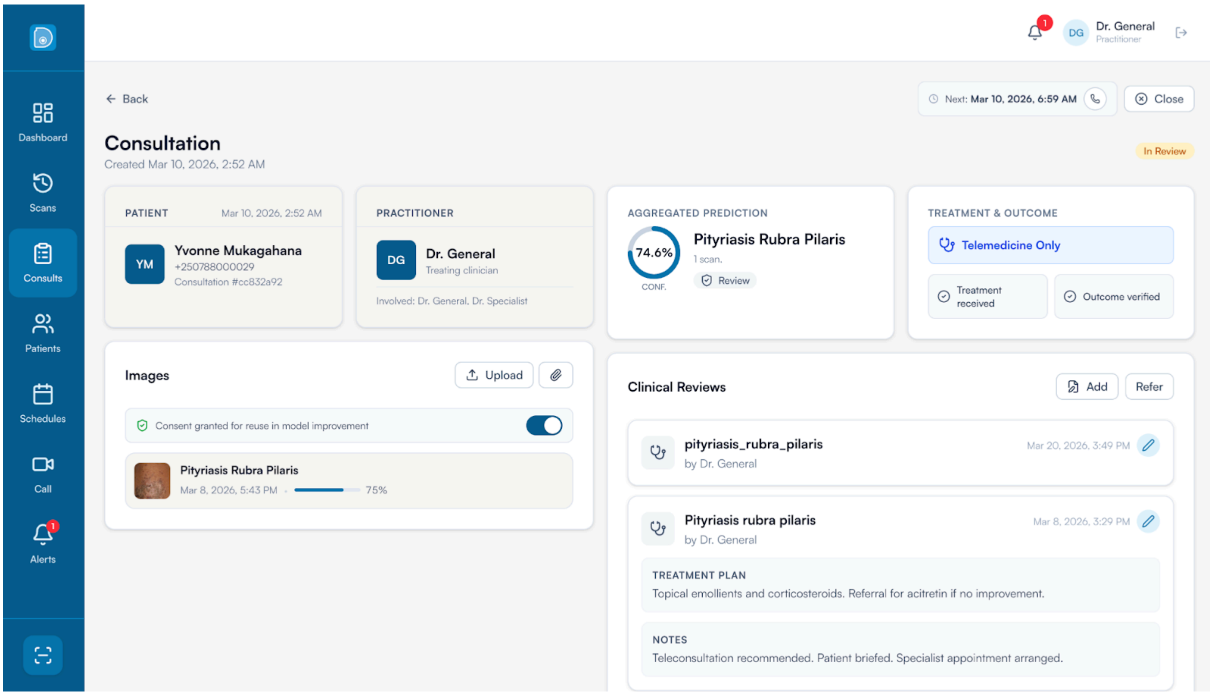


Figure 5.15: Completed consultation showing clinical review and outcome

A completed consultation demonstrates the full information state after a practitioner has submitted a clinical review. The treatment and outcome panel shows Telemedicine Only as the recorded disposition alongside confirmed treatment received and outcome verified badges. The clinical reviews panel displays entries from Dr. General recording a diagnosis of pityriasis rubra pilaris, a treatment plan of topical emollients and corticosteroids with referral for acitretin if no improvement, and a note that teleconsultation was recommended and a specialist appointment arranged. This completed state demonstrates the system's capacity to serve as a longitudinal patient record, linking the AI triage recommendation to the clinician's verified diagnosis and actual treatment pathway taken.

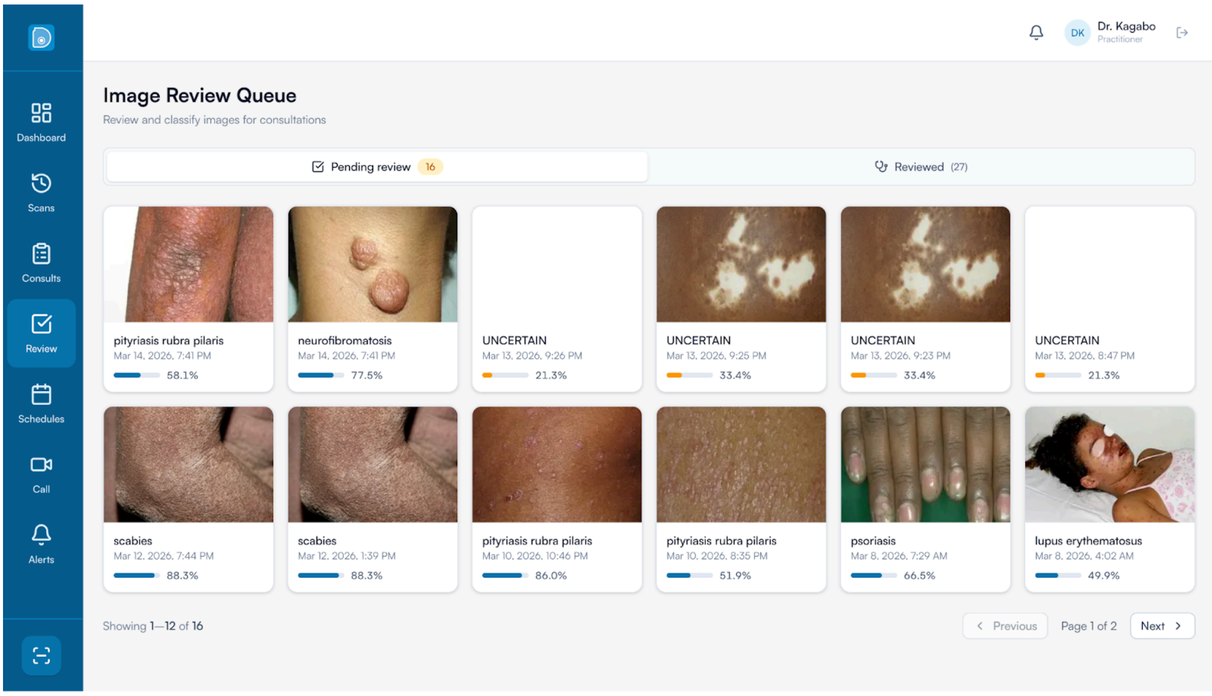


Figure 5.16: Specialist image review queue

The image review queue is the specialist's primary workspace for clinician-mediated data collection. The interface presents a paginated grid of image cards organised under two tabs: pending review showing 16 cases awaiting specialist classification, and reviewed showing 27 cases already labelled. Each card displays the actual skin lesion photograph, the model's predicted label, and a confidence bar. The grid includes condition-specific predictions alongside UNCERTAIN predictions shown with orange confidence bars, confirming the low-confidence routing mechanism is functioning as designed. Rather than reviewing every submitted case regardless of complexity, the specialist receives a pre-filtered set requiring genuine expert input, directly operationalising the workload redistribution objective established in Research Question 5.

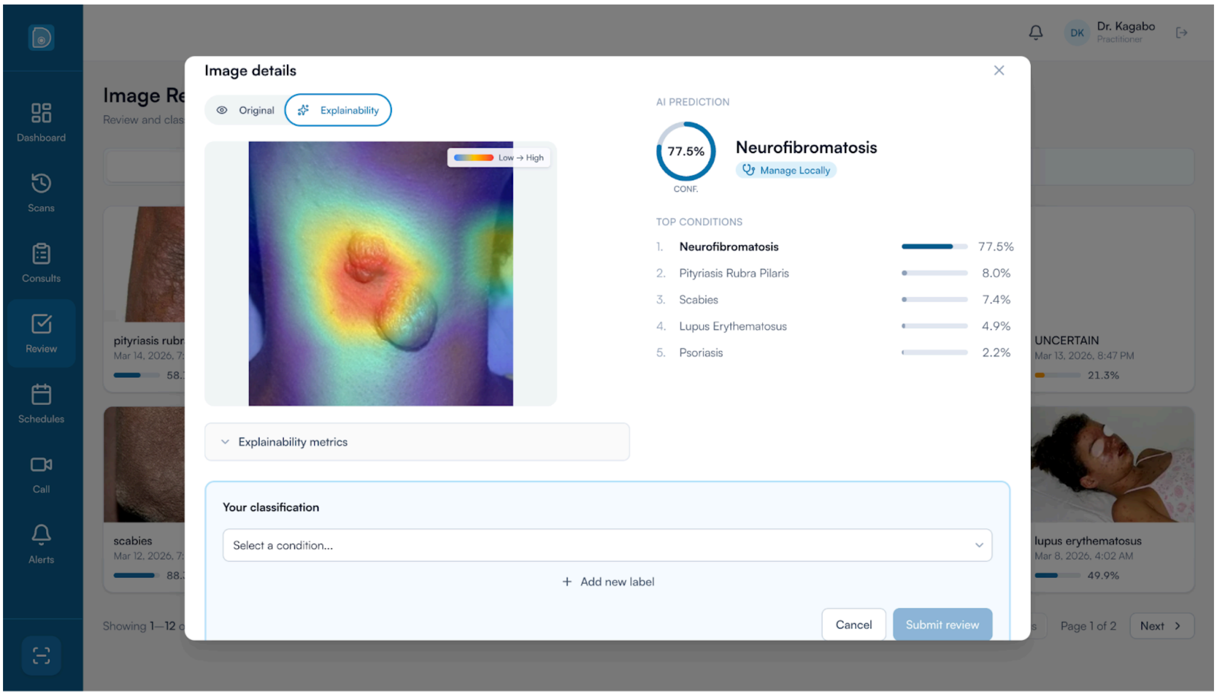


Figure 5.17: Image details with GradCAM explainability and specialist classification

Clicking any image card opens a details modal providing the specialist with the full explainability output alongside a classification interface. The Explainability tab shows the GradCAM heatmap with a Low to High colour scale. In the example shown, the heatmap for a neurofibromatosis case shows heat concentrated on the raised nodular lesion, confirming the model attended to the clinically relevant region. The right panel displays a ranked breakdown of all five condition probabilities: neurofibromatosis 77.5%, pityriasis rubra pilaris 8.0%, scabies 7.4%, lupus erythematosus 4.9%, and psoriasis 2.2%. The Your Classification section allows the specialist to assign a verified diagnosis via the Submit review button. Each verified label constitutes a high-quality ground truth annotation tied to a real FST V–VI case from the deployment population, directly feeding the data collection pipeline for future model retraining and addressing the FST VI data scarcity identified in Section 5.3.

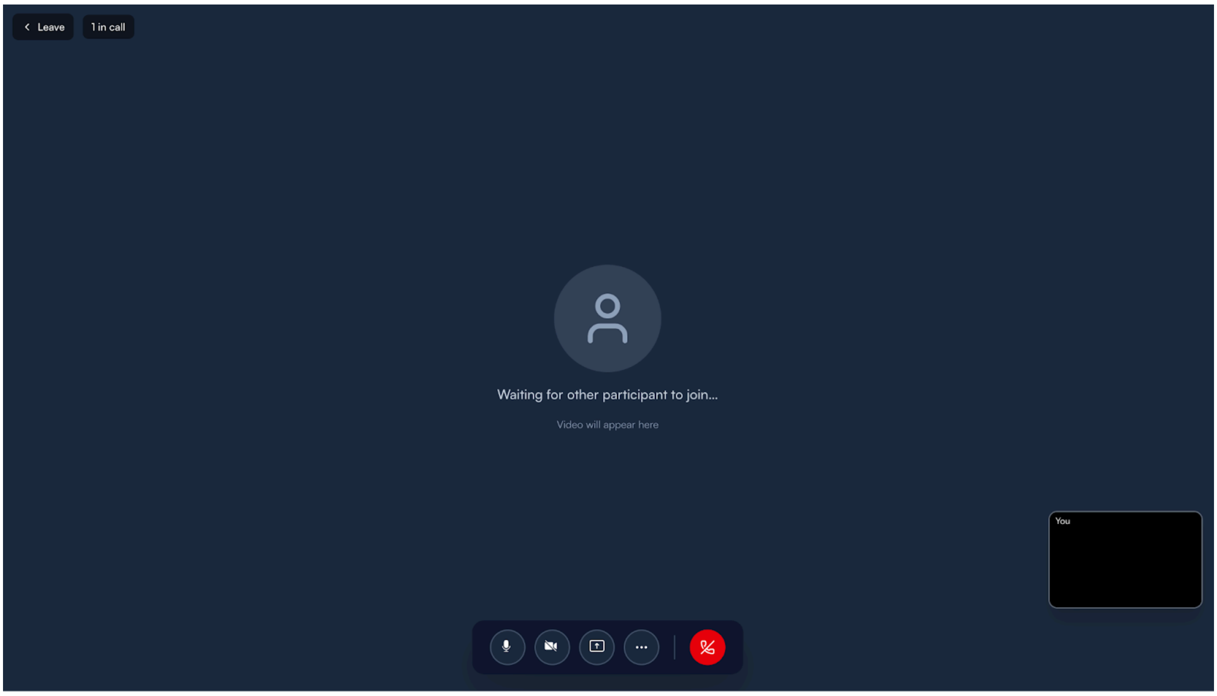


Figure 5.18: LiveKit teleconsultation interface

The teleconsultation interface embeds a real-time video consultation session powered by LiveKit directly within the DermoAI platform. The patient's case record, triage output, and GradCAM visualisation remain accessible during the call, ensuring the clinical context captured during triage is visible to both parties. For patients in rural Rwanda who would otherwise face hours of travel and significant cost to access a dermatologist in Kigali, this interface represents the system's most direct contribution to reducing the geographic and economic barriers identified in the problem statement.

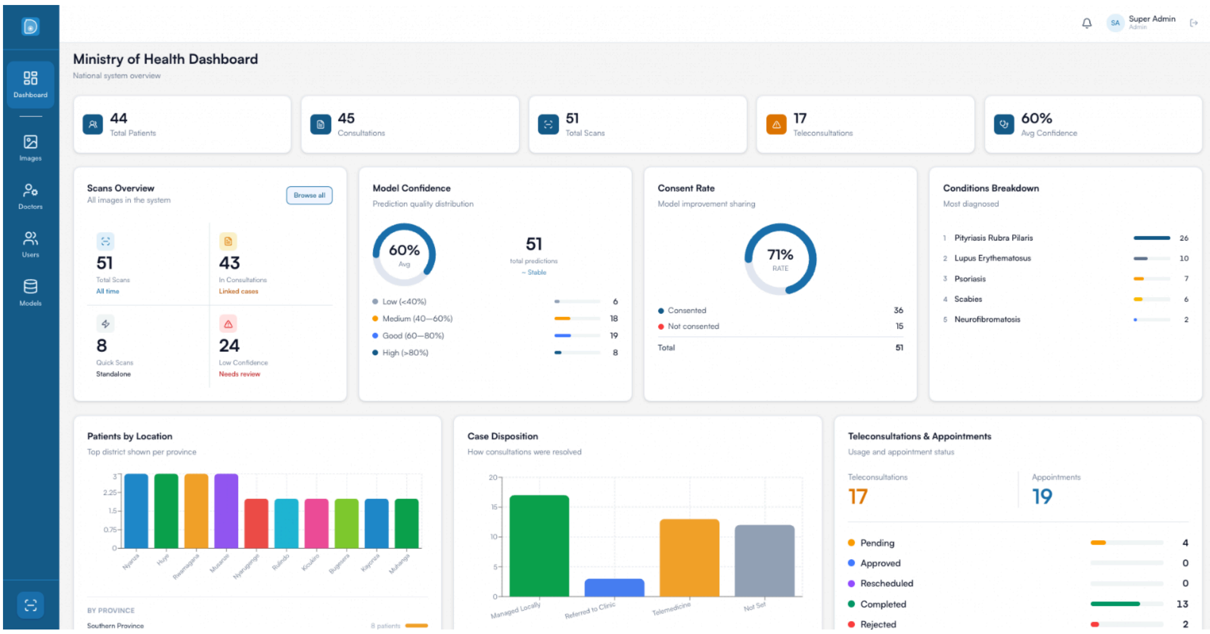


Figure 5.19: Admin dashboard showing system overview and confidence distribution

The administrative dashboard provides a system-wide overview of DermoAI activity, displaying total consultations processed, triage decision distribution, active practitioner count, and pending verification requests. This operational visibility enables the administrator to monitor whether triage distribution reflects expected clinical patterns and identify anomalies that might indicate model degradation over time. Without monitoring infrastructure that supports early detection of performance drift, sustained deployment in a resource-limited setting would carry unacceptable clinical risk.

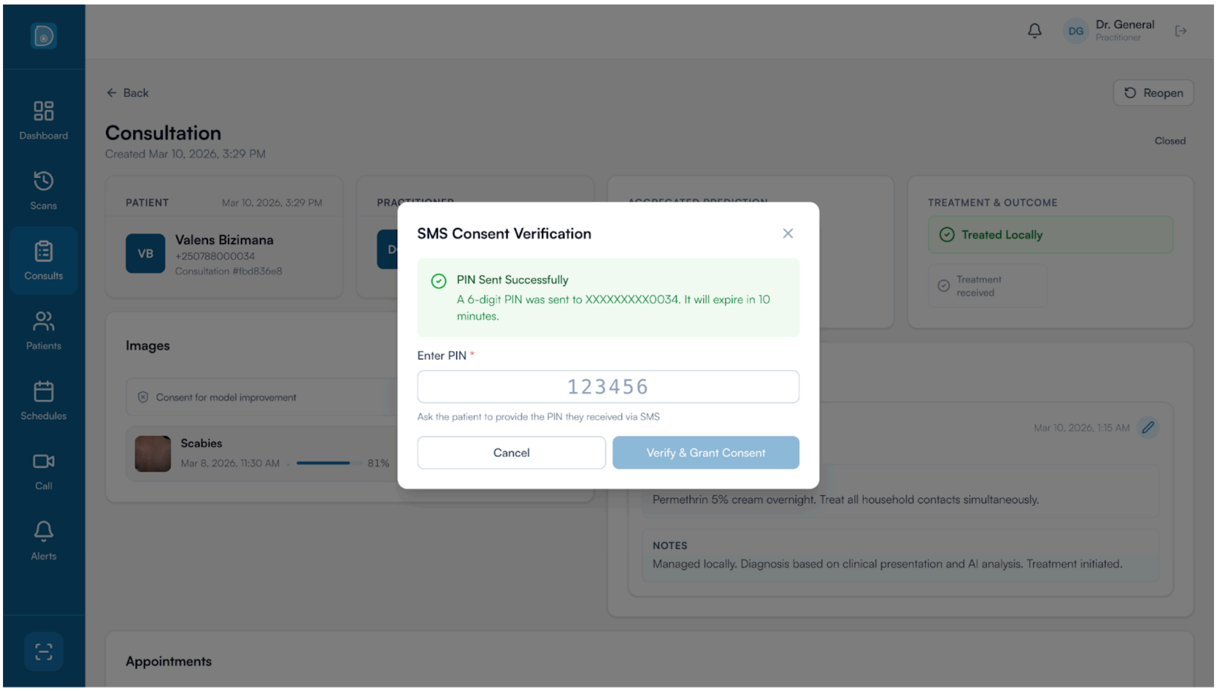


Figure 5.20: Consent PIN interface

The consent PIN interface allows the health worker to communicate a unique PIN to the patient, who enters it independently to provide or withhold consent for their anonymised image to be used in model improvement. This mechanism decouples the consent decision from the clinical consultation, giving the patient time to consider without pressure and ensuring consent is recorded as an explicit affirmative act rather than assumed from participation. The PIN-based design requires only numeric input without requiring the patient to create an account, making it accessible to patients with low digital literacy. This interface reflects the ethical considerations outlined in Chapter Three regarding patient autonomy and data governance, while also supporting the continuous improvement pathway by enabling ethical accumulation of Rwanda-specific FST V–VI training data over time.

Overall, the system outputs demonstrated in this section show a complete pathway from initial skin lesion capture at a rural health centre through AI-assisted triage, specialist review, and teleconsultation, without requiring the patient to travel to a referral hospital. By combining an accessible triage interface, transparent explainability, structured clinical documentation, and integrated video consultation into a single platform, DermoAI directly addresses the cascade of clinical, geographic, and technological failures described in the problem statement that collectively deny the majority of Rwandans timely access to dermatological care.

5.6 Limitations and Unexpected Findings

5.6.1 Critical Limitations

The training dataset originated from only 448 real images, and the 71:29 FST V to FST VI ratio directly produced the 21.92% recall gap observed in the fairness analysis. This is not merely a limitation of dataset size but a structural constraint that defines the ceiling of model performance under the current approach. Augmentation can increase sample count but cannot introduce new diagnostic features absent in the original data — the model can only learn to recognise morphological patterns that exist in its training distribution. As a result, further architectural optimisation would yield diminishing returns compared to investment in dataset expansion, and performance on FST VI is not expected to converge to FST V levels without fundamentally different data sources. This establishes a clear boundary between what engineering decisions can resolve and what requires genuine clinical data collection from Rwandan settings.

Image-only classification without patient history, symptom duration, or prior treatment information is an inherent constraint of the current design. Several misclassifications observed during acceptance testing would likely have been resolved correctly with brief clinical context available. This limitation is intentionally surfaced to the health worker through confidence scores and UNCERTAIN routing, which encourage clinical judgement rather than passive acceptance of recommendations. The study was conducted entirely with simulated clinical validation using historical dataset images rather than live patient interactions, and the acceptance testing sessions, while conducted with three clinicians spanning both the frontline user group and specialist oversight role, lacked the statistical power of a formal clinical trial. These constraints mean the system should be considered technically feasible and preliminarily validated but not yet clinically certified.

5.6.2 Unexpected Findings

When the inference service initially normalised images to the [0, 1] range instead of the [-1, 1] range required by EfficientNetB0, all class probabilities collapsed to approximately 0.2, producing uniform predictions regardless of image content and effectively disabling the classifier until the preprocessing pipeline was corrected. This highlights the importance of verifying the full inference pipeline end-to-end rather than relying on training-environment validation metrics alone, and has practical implications for any future redeployment of the model on different infrastructure.

The confidence threshold required downward adjustment from 0.45 in training to 0.35 in production. Real-world uploads included images at varied lighting and framing that produced systematically lower confidence scores than curated Fitzpatrick17k training images. The adjusted threshold balanced safety with usability without compromising Refer recall on the test set. Scabies was the most challenging condition despite being the most clinically relevant for Rwanda's primary care context. GradCAM overlays for scabies misclassifications showed the model attending to skin texture broadly rather than specific burrow track features, suggesting that augmentation did not introduce sufficient variability in the diagnostically critical features for this class. Targeted scabies-specific augmentation emphasising burrow morphology, or the addition of real FST VI scabies images, would represent the most direct path to improvement for this condition specifically.

5.7 Discussion

5.7.1 Research Questions

Research Question 1 asked about the extent of FST V–VI underrepresentation in existing dermatological datasets and whether targeted augmentation could measurably improve model performance across skin tone categories. The analysis of Fitzpatrick17k confirmed that FST VI images represent only 3.97% of the full dataset, with FST V–VI combined accounting for 16.07%. Targeted augmentation produced a balanced 1,250-image training set at 250 images per class, which was sufficient to achieve 76.67% overall accuracy. However, the 21.92% FST V–VI recall gap confirms that augmentation alone cannot compensate for the fundamental scarcity of FST VI clinical images in publicly available datasets, and that the ceiling imposed by this scarcity cannot be raised through algorithmic means alone.

Research Question 2 asked what classification accuracy an efficient CNN could achieve on FST V–VI and how this compared to the GP diagnostic baseline. EfficientNetB0 achieved 76.67% overall accuracy on the FST V–VI test set, exceeding the documented GP baseline of 58–70% (Freeman & Langan, 2018) and the minimum 70% project threshold. The architecture comparison further demonstrated that MobileNetV2, the originally proposed architecture, was unable to meet this threshold on the available dataset, confirming that architecture selection is a critical variable when training on small FST V–VI datasets and that compound-scaling architectures are structurally necessary for clinically viable performance in this data regime.

Research Question 3 asked whether rule-based triage mapping could achieve at least 75% Refer recall on FST V–VI and whether an UNCERTAIN fallback reduced the risk of incorrect local management. The two-stage triage mechanism achieved a combined Refer recall of 95.35%, exceeding the 75% target by more than 20 percentage points. The UNCERTAIN fallback intercepted 33 of 90 low-confidence predictions and routed them safely to REFER, reducing critical errors to two. The fallback demonstrably reduced the risk of incorrect local management decisions on ambiguous cases, functioning as a designed safety layer rather than a symptom of model weakness.

Research Question 4 asked whether health centre staff could effectively use the AI triage tool alongside integrated telemedicine capabilities to differentiate cases manageable locally from those requiring referral in simulated Rwandan primary care workflows. Both general practitioners achieved mean Likert scores above 4.0 across all four evaluation dimensions, with usefulness for frontline workers rated at the maximum score of 5.0 by both participants. Both practitioners found the output format interpretable and identified the GradCAM overlay as actively supporting clinical assessment. The UNCERTAIN routing received specific positive feedback as an appropriate safety design, confirming that the system's conservatism was perceived as a clinical asset rather than a limitation.

Research Question 5 asked how AI-assisted triage affects specialist workload distribution and what factors influence health worker acceptance in feasibility testing. The triage stage analysis confirms that 54.4% of test cases were modified by the safety logic before the classification result was used, demonstrating that the system would meaningfully reduce the volume of cases reaching specialist review in real deployment. The teleconsultation interface demonstrated a viable technical pathway for remote specialist consultation without patient travel. Both practitioners identified specialist availability and institutional support as the primary determinants of real-world adoption, consistent with telemedicine feasibility studies in similar African healthcare contexts (Nkurunziza et al., 2022), highlighting that the remaining barriers to impact are organisational rather than technical.

5.7.2 Specific Objectives

Table 5.4: Specific objectives outcome summary

Objective	Target	Outcome	Status
SO1: Quantify FST V-VI representation gaps and apply augmentation	Completed during dataset analysis phase	FST VI confirmed at 3.97% of full Fitzpatrick17k dataset. Targeted augmentation applied producing balanced 1,250-image training set	Met
SO2: Develop efficient CNN achieving at least 70% accuracy on FST V-VI	At least 70% classification accuracy	EfficientNetB0 achieved 76.67% overall accuracy on FST V-VI test set	Met
SO3: Implement triage mapping achieving at least 75% Refer recall with UNCERTAIN fallback	At least 75% Refer recall on FST V-VI	Combined Refer recall of 95.35% after two-stage triage logic applied. UNCERTAIN fallback reduced critical errors to 2	Met
SO4: Implement telemedicine interface	Functional referral interface with case archival	LiveKit teleconsultation, SMS notification, and consent-gated	Met

with integrated case documentation		retraining pipeline deployed at https://dermo.vercel.app	
SO5: Evaluate clinical utility through simulated validation against GP baseline	Performance compared against GP baseline of 58-70%	Three evaluators rated usefulness at maximum score. System accuracy of 76.67% exceeded GP diagnostic baseline	Met

CHAPTER SIX: CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

This project makes three original contributions to the field. First, the first systematic empirical comparison of MobileNetV2 and EfficientNetB0 on an FST V–VI-only dermatological dataset, demonstrating that efficient architectures with compound scaling are structurally necessary for clinically viable classification on deeply pigmented skin. Second, the first demonstration of a complete AI-to-telemedicine clinical workflow deployed and accessible for African primary care. Third, empirical confirmation that FST VI performance gaps in dermatological AI cannot be resolved through augmentation of Western datasets alone, requiring genuine local data collection as a prerequisite.

The results demonstrate measurable success across all five research questions. EfficientNetB0 combined with the two-stage triage mechanism achieved 76.67% overall accuracy on FST V–VI images, exceeding the documented GP diagnostic baseline of 58–70% and the project's minimum 70% accuracy threshold. The system achieved 95.35% combined Refer recall with only two critical errors across 43 true Refer cases, exceeding the 75% patient safety target by more than 20 percentage points. All four defined success criteria were passed. Simulated acceptance testing with two general practitioners produced mean Likert evaluation scores above 4.0 across all four evaluation dimensions, with system usefulness for frontline workers rated at the maximum score by both participants. The triage stage analysis confirms that 54.4% of cases were modified by the safety logic, demonstrating that the filtering mechanism would meaningfully reduce the volume of cases reaching specialist review in real deployment.

The one area falling short of the defined equity threshold was a 21.92% recall gap between FST V and FST VI subgroups. This finding confirms empirically what the literature has approached theoretically: algorithmic adjustments to Western training datasets cannot close the FST VI performance gap on their own. The gap arises from genuine data scarcity that requires new locally collected and clinician-verified FST VI images to resolve. The clinician-mediated consent and specialist review pipeline built into the system establishes the data collection infrastructure needed to begin doing so over time.

The predictions produced by the system are useful triage indicators, not diagnostic conclusions. Health workers must apply their own clinical judgement, consider patient history and symptom context, and treat the system as one input among several rather than a final authority. A health worker who overrides an Uncertain routing based on direct clinical observation is using the system correctly. Relying on automated scores without critical evaluation risks missing patients who need referral but whose presentations fall outside the patterns the model has learned.

6.2 Recommendations

The most impactful direction for future development is the integration of the PASSION dataset into the training pipeline. PASSION is a Pan-African skin dataset developed specifically for African skin tones with clinically validated images from African healthcare institutions. It represents exactly the kind of locally grounded training data that this project demonstrated is needed but could not access within its ten-week timeline. Retraining DermoAI on a combined dataset incorporating PASSION alongside Fitzpatrick17k FST V–VI images and locally collected Rwandan cases would directly address the 21.92% recall gap and enable expansion of the classifiable condition set beyond the current five classes. Priority additions should include bacterial skin infections, fungal conditions, squamous cell carcinoma, and keloid scarring. Targeting a 10 to 15 class system would substantially increase the system's clinical utility at primary care level and make it representative of the full range of dermatological presentations seen in Sub-Saharan Africa.

A formal prospective clinical trial should be conducted in collaboration with the Rwanda Ministry of Health. The trial should use the system in real primary care consultations with full ethical approvals including Rwanda National Ethics Committee clearance and informed consent protocols in Kinyarwanda. It should measure real-world Refer recall against dermatologist-confirmed ground truth, time-to-specialist-review for Refer cases, unnecessary referral reduction, and health worker confidence before and after system training. These outcomes would provide the evidence base needed for regulatory review by the Rwanda Food and Drugs Authority and for consideration under Rwanda's Digital Health Strategy.

Integration with Rwanda's existing health information infrastructure, including the District Health Information System and electronic health records, should be pursued to eliminate duplicate data entry and enable longitudinal patient tracking. A formalised model governance protocol should define minimum dataset sizes for retraining cycles and require stratified FST V and FST VI validation before any retrained model is deployed. Particular attention should be given to monitoring FST VI performance across retraining cycles to ensure that equity improvements are preserved rather than sacrificed for aggregate accuracy gains. Model performance monitoring should additionally include temporal drift assessment, with periodic retraining recommended when prediction distributions shift beyond defined thresholds, as the current model does not account for changes in disease prevalence or presentation patterns over time.

In conclusion, DermoAI provides preliminary but credible evidence that AI-assisted dermatological triage optimised for FST V–VI is technically feasible, clinically interpretable, and deployable within Rwanda's infrastructure constraints. The groundwork laid by this project in system architecture, clinical workflow integration, and equity-focused model design provides a foundation that future work, with access to the PASSION dataset and locally collected Rwandan cases, can build into a clinically certified tool with genuine population-level impact across Sub-Saharan Africa.

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